# Expectations Formation with Fat-Tailed Processes: Evidence and Theory*

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**Abstract**

Using a large sample of forecasts of 22 firm-level variables, we document three facts: (i) the relationship between forecast revisions and future forecast errors is strongly non-linear, indicating more overreaction to large shocks, (ii) the distributions of the underlying processes have fat tails, and (iii) extreme realizations tend to mean-revert. Next, we formally show that these three facts are consistent with a model in which the underlying process is non-Gaussian, but forecasters do not fully recognize this. Finally, we provide additional evidence in support of our theory by showing that it can explain evidence from an online forecasting experiment, that the non-linear relationship between errors and revisions is also present in macroeconomic forecasts, and that it provides an explanation for the presence of non-linearity in the momentum of stock returns.

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The recent literature on expectations formation has accumulated evidence that subjective forecast errors are predictable. To explain this predictability, theories of expectations formation have been proposed with a variety of ingredients such as information frictions (Coibion and Gorodnichenko, 2015), sluggish updating (Bouchaud et al., 2019; Ma et al., 2020), representativeness bias (Bordalo et al., 2019, 2020a), and imperfect memory (Afrouzi et al., 2023). A common feature of these theories is that they assume that the underlying data-generating process (DGP) is Gaussian. However, in many economic applications, tails are "fat", meaning that extreme outcomes occur much more frequently than in a Gaussian distribution (Gabaix, 2009). Such extreme shocks are common in dynamic processes about which agents need to continuously update their expectations, such as stock returns (Fama, 1965), firm growth (Stanley et al., 1996), and income growth (Guvenen et al., 2014). Therefore, understanding how large non-Gaussian shocks shape expectations formation is important for studying many dynamic financial decisions, such as corporate investment, household consumption-savings, or portfolio choice.

In this paper, we show that large shocks affect expectations formation differently from small shocks. We begin with an empirical analysis of forecasts made by sell-side analysts about over twenty firm outcomes, including sales growth and earnings per share. Using these data, we document three robust empirical facts: (i) the relationship between forecast revisions and future forecast errors is strongly non-linear for all firm-level outcomes, with more overreaction to extreme shocks; (ii) the distributions of the underlying processes have fat tails; and (iii) the (objective) expectation of future realizations is non-linear in current realizations, with persistence in the bulk and mean reversion in the tails. Next, we build an expectations formation model that connects these facts. The key ingredients in our model are that (i) large shocks come from a fat-tailed process and are less persistent than small shocks, and (ii) forecasters fail to recognize this and incorrectly respond to extreme shocks the same way they respond to non-extreme ones. We provide one microfoundation based on forecasters' insensitivity to signal strength, generalizing the theory in Augenblick et al. (2025) to a continuous-state time-series setting. Finally, we show that the non-linear relationship between errors and revisions is also present in macroeconomic forecasts, that our theory explains expectations formation in an online forecasting experiment, and that it provides an explanation for the presence of non-linearity in the momentum of stock returns. Overall, our analysis suggests that forecasters do not fully appreciate the differences between large and small shocks, which generates size-dependent overreaction—an insight that could be leveraged in models of household and firm behavior with fat tails (Guvenen et al., 2021; Boar et al., 2022).

Our main analysis uses a panel of analyst forecasts and realizations of 22 firm-level variables from IBES that cover the most important financial statement items, such as revenue, earnings, cash flow, and debt levels. The main benefit of these data is that the sample is large, with over 3 million

observations covering more than 20 years, which allows us to study the tails of the distribution. Using these data, the first and most important fact that we document is that the relationship between current forecast revisions and future forecast errors is strongly non-linear. For large revisions, there is a strong and robust *negative* relationship between revisions and future errors, indicating overreaction to news. For more moderate revisions, the relationship between revisions and future errors is less negative and sometimes positive, indicating much less overreaction (and sometimes underreaction). This non-linear relation between errors and revisions is present for all 22 firm-level variables in our sample, and survives a battery of robustness checks. For example, we show that this non-linearity is present in both individual *and* consensus forecasts, continues to be strong when we include firm fixed effects, does not vary with analysts' forecasting experience, and is robust to adjustments for firm-specific and time-varying volatility.

Our second empirical fact is that the realizations of all 22 firm-level variables in our sample have fat tails. To detect the presence of fat tails, we measure the kurtosis of each variable and the extent to which it is well-approximated by a power law in the tails of the distribution (Gabaix, 2009). We find that most variables in our sample have kurtosis above six and power law tail parameters below three (with $R^2$'s of log-log regressions above $0.99$), both of which indicate the presence of fat tails relative to a normal distribution. These fat tails do not arise from heterogeneity in volatility across firms (Wyart and Bouchaud, 2003) and do not arise from time-varying aggregate volatility.

Our third and final fact is that the conditional expectation of future realizations given the current realization is non-linear in a way that is reminiscent of the non-linearity in our first fact. In particular, we find that the relationship between current and one-year-ahead realizations is increasing and linear in the bulk of the distribution for most of the variables in our sample. However, in the tails of the distribution, this persistence is weaker for all variables and often changes sign, indicating that extreme values tend to mean-revert. As described below, this non-linearity helps inform the exact way the DGP in our model deviates from that of standard Gaussian models.

In the next part of the paper, we develop a model of expectations formation that is designed to connect the above three facts, while also being simple and portable. In the model, the DGP for the forecasting variable contains a persistent and a transitory component, which is common in models of dividend growth (Bansal and Yaron, 2004; Lettau and Wachter, 2007) and income dynamics (Guvenen et al., 2014). Our two key assumptions are that the transitory component follows a power law distribution, a tractable way of characterizing processes with fat tails (Gabaix, 2009), and that the persistent component has a thinner tail than the transitory component.[1]

[1]The latter assumption is consistent with our third fact: if the persistent component had the thicker tail, we would not observe mean reversion in the tails of the distribution. Additionally, our results generalize to the case in which the

We show that this model of the DGP is consistent with our second and third facts. The model generates fat tails by assumption, consistent with our second fact. To show that it generates the non-linear conditional expectation of future realizations conditional on current realizations—our third fact—we leverage a result from empirical Bayes theory known as Tweedie's formula (Efron, 2011). Although this expectation cannot be expressed in closed-form due to the non-normality, this result allows us to characterize it as a function of the observable density of realizations. We show that this result implies that the conditional expectation is locally linear when the density is locally Gaussian, as in the bulk of the distribution of most variables. However, when the density is locally non-Gaussian, as in the tails, the conditional expectation is no longer linear and is asymptotically decreasing in the current realization. Intuitively, very large realizations are likely due to the transitory component and, hence, are not likely to persist.

Given this DGP, consistent with our second and third facts, we can also explain the non-linear relationship between forecast errors and revisions—our first fact—with a single assumption: agents construct their forecasts ignoring fat tails. Formally, we assume that agents form forecasts according to the Kalman filter, which would be the rational expectation in our model if the transitory and persistent components were both normally distributed, and is equivalent to assuming that forecasters use the linear forecast with the lowest possible mean-squared error. Using results from the theory of regularly varying functions (Bingham et al., 1987; Geluk and de Vries, 2006), we prove that this assumption is enough to generate a non-linear error-revision relationship with more overreaction in the tails. This is because large revisions are driven by large realizations that asymptotically come from the transitory component of the DGP. While a rational forecaster would recognize that these extreme shocks are unlikely to persist, our agents who ignore fat tails do not and, therefore, overreact. In contrast, smaller revisions are more likely to come from the persistent component of the DGP, so our agents' responses are closer to those of a rational forecaster, generating less overreaction.

While ignoring fat tails can be viewed as a form of bounded rationality in which forecasters use a simple misspecified model (Fuster et al., 2010; Gabaix, 2019), an alternative microfoundation is that forecasters are insensitive to the strength of the signals that they observe. Insensitivity to signal strength is the key element of the theory in Augenblick et al. (2025) and is consistent with evidence in Ba et al. (2024) and Bastianello and Imas (2026). We show that this theory provides a natural microfoundation for why agents may partially ignore the presence of fat tails.

In the final part of the paper, we provide additional evidence in support of our theory. First, we show that our first fact—the non-linear relationship between forecast errors and revisions—

persistent component is regularly varying (Bingham et al., 1987).

replicates "out-of-sample" in three different settings. We start with an online forecasting experiment similar to Afrouzi et al. (2023). When we use a data-generating process with fat tails similar to those in our empirical analyses, we find that the relationship between errors and revisions is non-linear. In contrast, when forecasters forecast a similar process with no fat tails (as in Afrouzi et al. 2023), we find no evidence of a non-linear relationship between errors and revisions. These findings provide direct evidence that the non-linear relationship between errors and revisions in the data is driven by the fat tails of the DGP, which is the key result of our theory.

Our second setting consists of forecasts of macroeconomic outcomes from the Survey of Professional Forecasters (1968–2025). Of the 14 variables that are available, we find that the relationship between individual-level forecast errors and revisions is non-linear, with a more negative slope in the tails of the distribution for all but one variable. For most variables, the slope is positive in the bulk of the distribution and negative in the tails of the distribution, similar to our findings from firm-level forecasts. These findings suggest that the prevalence of more overreaction in the tails of the distribution is more general than our sample of firm-level forecasts.

Third, we show that our model makes predictions for return momentum that are supported by the data. We translate expectations of sales growth, one of the variables in our firm-level sample, into returns by applying the Campbell (1991) return decomposition with a constant subjective discount rate (as in Bouchaud et al. 2019 and Nagel and Xu 2022). Our model predicts that the relationship between past and future returns should be positive in the bulk of the distribution, where underreaction to news is more prevalent. In contrast, it predicts mean-reversion of returns in the tails, where agents do not fully recognize that the extreme shocks are not persistent. We find some support for this prediction in the universe of smaller stocks: for these stocks, momentum tends to mean-revert for extreme losers and winners.

Finally, we test whether, in the cross-section of *outcomes*, the degree of non-linearity in the error-revision relationship is greater for variables that have fatter tails. Holding all other parameters of the data-generating process constant, our theory predicts that this should be the case. Across both the firm-level and macroeconomic variables, we find that this tends to be the case: there is more overreaction to large shocks when a variable has higher kurtosis or a lower power law tail parameter, both of which indicate fatter tails. While this evidence is only based on the set of variables that we study (22 firm-level and 14 macroeconomic ones), it supports the key implication of our theory: fat tails can be the source of overreaction to extreme shocks.

**Related literature.** A longstanding literature has used survey data on expectations to study economic behavior (Manski, 2004), but recent literature has shifted the emphasis to the study of forecast error predictability (e.g., Coibion and Gorodnichenko 2015). This paper contributes to this branch of literature in two ways. First, we contribute to the empirical literature that has documented under- and overreaction across many forecasting variables and horizons. Broadly speaking, this literature tends to find evidence of underreaction when looking at shorter-term or consensus forecasts (Coibion and Gorodnichenko, 2015; Bouchaud et al., 2019), and overreaction when looking at longer-term or individual forecasts (Bordalo et al., 2019, 2020a; Wang, 2021). Relative to this empirical literature, our contributions are to provide field evidence of both under- and overreaction within the *same* forecasting variable and horizon, and lab evidence that the degree of overreaction can vary within a sample depending on the tail parameter in the DGP.

Second, we contribute to the literature that proposes models of belief formation that can generate under- and overreaction, including constant-gain learning (Nagel and Xu, 2022), selective recall (Bordalo et al., 2020b), and biased perceptions of autocorrelation (Wang, 2021). Most of this literature works with models in which data-generating processes are Gaussian, implying that conditional expectations are linear. In contrast, our empirical analyses highlight how this relationship can be quite non-linear, and we provide a theory that links this non-linearity to the presence of under- and overreaction within a given process. Our focus on non-Gaussian dynamics is similar to Kozlowski et al. (2020), and our assumption that forecasters (partially) ignore non-Gaussian dynamics is inspired by the literatures on bounded rationality (Fuster et al., 2010; Gabaix, 2019; Molavi et al., 2024) and insensitivity to signal strength (Augenblick et al., 2025; Ba et al., 2024; Bastianello and Imas, 2026). However, another possible microfoundation for this assumption is Bayesian learning, which can occur relatively slowly with one time-series (Farmer et al., 2024) and generate an in-sample relationship between errors and revisions even under full rationality (Singleton, 2021; Bianchi et al., 2022).[2] We choose not to focus on learning directly because, empirically, we find that the non-linearity in the error-revision relationship does not vary with analysts' forecasting experience.

Finally, given its focus, our paper is also related to recent work investigating non-Gaussianity and non-linearity in expectations formation. Like us, Kwon and Tang (2026) propose a model of belief formation with non-Gaussian dynamics. In their model, news events belong to categories with different power law distributions, and forecasters have diagnostic expectations, causing them to overreact to news from categories with fatter tails and underreact to news from categories with thinner tails. This prediction is similar to that of our model. An important difference is that our theory explains why over- and underreaction would vary with the *size* of shocks: overreaction

[2]See Dew-Becker et al. (2024) for a characterization of Bayesian learning with arbitrary non-Gaussian dynamics.

is more pronounced in response to extreme shocks because forecasters fail to realize their lower persistence.[3] Also related is Graeber et al. (2026), who analyze the S-shaped relationship between returns and earnings surprises, which loosely map into revisions and forecast errors. In their theory, the strong sensitivity between returns and surprises around zero comes from overreaction that occurs at a category boundary, while the lower sensitivity away from zero comes from dampening within a category due to noisy perceptions. Our main fact—a more negative error-revision relationship in the tails than in the bulk—points to a different mechanism in our setting.

Through our model of the DGP with fat tails, we connect the expectations formation literature with the literature on power laws. The omnipresence of power laws (Gabaix, 2009) suggests that the misperception of fat tails that we document is likely important for understanding subjective forecasts in other settings. The facts that we document about the data-generating process of firm-level realizations are consistent with the literature on firm dynamics. For example, we find that sales growth (rather than its level) has fat tails, which is a violation of Gibrat's law that was first documented by Stanley et al. (1996) and recently emphasized by Boar et al. (2022). Our third fact is consistent with Jaimovich et al. (2025), who find that revenue is more persistent in the bulk of the distribution than in the tails and exhibits fat tails.

**Outline.** Section 1 describes our data on forecasts of 22 firm-level variables. Section 2 documents our three main facts. Section 3 describes the model that we build to connect and explain these facts. Section 4 provides additional evidence in support of our theory from an online forecasting experiment, macroeconomic forecasts, and data on stock returns.

# 1 Firm-Level Forecast Data

## 1.1 Data Source and Sample

Our analysis primarily relies on an annual panel between 2000 and 2023 of forecasts issued by over 38,000 security analysts who follow around 36,000 U.S. and international firms. These forecasts cover 22 different annual firm-level variables at different horizons. We obtain these analyst-level forecasts from LSEG IBES Unadjusted Detail files (henceforth "Detail"). In robustness analysis, we also study consensus analyst forecasts, which come from the LSEG IBES Summary Statistics files (henceforth "Summary"). Both of these files contain current estimates as of the third Wednesday of each month. We extract the forecasts issued in the third month of each fiscal year $t+1$, which

[3] An alternative interpretation is that Kwon and Tang (2026) provide a theory for why over- and underreaction varies across categories, while our model provides a theory for why over- and underreaction varies *within* a category.

ensures that the results for fiscal year $t$ have been released. We focus on forecasts for two different horizons: fiscal years $t+1$ and $t+2$. Along with one and two-year forecasts, LSEG IBES also provides data on the realizations of the forecasted variables, which the provider ensures are measured consistently with the forecasts made by analysts.

## 1.2 Definition of Forecasting Variables

We study forecasts and realizations of all the variables available in the Summary and Detail files. The list of these variables is provided in Table 1. These variables fall into one of three categories: accounting items in local currency (e.g., Revenue), items in local currency on a "per-share" basis (e.g., Earnings Per Share), and accounting ratios in percentage points (e.g., Return on Assets). To avoid imposing unnecessary sample filters, we choose to include all 22 variables regardless of their data coverage, which we discuss below.

The only transformation we apply to the variables of interest is done to ensure they are stationary. Out of the 22 variables, three are accounting ratios that are already in percentage points, which are presumably already stationary and therefore not transformed. The remaining 19 variables are accounting items in local currency or on a per-share basis. To make them stationary, we convert them from levels to growth rates, as in Bordalo et al. (2020a). More specifically, let $X_{it+h}$ denote the realization of one of these variables for firm $i$ at $t+h$. We transform this variable into a log growth rate according to $g_{it+h} = \log X_{it+h} - \log X_{it+h-1}$, which requires both $X_{it+h}$ and $X_{it+h-1}$ to be positive. To simplify exposition, for the remaining three accounting ratios, we define $g_{it+h}$ as the untransformed realization for firm $i$ at $t+h$. Moving forward, we refer directly to $g_{it+h}$ as our realization of interest.

## 1.3 Definitions of Forecast Errors and Revisions

For each firm $i$ and each year $t$, we denote $F_t^j X_{it+h}$ as the forecast of $X_{it+h}$ made in year $t$ by analyst $j$ that comes from the Detail files. We denote the consensus forecast by $F_t X_{it+h}$, which is defined as the mean forecast across analysts from the Summary files. For both the analyst-level and consensus, we use forecasts that are active three months after the end of fiscal year $t$, which ensures that the results for fiscal year $t$ have been released by the time these forecasts are made. This also maximizes the chance that the forecast is not "stale". Similarly, we denote $F_{t-1}^j X_{it+h}$ and $F_{t-1} X_{it+h}$ as the analyst-level and consensus forecasts extracted three months after the end of fiscal year $t-1$.

For forecasts of the three untransformed accounting ratios, we can work with our raw forecasts

directly, so $F_t^j g_{it+h} = F_t^j X_{it+h}$. For the remaining 19 variables, we define forecasts of growth rates as $F_t^j g_{it+h} = \log F_t^j X_{it+h} - \log F_t^j X_{it+h-1}$, with the convention that $F_t^j X_{it} = X_{it}$. For consensus forecasts, variables are constructed analogously without the $j$ superscript. Note that the way we translate from forecasts of levels to growth rates implicitly ignores a Jensen's inequality term because $\log F_t^j X_{it+h} \neq F_t^j \log X_{it+h}$. We ignore this adjustment for simplicity, but we do not think it materially affects our analysis for two reasons. First, a constant Jensen's term would simply shift the unconditional level of forecasts, while our analysis focuses on the conditional properties of forecast errors. Second, we show that our results are robust to using percent (instead of log) growth.

Having defined $F_t^j g_{it+h}$ for each of the 22 variables, we then construct forecast errors and revisions following the literature on expectations formation (Coibion and Gorodnichenko, 2015; Bouchaud et al., 2019):

$$ERR_{it+1}^j = g_{it+1} - F_t^j g_{it+1}, \quad REV_{it}^j = F_t^j g_{it+1} - F_{t-1}^j g_{it+1}.$$

For consensus forecasts, we again use the analogous definitions without the $j$ superscript.

## 1.4 Summary Statistics

Table 1 shows summary statistics on the realizations, analyst-level forecast errors, and analyst-level forecast revisions for each of our 22 variables. Table IA.1 shows the same table for consensus-level forecasts. For the sample of each variable in the table, we remove outliers by dropping the top and bottom one percent of each distribution. The number of observations reported in the table is the number of firm-year observations for which we can compute all of the variables of interest. The size of our sample varies a lot across variables, but is in general quite large. Earnings Per Share (EPS), which is studied by most of the literature on analyst forecasts, is one of the most covered variables. However, many other variables have good coverage, such as Net Sales, which is even more covered than EPS with over 407k observations. ROE and Net Income are nearly as covered as EPS. These four variables, as well as EBITDA, EBIT, and Pre-Tax Profit, have more than 200k observations. 13 variables have more than 100k observations, and 19 more than 50k. Two variables, EPS Before Goodwill and Funds From Operations Per Share, have fewer than 10k observations. In the following section, we report analysis results for all variables, ranking them by their number of observations.

In the main text of this paper, we focus on analyst-level forecasts to avoid the possibility that our results are affected by aggregating across forecasters (Bordalo et al., 2020a), but also to be consistent with our experimental evidence in Section 4.1. However, after presenting our main

Table 1: Summary Statistics: Analyst-Level Forecasts

| | Transformation | Realizations | | Errors | | Revisions | | | |
|---|---|---|---|---|---|---|---|---|---|
| Variable | (1) | Mean (2) | SD (3) | Mean (4) | SD (5) | Mean (6) | SD (7) | # of Analysts (8) | # of Obs. (9) |
| Net Sales | Growth | 0.06 | 0.16 | -0.00 | 0.10 | -0.02 | 0.13 | 6.33 | 399809 |
| Earnings Per Share | Growth | 0.06 | 0.44 | -0.04 | 0.31 | -0.07 | 0.34 | 6.35 | 360011 |
| Return On Equity | Level | 0.13 | 0.13 | -0.01 | 0.07 | -0.01 | 0.07 | 5.17 | 345082 |
| Net Income | Growth | 0.06 | 0.43 | -0.03 | 0.31 | -0.06 | 0.34 | 6.24 | 325230 |
| EBITDA | Growth | 0.06 | 0.29 | -0.02 | 0.20 | -0.04 | 0.23 | 5.82 | 280111 |
| Pre-tax Profit | Growth | 0.06 | 0.41 | -0.04 | 0.30 | -0.06 | 0.32 | 5.71 | 269883 |
| EBIT | Growth | 0.06 | 0.37 | -0.02 | 0.27 | -0.07 | 0.31 | 6.32 | 231455 |
| GAAP EPS | Growth | 0.06 | 0.55 | -0.08 | 0.40 | -0.03 | 0.41 | 4.85 | 171523 |
| Book Value Per Share | Growth | 0.05 | 0.17 | -0.01 | 0.16 | -0.02 | 0.13 | 3.82 | 170561 |
| Return On Assets | Level | 0.05 | 0.06 | -0.01 | 0.03 | -0.01 | 0.03 | 3.47 | 157063 |
| Gross Margin | Level | 0.41 | 0.21 | -0.00 | 0.04 | -0.01 | 0.03 | 4.56 | 150220 |
| Net Asset Value | Growth | 0.06 | 0.18 | -0.00 | 0.17 | -0.02 | 0.14 | 3.12 | 119885 |
| Cash Flow Per Share | Growth | 0.04 | 0.48 | -0.03 | 0.41 | -0.07 | 0.48 | 3.78 | 112323 |
| Net Debt | Growth | 0.04 | 0.41 | 0.07 | 0.41 | 0.10 | 0.45 | 3.69 | 90126 |
| Enterprise Value | Growth | 0.04 | 0.32 | 0.04 | 0.29 | 0.02 | 0.26 | 2.98 | 88791 |
| Dividend Per Share | Growth | 0.06 | 0.37 | -0.02 | 0.31 | -0.04 | 0.32 | 2.82 | 63840 |
| Capital Expenditures | Growth | 0.04 | 0.41 | -0.06 | 0.45 | 0.10 | 0.52 | 2.89 | 62099 |
| Operating Profit | Growth | 0.06 | 0.40 | -0.05 | 0.28 | -0.05 | 0.30 | 3.57 | 57756 |
| EBITDA Per Share | Growth | 0.05 | 0.31 | -0.02 | 0.21 | -0.05 | 0.25 | 1.44 | 16824 |
| Cash Earnings Per Share | Growth | 0.05 | 0.31 | -0.02 | 0.27 | -0.06 | 0.28 | 1.85 | 11578 |
| Funds From Operations Per Share | Growth | 0.03 | 0.14 | -0.00 | 0.08 | -0.04 | 0.11 | 4.74 | 9279 |
| EPS Before Goodwill | Growth | 0.09 | 0.36 | -0.01 | 0.27 | -0.06 | 0.30 | 2.19 | 2558 |

*Notes*: This table reports the mean, standard deviation, and number of observations for each one of the 22 variables forecasted by analysts in our sample. The sample includes all observations for which an analyst's forecast error and revision are available for a given variable. Variables are ranked by decreasing number of observations. 19 variables are transformed into log growth as explained in the text; 3 are left untransformed. Column (8) reports the average number of analysts forecasts available per firm-year, and column (9) reports the total number of observations.

results in Section 2.1, we discuss the results from using consensus forecasts, which are presented in the Appendix. Additionally, we also conduct a series of robustness checks that are also presented in the Appendix, including showing that our results do not depend on our particular log growth transformation.

# 2 Three Facts

In this section, we document the three key facts that motivate the model of expectations formation that we develop in Section 3.

## 2.1 Fact #1: Non-Linear Relationship Between Forecast Errors and Revisions

Our first and central empirical fact is based on a regression of forecast errors on forecast revisions. This regression was introduced by Coibion and Gorodnichenko (2015) (CG) and takes the following

form, for forecaster $j$, firm $i$ and year $t$:

$$ERR^j_{it+1} = \alpha + \beta REV^j_{it} + e^j_{it+1} \tag{1}$$

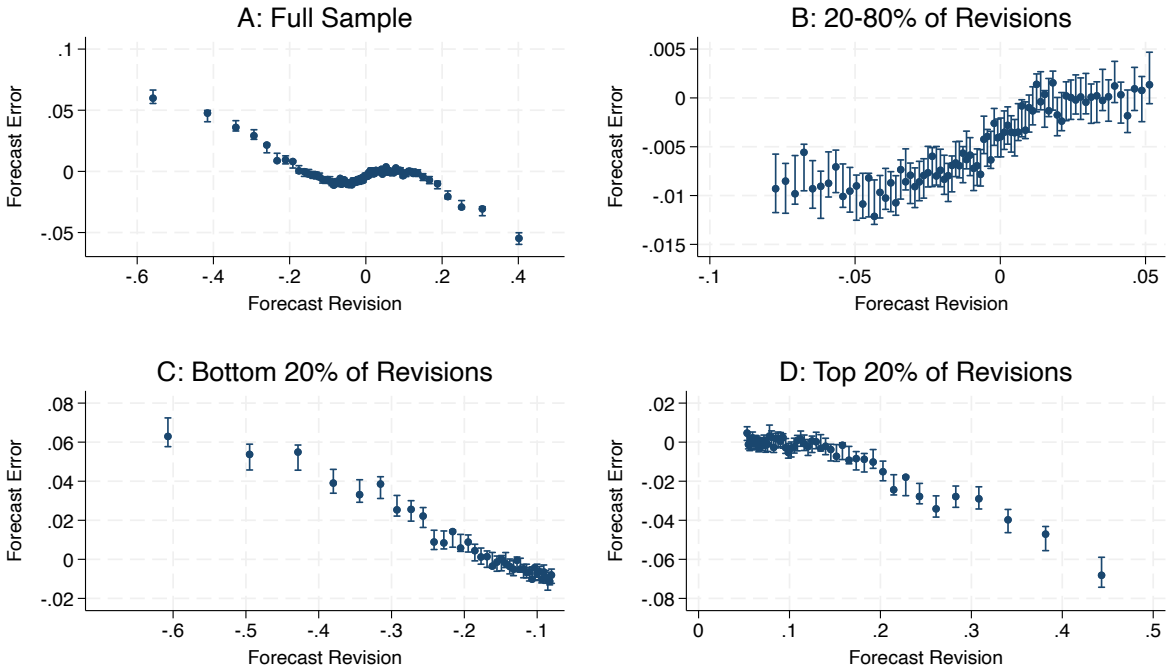
where the forecast error $ERR^j_{it+1}$ and revision $REV^j_{it}$ for each variable are defined in Section 1. The slope coefficient is typically used to distinguish between different models of expectations formation, requiring only panel data on expectations and realizations. The rational expectations hypothesis predicts $\beta = 0$, since forecast errors cannot be predicted using data in the forecaster's information set. When this regression is run using consensus forecasts, $\beta > 0$ is typically interpreted as evidence of information frictions, as in models of sticky or noisy information (Coibion and Gorodnichenko, 2015). With individual forecasts, $\beta > 0$ is interpreted as non-Bayesian underreaction (Bouchaud et al., 2019), while $\beta < 0$ is interpreted as overreaction (Bordalo et al., 2020a). A key feature of prior literature is that it restricts analysis to linear functional forms, as in equation (1). While this is a natural starting point, especially in settings with small sample sizes, this section uses our large sample of firm-level expectations to provide evidence that this relationship is non-linear.

**Graphical evidence.** Figure 1 provides a non-parametric look at the relationship between forecast errors and revisions for Net Sales, the best-populated variable with more than 400k observations. The figure shows that the relationship between forecast errors $ERR^j_{it+1}$ and forecast revisions $REV^j_{it}$ is highly non-linear. Panel A shows a binned scatterplot on the full sample with 100 bins; Panels B, C, and D show the same plot separately for observations in the 20-80th, 0-20th, and 80-100th percentiles of revisions. For revisions in the bulk of the distribution, Panel B shows that errors are increasing in revisions, consistent with forecasters underreacting to news that causes moderately-sized revisions in forecasts. This finding is not novel to our paper: it is consistent with the underreaction in analysts' EPS forecasts for US firms (Bouchaud et al., 2019), as well as managers' revenue forecasts in the US and Italy (Ma et al., 2020).

Panels C and D of Figure 1 show that for large (positive or negative) revisions, the positive relationship between forecast errors and revisions reverses and becomes negative: large positive (negative) forecast revisions are predictive of negative (positive) future forecast errors. Unlike the relationship in the bulk of the distribution, which is consistent with underreaction, this finding is consistent with *overreaction* to news. In other words, forecasters appear to overreact in response to news that generates large revisions, while underreacting to more moderate news.

**Regression evidence.** While the results in Figure 1 are specific to Net Sales, we now show that this same non-linearity is present for almost all of the other variables in Table 1. To do so, we

Figure 1: Non-Linear Error-Revision Relationship: Analyst-Level Net Sales Forecasts




*Notes*: This figure shows binned scatterplots of analyst-level forecast errors on forecast revisions for Net Sales. Panel A shows the results for the entire sample; Panel B restricts the sample to the 20-80th percentiles of revisions; Panel C restricts the sample to below the 20th percentile of revisions; Panel D restricts the sample to above the 80th percentile of revisions. Vertical bars represent 95% confidence intervals, assuming the relationship is piecewise linear and continuous (option "ci(1 1)" in Stata command "binsreg").

estimate the following variant of the regression in (1) separately for each variable with OLS:

$$ERR^j_{it+1} = \alpha + \beta REV^j_{it} + \gamma REV^j_{it} \times (P^+_{ijt} + P^-_{ijt}) + \delta^+ P^+_{ijt} + \delta^- P^-_{ijt} + e^j_{it+1} \tag{2}$$

where $P^+_{ijt}$ (resp. $P^-_{ijt}$) is equal to 1 if $REV^j_{it}$ belongs to the top (resp. bottom) 20% of all revisions for this variable. In this specification, $\beta$ is the error-revision coefficient for revisions between the 20th and 80th percentiles of the distribution of revisions, while $\gamma$ captures the change in this slope in the tails of the distribution (i.e., below the 20th and above the 80th percentiles). Equation (2) crucially allows intercepts to differ in the two tails, as is the case in Figure 1, but estimates a pooled slope in the tails of the distribution. We choose to estimate a pooled slope to improve power and because our focus is on the non-linearity of the error-revision relationship rather than any asymmetries.[4]

[4]Table IA.2 shows the results of a specification that allows for asymmetric slopes. While the strong non-linearity shown here is present for all variables, some variables exhibit asymmetry.

Table 2: Non-Linear Error-Revision Relationship: Analyst-Level Forecasts

| $\text{Error}_{it+1}$ | $\text{Revision}_{it}$ | | $\text{Revision}_{it} \times$ Top/Bottom 20% of $\text{Revision}_{it}$ | | Top 20% of $\text{Revision}_{it}$ | | Bottom 20% of $\text{Revision}_{it}$ | | |
|---|---|---|---|---|---|---|---|---|---|
| Variable | $\widehat{\beta}$ (1) | $t$-stat (2) | $\widehat{\gamma}$ (3) | $t$-stat (4) | $\widehat{\delta^+}$ (5) | $t$-stat (6) | $\widehat{\delta^-}$ (7) | $t$-stat (8) | # of Obs. (9) |
| Net Sales | 0.11 | 6.31 | -0.26 | -14.99 | 0.02 | 5.86 | -0.02 | -5.09 | 399809 |
| Earnings Per Share | 0.08 | 2.97 | -0.26 | -9.44 | 0.02 | 1.48 | -0.07 | -5.09 | 360011 |
| Return On Equity | 0.23 | 9.75 | -0.27 | -11.70 | 0.00 | 2.56 | -0.02 | -5.99 | 345082 |
| Net Income | 0.04 | 1.53 | -0.25 | -11.08 | 0.02 | 1.63 | -0.06 | -4.91 | 325230 |
| EBITDA | -0.04 | -1.51 | -0.15 | -5.28 | 0.01 | 1.72 | -0.02 | -2.00 | 280111 |
| Pre-tax Profit | 0.01 | 0.47 | -0.22 | -9.05 | 0.03 | 2.38 | -0.05 | -3.07 | 269883 |
| EBIT | -0.05 | -1.60 | -0.22 | -8.12 | 0.03 | 2.94 | -0.03 | -2.06 | 231455 |
| GAAP EPS | -0.02 | -0.73 | -0.14 | -4.99 | 0.01 | 0.31 | -0.04 | -1.91 | 171523 |
| Book Value Per Share | -0.06 | -1.52 | -0.42 | -8.53 | 0.01 | 2.44 | -0.04 | -6.90 | 170561 |
| Return On Assets | 0.41 | 10.35 | -0.45 | -10.15 | -0.01 | -5.32 | -0.01 | -8.49 | 157063 |
| Gross Margin | 0.09 | 3.90 | -0.15 | -6.23 | 0.00 | 1.70 | -0.01 | -6.62 | 150220 |
| Net Asset Value | 0.04 | 0.99 | -0.56 | -11.26 | 0.03 | 7.75 | -0.03 | -4.61 | 119885 |
| Cash Flow Per Share | -0.28 | -15.86 | -0.06 | -3.27 | 0.05 | 4.05 | -0.00 | -0.36 | 112323 |
| Net Debt | -0.17 | -8.31 | -0.12 | -5.46 | 0.12 | 10.06 | 0.12 | 9.14 | 90126 |
| Enterprise Value | -0.20 | -5.82 | -0.26 | -8.25 | 0.11 | 11.91 | -0.01 | -0.82 | 88791 |
| Dividend Per Share | -0.12 | -4.54 | -0.17 | -6.15 | -0.01 | -1.13 | -0.03 | -3.08 | 63840 |
| Capital Expenditures | -0.36 | -21.93 | -0.16 | -8.53 | 0.06 | 5.20 | -0.05 | -4.32 | 62099 |
| Operating Profit | 0.12 | 3.07 | -0.35 | -7.83 | 0.04 | 3.40 | -0.06 | -4.54 | 57756 |
| EBITDA Per Share | -0.09 | -2.84 | -0.13 | -3.14 | 0.02 | 2.24 | -0.03 | -1.85 | 16824 |
| Cash Earnings Per Share | -0.19 | -4.95 | -0.20 | -3.83 | 0.03 | 1.98 | -0.02 | -1.01 | 11578 |
| Funds From Operations Per Share | -0.01 | -0.25 | -0.09 | -1.72 | 0.01 | 1.61 | -0.02 | -2.00 | 9279 |
| EPS Before Goodwill | -0.13 | -2.29 | -0.11 | -1.36 | 0.04 | 2.82 | -0.09 | -3.21 | 2558 |

*Notes*: This table reports the OLS estimates of equation (2), for each one of the 22 variables. We report the estimates of $\beta$ (column 1), $\gamma$ (column 3), $\delta^+$ (column 5), and $\delta^-$ (column 7), along with the corresponding $t$-statistics with standard errors clustered by firm and time. Variables are ranked by number of observations.

Table 2 shows the estimated coefficients in (2) for all of the 22 variables in our sample.[5] Starting with Net Sales in the first line, we find $\widehat{\beta} = 0.1$ ($t = 6.3$) and $\widehat{\gamma} = -0.26$ ($t = -15$). This is consistent with Figure 1: in the bulk of the distribution, errors are positively related to revisions with slope $\widehat{\beta} = 0.1$, consistent with underreaction, while in the tails of the distribution, errors are negatively related to revisions with slope $\widehat{\beta} + \widehat{\gamma} = 0.1 - 0.26 = -0.16$, consistent with overreaction.

The remaining rows in Table 2 show that the non-linear relationship between forecast errors and revisions that we find for sales is present for all 22 variables in our data. For 20 out of 22 variables,

[5] Table IA.3 shows that the results are similar without removiing outliers: $\widehat{\gamma}$ is negative for all variables and statistically significant at the 5% level for 19 of them, while the implied slope in the tails, $\widehat{\beta} + \widehat{\gamma}$, remains negative for 21 variables.

we find that $\widehat{\gamma}$ is negative and significant at the 1% level. $\widehat{\beta}$ varies more across variables, with some variables having underreaction in the bulk of the distribution (e.g., Net Sales, Earnings Per Share, Return On Equity), while others have overreaction (e.g., Enterprise Value, Dividend Per Share). Overall, across all variables available in our data, these results show that the error-revision relationship is highly non-linear.

**Robustness.** We now discuss a battery of robustness checks that we conduct to ensure that the non-linear relationship between forecast errors and revisions is robust. We discuss each one of these robustness checks in a different paragraph below. For each robustness check, we produce an analogous version of Table 2 that is referenced and shown in the Appendix.

First, we check that the relationship continues to hold for consensus forecasts. A potential issue with analyst-level forecasts is that they are sensitive to analyst-level noise, which introduces a negative bias in the estimate of $\beta$ in (1) (de Silva and Thesmar, 2024). Therefore, our results could be driven by the fact that expectation noise is larger for bigger revisions, inducing a stronger bias in the tails. However, Table IA.4 shows that this is not the case. In particular, when working with consensus forecasts, we find that $\widehat{\gamma}$ is statistically significantly negative for all but 5 variables, with magnitudes that are similar to the analyst-level results.

Second, we show that our results are not sensitive to our log growth transformation. Two concerns with this transformation are that we ignore a Jensen's inequality term when constructing $F_t^j g_{it+h}$ (the log of a forecast is not necessarily the forecast of the log), and that log growth is only defined for positive values. To address these concerns, we redefine growth as $g_{it+1} = \frac{X_{it+1}-X_{it}}{X_{it}}$ and forecasted growth at horizon $h$ as $F_t^j g_{it+h} = \frac{F_t^j X_{it+h}-F_t^j X_{it+h-1}}{F_t^j X_{it+h-1}}$ (for the 19 variables that are transformed). The benefit of this approach is that our construction of $F_t^j g_{it+1}$ is no longer affected by Jensen's inequality.[6] Table IA.5 shows that, when using percent growth, $\widehat{\gamma}$ is negative for 18 of the 19 transformed variables, and significantly negative for 15 variables with similar magnitudes.

Third, we argue later in the paper that the driver of the non-linear error-revision relationship is the presence of fat tails. One concern with this argument is that fat tails may instead reflect a mixture of thin-tailed processes with heterogeneous variances (Wyart and Bouchaud, 2003). To address this concern, we restrict our sample to firms for which we have at least 10 observations and compute the standard deviation of the realization of each variable. We then normalize realizations and forecasts by this firm-specific standard deviation. Table IA.6 shows that, even after adjusting for firm-specific

[6]We still need to ignore a Jensen's inequality term when computing $F_{t-1}^j g_{it+1}$, which is needed to compute forecast revisions. However, the fact that our results are similar when using percent growth to remove this effect on $F_t^j g_{it+1}$ makes us less concerned that this is an important issue for our results.

volatility, our results are quantitatively similar with $\widehat{\gamma}$ estimated to be significantly negative for all but one variable. In Table IA.7, we perform a similar analysis but instead normalize by the annual cross-sectional standard deviation of each realization. Again, we find quantitatively similar results, suggesting our findings are not driven by time-varying aggregate volatility.

Fourth, one concern is that the non-linear relationship between forecast errors and revisions may be an artifact of pooling observations across firms that have heterogeneous data-generating processes. Table IA.8 reproduces Table 2 after including firm fixed effects in (2).[7] We find that $\widehat{\gamma}$ is statistically significantly negative for 20 variables, with magnitudes that are similar to those in Table 2, suggesting that the non-linear error-revision relationship is not driven by firm-level differences.

Finally, part of the literature on expectation formation has emphasized that rational learning can occur slowly when the data-generating process is unknown (Farmer et al., 2024). While difficult to test directly, we attempt to detect the importance of learning by splitting our sample into four quartiles of analyst-firm experience, measured by the number of prior firm-years in which the analyst issued forecasts for the same firm.[8] Table IA.9 shows that the non-linear pattern is quantitatively similar in all quartiles, suggesting that learning is not a first-order driver of our results. Nevertheless, we acknowledge that learning could provide a microfoundation for the assumption in Section 3 that forecasters do not fully recognize the presence of fat tails.

## 2.2 Fact #2: Fat Tails in Distribution of Realizations

The strong evidence in Section 2.1 that the relationship between forecast errors and revisions is non-linear points towards a different treatment of large versus small shocks. This evidence is hard to square with established models of expectation formation, which feature linear DGPs (e.g., AR1s) and linear expectations models. Next, we document two additional facts about the realizations of the variables being forecasted that guide our theory by informing our model of the DGP.

Our second fact is that the distributions of realizations have thick tails. It is well known that many financial and economic variables have tails that are fatter than a normal distribution (Gabaix, 2009). For example, the distribution of firm sizes follows a power law distribution with a tail coefficient of one (Axtell, 2001), which is known as Zipf's law, and the distribution of growth rates in COMPUSTAT follows a Laplace distribution, which has fatter tails than a normal distribution (Stanley et al., 1996; Bottazzi and Secchi, 2006). To measure the extent to which our variables are

[7] We can include firm fixed effects even with a short panel because we have analyst-level variation. The results are quantitatively similar if we run the regression among firms for which we have at least 10 observations.

[8] Our results are quantitatively similar if we instead measure experience aggregating across firms.

Gaussian, we use two metrics on the set of realizations at the firm-year level. The first is kurtosis, which would be three if the variables are normal. The second is the slope coefficient in a log rank-log size regression, which is a standard way of estimating the tail parameter of a power law distribution that would be infinity for a normal distribution (Gabaix, 2009). Specifically, for each variable $g_{it}$, we regress the log rank of $|g_{it}|$ on $\log |g_{it}|$ using observations between the 90th and 99th percentiles of $|g_{it}|$. We then report the negative of the slope coefficient, which corresponds to our estimate of the tail parameters.

Table 3 shows that the realizations of all 22 forecasting variables in our sample have tails that are much fatter than those of a Gaussian distribution. Apart from Return on Assets and Gross Margin, all variables have a tail coefficient that is below three. These imply very fat tails: a tail parameter of two or lower implies that the distribution has an infinite variance. Apart from Capital Expenditures, Enterprise Value, and Gross Margin, all variables have a kurtosis of five or more. These estimates are very far from normal, and formal tests based on kurtosis reject the null of a normal distribution at conventional significance levels.

**Robustness.** In the Appendix, we show that the presence of fat tails documented in Table 3 is robust to the alternative variable definitions discussed in the robustness checks in Section 2.1. First, Table IA.10 shows that fat tails are still present after dividing by firm-specific estimates of volatility, which suggests that they do not arise from a mixture of thin-tailed processes with heterogeneous variances (Wyart and Bouchaud, 2003). Second, Table IA.11 shows that adjusting for time-varying aggregate volatility does not change the results. Finally, Table IA.12 shows that tail coefficients remain low when using percent instead of log growth.

## 2.3 Fact #3: Non-Linear Conditional Expectation

The third and final fact that informs the specification of the data-generating process in our model is that the conditional expectation of the current realization is a non-linear function of the lagged realization. Figure 2 illustrates this point graphically for Net Sales, the best-populated variable. In Panel A, we plot the relationship between Net Sales at $t+1$ and Net Sales at $t$ using a binned scatter plot. As is evident from the figure, the conditional expectation of Net Sales conditional on its lag exhibits significant non-linearity reminiscent of the non-linearity in the relationship between forecast errors and revisions in Figure 1. Panels B, C, and D of Figure 2 zoom in on the three parts of the distribution of past growth. These panels show that Net Sales is approximately increasing and linear in the bulk of the distribution, while the slope is severely attenuated, or even reversed, in the tails of the distribution. This evidence is consistent with Jaimovich et al. (2025), who find that

Table 3: Fat Tails in the Distribution of Realizations

| | Transformation | Kurtosis | Log-Log Regression | | # of Obs. |
|---|---|---|---|---|---|
| | | | Tail Parameter | $R^2$ | |
| Variable | (1) | (2) | (3) | (4) | (5) |
| Net Sales | Growth | 5.86 | 2.67 | 0.99 | 140482 |
| Earnings Per Share | Growth | 6.41 | 2.39 | 0.99 | 129735 |
| Return On Equity | Level | 9.49 | 2.61 | 1.00 | 127268 |
| Net Income | Growth | 6.38 | 2.40 | 0.99 | 113862 |
| Pre-tax Profit | Growth | 6.25 | 2.45 | 0.99 | 102386 |
| EBITDA | Growth | 6.16 | 2.52 | 0.99 | 99731 |
| Book Value Per Share | Growth | 8.60 | 2.09 | 0.99 | 89884 |
| EBIT | Growth | 6.44 | 2.42 | 0.99 | 82100 |
| Return On Assets | Level | 6.46 | 3.40 | 0.99 | 79479 |
| GAAP EPS | Growth | 5.91 | 2.56 | 0.99 | 74210 |
| Net Asset Value | Growth | 7.92 | 2.16 | 0.99 | 71494 |
| Gross Margin | Level | 2.44 | 8.00 | 0.98 | 64483 |
| Cash Flow Per Share | Growth | 5.56 | 2.63 | 0.99 | 62446 |
| Dividend Per Share | Growth | 7.07 | 2.45 | 0.99 | 49402 |
| Net Debt | Growth | 6.57 | 2.33 | 0.99 | 48338 |
| Enterprise Value | Growth | 4.58 | 3.00 | 0.99 | 48140 |
| Capital Expenditures | Growth | 4.98 | 2.82 | 0.99 | 44623 |
| Operating Profit | Growth | 6.23 | 2.54 | 0.99 | 33033 |
| EBITDA Per Share | Growth | 6.26 | 2.41 | 0.99 | 15343 |
| Cash Earnings Per Share | Growth | 6.04 | 2.40 | 0.99 | 10013 |
| Funds From Operations Per Share | Growth | 8.39 | 2.38 | 0.99 | 3399 |
| EPS Before Goodwill | Growth | 6.25 | 2.87 | 0.99 | 2413 |
| Normal Distribution | · | 3 | $\infty$ | · | · |

*Notes*: This table reports the kurtosis and tail parameters of the 22 variables in our data. Tail parameters, reported in column (3), are obtained through the following procedure. For each variable $g_{it}$, we regress the log rank of $|g_{it}|$ on $\log |g_{it}|$, restricting ourselves to the centiles of $|g_{it}|$ between 90 and 99. We report the $R^2$ of these regressions in column (4). We restrict ourselves to realizations available in LSEG IBES. Variables are ranked by number of observations in the final column.

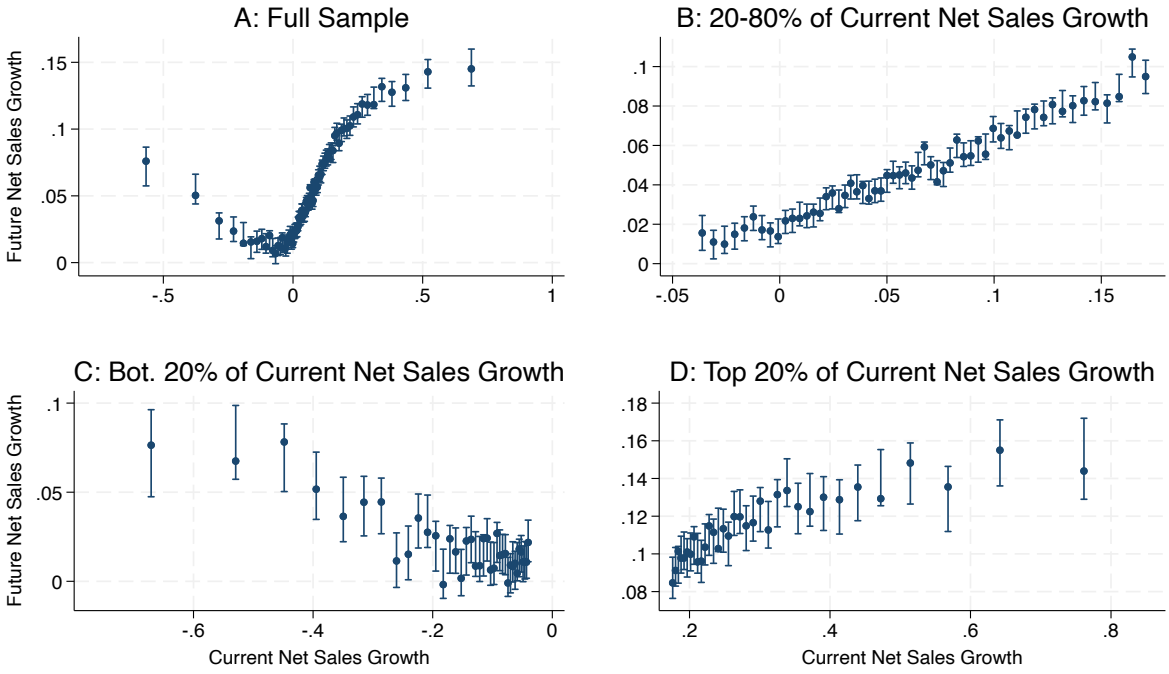
sales growth is more persistent in the bulk of the distribution than in the tails and exhibits fat tails.

While the evidence in Figure 2 is specific to Net Sales, we now show that this non-linear relationship holds for all other 22 variables. For each variable, $g_{it}$, we estimate the following regression with OLS:

$$g_{it+1} = \alpha_g + \beta_g g_{it} + \gamma_g g_{it} \times (M_{it}^+ + M_{it}^-) + \delta_g^+ M_{it}^+ + \delta_g^- M_{it}^- + e_{it+1} \tag{3}$$

where $M_{it}^+$ (resp. $M_{it}^-$) is equal to 1 if $g_{it}$ belongs to the top (resp. bottom) 20% of its distribution. This specification is analogous to equation (2), but for realizations instead of forecasts: $\beta_g$ is the

Figure 2: Non-Linear Conditional Expectation of Realizations: Net Sales




*Notes*: This figure shows binned scatter plots of current Net Sales on past Net Sales. Panel A shows the results for the entire sample; Panel B restricts the sample to the 20-80th percentiles of past Net Sales; Panel C restricts the sample to below the 20th percentile of past Net Sales; Panel D restricts the sample to above the 80th percentile of past Net Sales.

coefficient for the relationship between $g_{it+1}$ and $g_{it}$ in the bulk of the distribution, while $\gamma_g$ captures the change in this slope in the tails. As in (2), we allow the intercepts in the two tails to differ, but impose that the slopes are equal.

Table 4 shows that $\widehat{\beta}_g > 0$ in 20 out of 22 cases, and is statistically significant in 18 cases. Additionally, we find that $\widehat{\gamma}_g < 0$ in all cases, and it is statistically significant in all but one case. Overall, this evidence shows that the relationship between current and lagged values for our variables is highly non-linear, with persistence in the bulk and more mean-reversion in the tails of the distribution.

**Robustness.** In the Appendix, we show that this non-linear conditional expectation is robust to the alternative variable definitions discussed in the robustness checks in Section 2.1: Table IA.13 adjusts by firm-level volatility; Table IA.14 adjusts by aggregate volatility; and Table IA.15 uses percent instead of log growth. In all cases, the non-linearity is strong and pervasive.

Table 4: Non-Linear Conditional Expectation of Realizations

| $g_{it+1}$ | $g_{it}$ | | $g_{it} \times$ Top/Bottom 20% of $g_{it}$ | | Top 20% of $g_{it}$ | | Bottom 20% of $g_{it}$ | | |
|---|---|---|---|---|---|---|---|---|---|
| Variable | $\widehat{\beta}_g$ (1) | $t$-stat (2) | $\widehat{\gamma}_g$ (3) | $t$-stat (4) | $\widehat{\delta_g^+}$ (5) | $t$-stat (6) | $\widehat{\delta_g^-}$ (7) | $t$-stat (8) | # of Obs. (9) |
| Net Sales | 0.43 | 12.40 | -0.44 | -8.12 | 0.10 | 6.28 | 0.00 | 0.23 | 140482 |
| Earnings Per Share | 0.20 | 6.18 | -0.49 | -13.11 | 0.23 | 9.63 | -0.03 | -1.19 | 129735 |
| Return On Equity | 0.86 | 68.21 | -0.28 | -16.55 | 0.08 | 14.23 | 0.01 | 3.44 | 127268 |
| Net Income | 0.18 | 5.60 | -0.47 | -11.48 | 0.24 | 9.39 | -0.02 | -0.66 | 113862 |
| Pre-tax Profit | 0.22 | 5.92 | -0.48 | -11.61 | 0.23 | 8.96 | 0.00 | 0.03 | 102386 |
| EBITDA | 0.23 | 7.03 | -0.40 | -10.66 | 0.15 | 8.65 | -0.00 | -0.15 | 99731 |
| Book Value Per Share | 0.47 | 13.87 | -0.53 | -13.11 | 0.08 | 10.37 | -0.03 | -6.04 | 89884 |
| EBIT | 0.25 | 6.48 | -0.46 | -10.67 | 0.18 | 6.85 | -0.01 | -0.55 | 82100 |
| Return On Assets | 0.90 | 89.07 | -0.28 | -18.41 | 0.04 | 13.39 | 0.00 | 6.29 | 79479 |
| GAAP EPS | 0.01 | 0.28 | -0.36 | -7.56 | 0.22 | 8.84 | 0.01 | 0.44 | 74210 |
| Net Asset Value | 0.46 | 12.83 | -0.44 | -11.91 | 0.08 | 13.88 | -0.03 | -3.76 | 71494 |
| Gross Margin | 0.96 | 209.91 | -0.05 | -4.53 | 0.03 | 4.77 | 0.01 | 4.58 | 64483 |
| Cash Flow Per Share | -0.18 | -5.43 | -0.14 | -4.80 | 0.14 | 9.39 | 0.05 | 2.84 | 62446 |
| Dividend Per Share | 0.44 | 7.82 | -0.66 | -9.91 | 0.17 | 8.37 | -0.03 | -1.67 | 49402 |
| Net Debt | 0.09 | 4.35 | -0.27 | -8.79 | 0.16 | 9.73 | 0.00 | 0.13 | 48338 |
| Enterprise Value | -0.05 | -0.98 | -0.10 | -1.69 | 0.06 | 2.34 | -0.01 | -0.52 | 48140 |
| Capital Expenditures | 0.05 | 1.49 | -0.22 | -5.51 | 0.07 | 4.21 | -0.06 | -3.45 | 44623 |
| Operating Profit | 0.24 | 4.56 | -0.51 | -8.79 | 0.25 | 9.60 | -0.01 | -0.69 | 33033 |
| EBITDA Per Share | 0.13 | 3.28 | -0.27 | -4.76 | 0.08 | 3.12 | 0.00 | 0.05 | 15343 |
| Cash Earnings Per Share | 0.12 | 2.83 | -0.36 | -5.06 | 0.10 | 3.64 | -0.00 | -0.19 | 10013 |
| Funds From Operations Per Share | 0.37 | 3.48 | -0.35 | -2.78 | 0.04 | 2.46 | -0.02 | -1.54 | 3399 |
| EPS Before Goodwill | 0.13 | 2.02 | -0.34 | -7.87 | 0.13 | 4.13 | -0.03 | -1.50 | 2413 |

*Notes*: This table reports the OLS estimates of equation (3), for each one of the 22 variables. We report the estimates of $\beta_g$ (column 1), $\gamma_g$ (column 3), $\delta_g^+$ (column 5), and $\delta_g^-$ (column 7), along with the corresponding $t$-statistics with standard errors clustered by firm and time. Variables are ranked by number of observations. The sample includes one observation per firm-year.

# 3 Model

In this section, we develop a parsimonious model that ties the non-linear relationship between forecast errors and revisions, our Fact #1, to Facts #2 and #3, which are about the data-generating process. We start by describing a model of the DGP that is consistent with our latter two facts, and then turn to a model of belief formation that, given this model of the DGP, generates the first fact. We then discuss a potential microfoundation based on Augenblick et al. (2025).

## 3.1 Data-Generating Process

The first piece of the model is the data-generating process (DGP) for $g_t$, our forecasting variable of interest. Our model does not have any firm-specific heterogeneity, and we therefore omit the firm index $i$ for brevity.[9] Without loss of generality, we normalize the unconditional mean of $g_t$ to zero. We then assume that the DGP for $g_t$ takes the following form

$$g_{t+1} = g^*_{t+1} + \sigma_\epsilon \epsilon_{t+1}, \quad \epsilon_t \sim f_\epsilon(\cdot), \tag{4}$$

$$g^*_{t+1} = \rho g^*_t + \sigma_u u_{t+1}, \quad u_t \sim f_u(\cdot), \tag{5}$$

where $\epsilon_t$ and $u_t$ are IID shocks with a zero mean and unit variance. The DGP for $g_t$ therefore consists of two components: (i) a persistent component, $g^*_t$, that follows an AR1 process with shocks, $u_t$ that have PDF $f_u(\cdot)$, and (ii) a transitory shock, $\epsilon_t$, with PDF $f_\epsilon(\cdot)$. Throughout, we assume that $g^*_t$ is an *unobservable* latent state. We denote its unconditional variance by $\sigma^2_{g^*} = \frac{\sigma^2_u}{1-\rho^2}$, and denote its PDF by $\Phi(\cdot)$. We also denote the marginal PDF of $g_t$ by $h(\cdot)$, which is given by:

$$h(g) = \int_{-\infty}^{+\infty} \Phi(g - \sigma_\epsilon \epsilon) f_\epsilon(\epsilon) d\epsilon.$$

and denote $\sigma^2_g = \sigma^2_\epsilon + \sigma^2_{g^*}$ its unconditional variance.

At this point, if we assumed that $\epsilon_t$ and $u_t$ were Gaussian, our model of the DGP would be identical to the models of dividend growth in Bansal and Yaron (2004) and Lettau and Wachter (2007). However, unlike these models and existing literature on belief formation, we instead assume the PDF of $\epsilon_t$, $f_\epsilon(\cdot)$, has heavy tails in the sense that it is well approximated by a power law with tail parameter $\nu$ for large values of $\epsilon_t$. Formally, we make the following assumption, where we use the notation $a(x) \sim b(x)$ to denote asymptotic equivalence in the sense that $\lim_{x\to\infty} \frac{a(x)}{b(x)} = 1$.

**Assumption 1.** *The distribution of $\epsilon_t$ is symmetric and follows a power law:*

$$P(\epsilon_t > x) = \int_x^\infty f_\epsilon(\epsilon) d\epsilon \sim \Theta x^{-\nu} \text{ as } x \longrightarrow \infty$$

*where $\nu > 2$ is the tail index and $\Theta$ is a constant of proportionality.*[10]

An example of a case in which Assumption 1 holds would be if $f_\epsilon(\cdot)$ is the PDF of a $t$-distribution,

[9] The lack of firm-specific heterogeneity in the model is consistent with the fact that our results are unchanged when standardizing at the firm-level.

[10] We prove all of our results in Appendix B under a more general tail condition that $\epsilon_t$ is a regularly varying random variable such that $P(\epsilon_t > x) \sim x^{-\nu} L(x)$, where $L(x)$ is a slowly varying function, combined with two mild regularity conditions on $f_\epsilon(\cdot)$ that are satisfied by standard fat-tailed distributions, including the $t$-distribution.

where $\nu$ would be the degrees of freedom of the distribution. Finally, we need to make an assumption about the distribution of $u_t$. In all of our theoretical results, we assume that $u_t$ has "thinner" tails than $\epsilon_t$, which is formalized in the following assumption.

**Assumption 2.** *The distribution of $u_t$ is symmetric and exhibits tail behavior that is asymptotically negligible compared to $\epsilon_t$:*

$$P(u_t > x) = \int_x^\infty f_u(u)du = o(x^{-\nu}).$$

For our theoretical results that characterize the properties of beliefs, Assumptions 1 and 2 are sufficient. However, to obtain the analytic characterization of the data-generating process for $g_t$ below, we will make a stronger assumption than Assumption 2 that $u_t$ is normally distributed.

## 3.2 Replicating Facts #2 and #3: Fat Tails and Non-Linear $E(g_{t+1}|g_t)$

Before turning to our model of beliefs, we show that our model of the DGP can generate our second and third facts. Our second fact makes clear that we need a model of the DGP with fat tails. The fact that our model generates this follows from the properties of power laws: the tail parameter of a sum of independent random variables is the minimum of the tail parameters (Jessen and Mikosch, 2006). Given that $g_t^*$ is the sum of random variables that have thinner tails than $\epsilon_t$ and $g_t$ is the sum of $g_t^*$ and $\epsilon_t$, the tail parameter of $g_t$ will be the same as that of $\epsilon_t$, which we denoted by $\nu$.

Showing our model replicates our third fact—the non-linearity in $E(g_{t+1}|g_t)$—is more difficult because there is no closed-form expression for $E(g_{t+1}|g_t)$. Nevertheless, the following proposition leverages a result from empirical Bayes theory known as Tweedie's formula (Efron, 2011) to characterize it.

**Proposition 1** (Tweedie's formula)**.** *If $u_t$ is normally distributed, the expectation of future growth conditional on current growth takes the following form:*

$$E(g_{t+1}|g_t) = -\rho\sigma^2_{g^*}\frac{d}{dg}\log h(g_t). \tag{6}$$

*In particular, it is a function of the **observable** distribution of growth, $g_t$.*

*Proof.* See Appendix B.1. □

Proposition 1 shows that $E(g_{t+1}|g_t)$ is a function of $h(\cdot)$, the marginal PDF of $g_t$. This result

is useful because $h(\cdot)$ is observable, which means we can characterize the shape of $E(g_{t+1}|g_t)$ even without a closed form. For example, suppose that this distribution is approximately Gaussian in the bulk of the distribution: $\log h(g) \approx -\frac{g^2}{2\Sigma_g} + \text{constant}$, which Table IA.16 shows is a good approximation, especially for Net Sales. Given this approximation, Proposition 1 implies

$$E(g_{t+1}|g_t) \approx \rho \frac{\sigma^2_{g^*}}{\Sigma_g} g_t, \tag{7}$$

which is linear in $g_t$. This is consistent with our third fact: in the bulk of the distribution, the expectation of $g_{t+1}$ is linearly increasing in $g_t$. Intuitively, in the bulk of the distribution, $g_t$ is dominated by $g^*_t$ and $g^*_t$ is persistent, implying that $g_t$ is a good approximation of $g_{t+1}$. Note that if $g_t$ were globally Gaussian, then this equation would hold with equality, and we would recover the well-known result that the rational expectation given a signal of a state with additive Gaussian noise is linear. The benefit of Proposition 1 is that it allows us to extend this result to distributions that are locally Gaussian.

In the tails of the distribution, Table 3 shows that the growth distribution is well approximated by a power law, consistent with Assumption 1. Under Assumption 1, we can use the properties of regularly varying functions to derive the following result that characterizes $E(g_{t+1}|g_t)$.

**Proposition 2.** *If $u_t$ is normally distributed, Assumption 1 implies that:*

$$E(g_{t+1}|g_t) \sim \frac{\rho \sigma^2_{g^*}(\nu+1)}{g_t} \text{ as } |g_t| \to \infty. \tag{8}$$

*Proof.* See Appendix B.2. □

Proposition 2 shows that, in the tails of the growth distribution, the expectation of $g_{t+1}$ is *decreasing* in $g_t$, unlike in the bulk of the distribution. The intuition is that extreme values of $g_t$ are likely driven by $\epsilon_t$. However, because $\epsilon_t$ is not persistent, this value of $g_t$ is unlikely to persist at $t+1$. In the limit of $|g_t| \to \infty$, $g_t$ reflects $\epsilon_t$ with probability one, and $E(g_{t+1}|g_t)$ converges to the unconditional expectation of zero.[11] While it is clear that our model of the DGP needs to have fat tails, Proposition 2 also explains why an alternative model in which the distribution of permanent shocks, $u_t$, has fatter tails than that of transitory shocks, $\epsilon_t$, would be inconsistent with our third fact.[12] With fatter-tailed permanent shocks, large shocks would be more likely to persist, and the conditional expectation of $g_{t+1}$ given $g_t$ would be *increasing* in $g_t$ in the tails of the distribution.

[11] A related result is developed in Chambers and Healy (2011), who show that when a distribution has sufficiently fat tails such that MLRP does not hold, a positive signal may generate a negative update about the underlying parameter.

[12] Our specification of the DGP is reminiscent of the recent literature on income dynamics (Guvenen et al., 2014, 2021), which models income processes as the sum of persistent and transitory components. However, this literature

## 3.3 Expectations Formation

Given our model of the DGP, we now describe our model of subjective expectation formation. We assume that expectations do not have "full information," in the sense that forecasters only observe realizations of $g_t$ but not $g_t^*$. However, this assumption alone cannot explain our first fact, given it would imply that forecast errors should not be predictable by revisions, which are in forecasters' information set. Therefore, we also assume that forecasts are not rational given this information set. In particular, our core assumption is that forecasters incorrectly perceive the distributions of $\epsilon_t$ and $u_t$ to be Gaussian, inconsistent with Assumptions 1 and 2. This assumption could be viewed as a standard form of bounded rationality in which economic agents use simplified, or "sparse", models of reality to form their beliefs (Fuster et al., 2010; Gabaix, 2019). Alternatively, in Section 3.5, we show that this assumption can be microfounded based on insensitivity to signal strength. Because of this model misspecification, forecast errors will be (conditionally) predictable.

Since only past values of $g_t$ are observable, agents have to solve a filtering problem to compute their expectations about $g_t^*$ given $g_0, ..., g_t$, which they in turn use to forecast future growth. Under the assumption that agents perceive $\epsilon_t$ and $u_t$ as Gaussian, the solution to this filtering problem implies that their expectations will be characterized by the Kalman filter. Given the DGP, our assumption on belief formation is equivalent to assuming that forecasters use the *linear* forecast with the lowest possible mean squared error. For our theoretical results, we assume that agents are in a steady state in the sense that the posterior variance of the Kalman filter and, hence, the Kalman gain is constant. Denoting this steady state Kalman gain as $K$, agents' expectations at horizon $k$, $F_t g_{t+k}$, are characterized by:

$$F_t g_{t+k} = \rho^k K \sum_{s \geq 0} \rho^s (1-K)^s g_{t-s}. \tag{9}$$

The key property of (9) is that forecast revisions are *linear* in the size of the shock, meaning that forecasters apply the same gain to innovations of different sizes. Like us, Kwon and Tang (2026) also provide a theory of expectations formation with fat tails that generates differences in over- and underreaction. However, the key difference is that their theory generates differences across categories but no differences within a category, while our theory generates differences within a given process depending on the size of shocks.

Our assumption that forecasters use exactly the Kalman filter is not crucial for our results. For

emphasizes the importance of non-normal persistent shocks, while the key ingredient in our model is non-normal *transitory* shocks. Another difference is that our focus is primarily on generating the kurtosis in the data rather than skewness. The latter is a pervasive feature of income data due to extreme negative events like job loss.

example, the results in the next section about the non-linear error-revision relationship would be present if forecasters linearly extrapolated current observations. What is important is that forecasters use a forecasting rule that does not fully capture the presence of fat tails. Aside from its tractability, we choose to use the Kalman filter specifically because it would correspond to rational expectations under the assumption that $\epsilon_t$ and $u_t$ were normal. This clarifies that our analytical results are entirely about ignoring the presence of fat tails, rather than an arbitrary assumption about the history or functional form that forecasts take.

## 3.4 Replicating Fact #1: Non-Linear Error-Revision Relationship

We now show that our model of belief formation, combined with our model of the DGP that replicates Facts #2 and #3, also generates Fact #1.

**Proposition 3.** *Define forecast errors and forecast revisions as follows:*

$$\begin{aligned} ERR_{t+1} &= g_{t+1} - F_t g_{t+1}, \\ REV_t &= F_t g_{t+1} - F_{t-1} g_{t+1}. \end{aligned}$$

*Then, under Assumptions 1 and 2, forecast errors are asymptotically linear in forecast revisions:*

$$E\left(ERR_{t+1} \mid REV_t\right) \sim C \times REV_t \quad \text{as } |REV_t| \to \infty, \tag{10}$$

*where* $C < 0$.

*Proof.* See Appendix B.3. □

Proposition 3 shows that in the tails of the distribution of revisions, the relationship between forecast errors and revisions is linear and negative. This is consistent with our first fact in Figure 1: there is overreaction in the tails of the distribution of revisions. The proof of this result relies on showing that revisions are driven by changes in $g_t$, which are asymptotically large (in absolute value) for one of two reasons: (i) a large realization of $\epsilon_t$ or (ii) a large realization of past $\epsilon_{t-h}$. The intuition for why these large revisions reflect overreaction is that the forecaster does not realize that they are less likely to be persistent because they are driven by the fat-tailed component of the process, $\epsilon_t$, which is transitory. If $\epsilon_t$ and $u_t$ were Gaussian—and even if $\epsilon_t$ had a higher variance—there would still be transitory shocks, but the change in $g_t$ would not be informative about the relative size of transitory and persistent shocks. In contrast, when $\epsilon_t$ has fatter tails than $u_t$, large values of $|g_t|$ asymptotically only reflect $\epsilon_t$.

The definition of the Kalman filter implies that it is the minimum linear mean squared error predictor. Combining this with Proposition 3 implies the following result.

**Corollary 1.** *Under Assumptions 1 and 2, there exists an* $\overline{R} > 0$ *such that:*

$$E\Big(ERR_{t+1} \times REV_t \,\Big|\, |REV_t| > \overline{R}\Big) < 0,$$
$$E\Big(ERR_{t+1} \times REV_t \,\Big|\, |REV_t| < \overline{R}\Big) > 0.$$

*Proof.* See Appendix B.4. □

Corollary 1 shows that our model can generate the non-linear error-revision relationship in Figure 1. While Proposition 3 shows there is overreaction in the tails, Corollary 1 shows that errors and revisions are on average positively correlated in the bulk of the distribution of revisions. The intuition for why forecasters underreact in the bulk is similar to the intuition for overreaction in the tails: the forecasters do not realize that intermediate values of $g_t$, which generate smaller revisions, are more likely to reflect $u_t$ than $\epsilon_t$, and hence are more likely to be persistent.

An important caveat to Corollary 1 is that, while errors and revisions are positively correlated on average in the bulk where $|REV_t| < \overline{R}$, this may not be true for all $|REV_t| < \overline{R}$ without further assumptions on $f_\epsilon(\cdot)$ and $f_u(\cdot)$.[13] Therefore, our theory does not imply that there must be underreaction within what we define as the bulk of the distribution empirically. For this reason, we view our theory as primarily an explanation for why overreaction *varies* with shock size rather than for the overall *level* of overreaction. For example, our theory could be combined with a model of diagnostic expectations to generate more overreaction on average.

## 3.5 A Microfoundation Based on Insensitivity to Signal Strength

We now provide a microfoundation for the assumption that forecasters ignore fat tails and use the Kalman filter. The microfoundation extends the mechanism in Augenblick et al. (2025) (ALT) from a static binary-state environment to our continuous-state time-series environment. The key idea in ALT is that agents know the direction to update in response to a signal, but do not know how much. ALT assume that agents receive a noisy estimate of the strength of the signal that they observe and, therefore, shrink their estimate of signal strength to their prior. As a consequence,

[13] While $\overline{R}$ does not admit a closed-form expression, the condition that characterizes it is $E\left(ERR_{t+1} \mid REV_t = \overline{R}\right) = 0$. Intuitively, the switch between underreaction and overreaction occurs at the revision that coincides with the rational revision.

agents underestimate the strength of strong signals and overestimate it for weak signals, which in turn generates overreaction (underreaction) to weak (strong) signals.[14]

Formally, signal strength is defined as the amount by which a Bayesian agent would update their prior beliefs. While signal strength is a primitive in ALT, it is endogenous to the realization of the process in our continuous-state time-series setting. As Bastianello and Imas (2026) emphasize, signal strength in forecasting problems, $E\left(g_{t+1}|g_0,\ldots,g_t\right) - E\left(g_{t+1}|g_0,\ldots,g_{t-1}\right)$, depends on multiple interacting features of the information environment. Given our non-Gaussian DGP, we cannot characterize this definition of signal strength in closed form. Therefore, we start with a simplified one-period forecasting problem where the only signal about $g_{t+1}$ is $g_t$, in which case we can characterize signal strength more explicitly.[15]

**Signal strength in the one-period problem.** In a one-period forecasting problem, the strength of the signal $g_t$ is:

$$S(g_t) \equiv \frac{E(g_t^*|g_t)}{g_t}, \quad g_t \neq 0. \tag{11}$$

This is the natural definition of signal strength because the rational forecast is $E(g_{t+1}|g_t) = \rho E(g_t^*|g_t) = \rho S(g_t) g_t$. Therefore, $S(g_t)$ provides a unit-free measure of how informative the realization $g_t$ is about the persistent component $g_t^*$. This, in turn, determines how much the forecast should update in response to $g_t$. Applying Proposition 1, signal strength takes the following form:

$$S(g_t) = -\sigma_{g^*}^2 \frac{1}{g_t} \frac{d}{dg} \log h(g_t). \tag{12}$$

Equation (12) shows that signals are strong in the bulk of the distribution where $\log h(g) \approx -\frac{g^2}{2\Sigma_g} + \text{constant}$, implying $S(g_t) \approx \frac{\sigma_{g^*}^2}{\Sigma_g}$. In contrast, signals are weak in the tails of the distribution where $S(g_t) \sim \frac{\sigma_{g^*}^2(\nu+1)}{g_t^2}$, implying $S(g_t) \longrightarrow 0$ as $|g_t| \longrightarrow \infty$.

**Microfoundation for ignoring fat tails.** We now show how insensitivity to signal strength provides one microfoundation for ignoring fat tails. As in ALT, suppose that forecasters observe the sign of the signal, $\text{sign}(g_t)$, but observe a noisy estimate of signal strength, $\mathbb{S}(g_t)$. Additionally, we follow ALT and assume that forecasters shrink their noisy estimate to a default signal strength, $\bar{S}$. A natural choice for $\bar{S}$ is the following, which is the signal strength that generates the best linear

[14] Ba et al. (2024) (BBI) present an alternative model that makes the same prediction for the relationship between underreaction and overreaction and signal strength as ALT. However, the mechanism in BBI is quite different, which leads to other differing predictions, such as how overreaction responds to complexity. See Appendix C of BBI for additional details.

[15] In Appendix B.5, we discuss the definition of signal strength and the microfoundation in the dynamic model.

forecast:

$$\bar{S} \equiv \arg\min_{s} E\left[(g_t^* - s g_t)^2\right] = \frac{\sigma_{g^*}^2}{\sigma_g^2}.$$

Finally, we need to specify how forecasters form a posterior estimate of signal strength, denoted by $\widehat{S}(g_t)$, given their $\mathbb{S}(g_t)$ and $\bar{S}$. As in ALT, the key assumption we need is "updating toward the signal" (Chambers and Healy, 2012): the posterior $\widehat{S}(g_t)$ lies between the prior, $\bar{S}$, and the noisy estimate, $\mathbb{S}(g_t)$. Given this posterior, forecasters then form forecasts as follows: $F_t g_{t+1} = \rho \widehat{S}(g_t) g_t$.

Under these assumptions, our reduced-form assumption in (9) that forecasters use a constant Kalman gain corresponds to the special case in which forecasters do not update in response to their noisy estimate, setting $\widehat{S}(g_t) = \bar{S}$. In this case, the relationship between forecast errors and revisions is

$$E\left(ERR_{t+1} \mid REV_t\right) = \rho\left(S(g_t) - \bar{S}\right) g_t. \tag{13}$$

Therefore, by updating as if signal strength is constant, forecasters give too much weight to large uninformative shocks in the tails of the distribution where $S(g_t) - \bar{S} < 0$, generating a forecast error with the opposite sign from the revision. In the bulk of the distribution, $S(g_t) - \bar{S} > 0$, forecasters give too little weight to moderate informative shocks, generating a forecast error with the same sign as the revision.[16]

More generally, forecasters may update $\widehat{S}(g_t)$ away from $\bar{S}$, in which case the same intuition applies. For example, consider the following parametric updating rule (similar to the log-normal example in ALT): $\widehat{S}(g_t) = \bar{S}^{1-\lambda} S(g_t)^{\lambda}$. In this case, we have

$$E\left(ERR_{t+1} \mid REV_t\right) = \rho S(g_t)^{\lambda}\left(S(g_t)^{1-\lambda} - \bar{S}^{1-\lambda}\right) g_t.$$

As long as $\lambda < 1$, underreaction still occurs in the bulk of the distribution where $S(g_t) - \bar{S} > 0$, while overreaction occurs in the tails where $S(g_t) - \bar{S} < 0$. However, without a constant Kalman gain, we cannot derive a result like Proposition 3 that characterizes the relationship between errors and revisions in a dynamic problem.

# 4 Additional Evidence from Other Settings

This section presents additional evidence in support of our theory. First, we examine whether our first fact—the non-linear error-revision relationship—replicates "out-of-sample" in three different

[16] This relies on the fact that $\bar{S} = E\left(\frac{g_t^2}{E g_t^2} S(g_t)\right)$ by applying the law of iterated expectations to the definition of $\bar{S}$. Given that $S(g_t) \longrightarrow 0$ as $|g_t| \longrightarrow \infty$ and $\bar{S} > 0$, there must exist an intermediate range of $g_t$ where $S(g_t) > \bar{S}$.

settings: an online forecasting experiment, macroeconomic forecasts, and stock returns. Second, we provide evidence that, in the cross-section of forecasted variables, fatter tails are associated with a more non-linear error-revision relationship.

## 4.1 Evidence from a Forecasting Experiment

**Experimental design.** The design of our online forecasting experiment is taken from Afrouzi et al. (2023) (AKLMT). Participants are asked to predict the outcome of a process, and their compensation depends on the accuracy of their forecasts. We recruit participants on Amazon MTurk, and they are reasonably representative of the general population. The experiment does not require participants to have any prior knowledge of statistics, and participants do not know the DGP (AKLMT show that this is not important). The interface is graphical and user-friendly: participants click with their mice to provide their forecasts at one- and two-period-ahead horizons. Prior to making their first forecast, they see 40 prior realizations of the process, and then sequentially provide both forecasts in 40 periods, seeing the realization of the process between each period. We refer the reader to AKLMT for further details about this design; see Figure IA.1 for an example of the interface our participants see for these parameters.

Starting with this design, we set the DGP of the process being forecasted to be:

$$g_{t+1} = g^*_{t+1} + 12.16\epsilon_{t+1}, \quad \epsilon_t \sim t(2.533) \tag{14}$$

$$g^*_{t+1} = 0.529 g^*_t + 12.62 u_{t+1}, \quad u_t \sim N(0,1) \tag{15}$$

This is the DGP in our theoretical model (equations (4)-(5)) with parameters to generate the key properties of the DGPs in our empirical analysis: fat tails and mean reversion in the tails.[17] We set $\sigma_u$ and $\sigma_\epsilon$, which simply scale the DGP, to be larger numbers so that values are easier for participants to interpret. As shown in Figure IA.1, the tail of our DGP is thick enough that participants typically observe a few large movements in the DGP in the past 40 realizations they are initially given. We are not in a situation where large shocks of $\epsilon$ rarely happen.

We ran the experiment in two waves with 201 participants on March 17, 2025 and 202 participants on April 29, 2025. Given that each participant makes 40 forecasts, our data have 16,120 observations. We compare our results using the DGP in (14) and (15) with those from a condition in AKLMT in which the process is an AR1 with Gaussian shocks and persistence parameter of 0.2.[18]

[17] These parameters come from calibrating a version of this DGP to match our second and third facts for Net Sales.

[18] We use 0.2 because it is the closest experimental condition in AKLMT to the regression coefficient of $g_{it+1}$ and $g_{it}$ when simulating from (14) and (15).

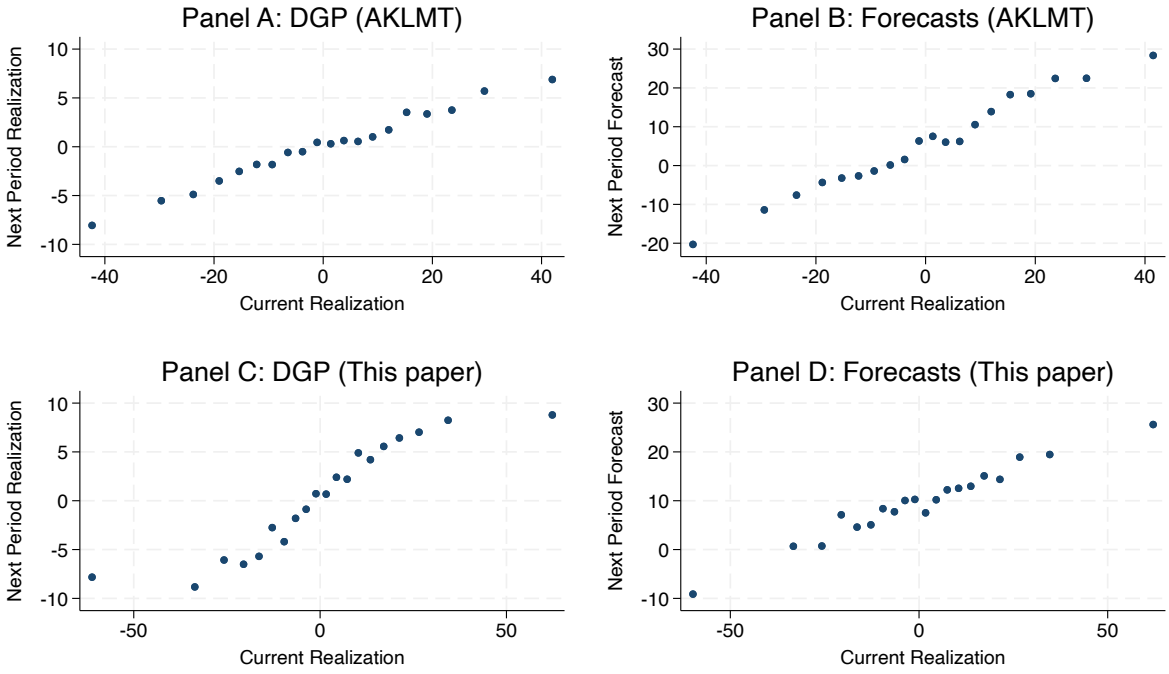
Before describing the results, we briefly discuss the relative benefits and costs of this experiment relative to our evidence in Section 2. The main benefit of running an experiment is that it allows us to directly control the parameters of the DGP. However, this benefit comes with several downsides. First, participants in the experiment are fundamentally different from the forecasters in our data, who are professional equity analysts with incentives aside from accuracy. Second, the professional forecasters in our data are likely to have a better understanding of the DGP, while we cannot control the priors of the participants in our experiment. Finally, the environment of forecasting a variable in our experiment is obviously quite different from forecasting in the real world. Despite these differences, we still view this experiment as useful because, by comparing our results with those in AKLMT, we can perform a direct test of our theory's key prediction that transitory fat-tailed shocks create a non-linear relationship between errors and revisions.

**Results.** Panel A of Figure 3 shows the relationship between $g_{it+1}$ and $g_{it}$ in the data from AKLMT, where $i$ is the participant and $t$ is the round of forecasting. As expected, this relationship is linear. In contrast, Panel C shows the same relationship in our experimental data, where the DGP is given by (14) and (15). Although our experimental data have one order of magnitude fewer observations than our firm-level data, these data still replicate the non-linear relationship in Fact #3.

Panels B and D of Figure 3 provide evidence that, despite the differences in the DGP, participants' one-period ahead forecasts, $F_{it}g_{it+1}$, are still linear in $g_{it}$, unlike the true DGP. This is consistent with our theory that forecasters ignore the fact that large shocks are likely to be transitory. Comparing Panels A and C with B and D also illustrates that forecasters substantially overestimate the persistence of the process: $E\left(F_{it}g_{it+1} \mid g_{it}\right)$ is steeper than $E\left(g_{it+1} \mid g_{it}\right)$ at all points. The combination of this extrapolation (a core result of AKLMT) and the likely presence of expectation noise (de Silva and Thesmar, 2024) leads to a strong negative correlation between errors and revisions on average. Columns (1) and (3) of Table 5 show that the relationship between forecast errors and revisions is significantly negative in our experiment and in AKLMT, with a similar magnitude.

Despite the presence of overreaction on average, the fact that $E\left(F_{it}g_{it+1} \mid g_{it}\right)$ is linear in Panel D while $E\left(g_{it+1} \mid g_{it}\right)$ is non-linear in Panel C provides preliminary evidence that forecasters are likely to overreact more in the tails of the distribution, consistent with our theory. We test this statistically by estimating (2), which is the same regression that we estimated on analyst forecast data in Section 2.1. For the data from AKLMT, whose DGP is fully linear and Gaussian, column (2) shows no difference in the amount of overreaction that is present in the bulk of the distribution and the tails: the extra slope for the top and bottom quintile of past revisions is small and not statistically different from zero. In contrast, column (4) shows that, in our experimental data, this extra slope

Figure 3: Data-Generating Process and Forecasts in Experimental Data




*Notes*: Panels A and C show binned scatterplots of $g_{it+1}$ versus $g_{it}$ in experimental data, where $i$ is the participant and $t$ the round of forecasting. Panels B and D show binned scatterplots of $F_{it}g_{it+1}$ against $g_{it}$. Panels A and B use data from Afrouzi et al. (2023), where the DGP is a Gaussian AR1 process with a persistence parameter of 0.2. Panels C and D use the experimental data described in the main text, where the DGP is given by equations (14) and (15).

is three times larger and has a $t$-stat of 3.3, which is lower than in the analyst data but consistent with the fact that the sample is an order of magnitude smaller. Collectively, these findings provide direct evidence for our theory's key prediction that transitory fat-tailed shocks create a non-linear relationship between $g_{t+1}$ and $g_t$ and, as a result, between errors and revisions.

## 4.2 Evidence from SPF Macro Forecasts

We next study forecasts of macroeconomic outcomes from the Survey of Professional Forecasters (1968–2025) (SPF), which have been studied extensively in the literature on belief formation (Bordalo et al., 2020a). While we find qualitatively similar results to our findings at the firm-level, we interpret these results as suggestive given the much smaller sample size. In constructing our sample, we attempt to follow the exact choices made in the replication package of Bordalo et al. (2020a) (BGMS) as closely as possible, and refer the reader to BGMS for additional details.

Table 5: Non-Linear Forecast Error-Revision Relationship: Experimental Data

| | Dependent Variable: $\text{Error}_{it+1}$ | | | |
|---|---|---|---|---|
| | Afrouzi et al. (2023) | | New Experiment | |
| | (1) | (2) | (3) | (4) |
| $\text{Revision}_{it}$ | -0.42 | -0.40 | -0.35 | -0.27 |
| | (-21.56) | (-10.24) | (-20.51) | (-6.73) |
| $\text{Revision}_{it} \times$ Top/Bottom 20% of $\text{Revision}_{it}$ | | -0.06 | | -0.17 |
| | | (-1.17) | | (-3.28) |
| Top 20% of $\text{Revision}_{it}$ | | -2.37 | | 7.64 |
| | | (-0.98) | | (3.55) |
| Bottom 20% of $\text{Revision}_{it}$ | | -7.57 | | -3.96 |
| | | (-2.94) | | (-1.53) |
| Constant | -6.98 | -5.22 | -8.40 | -9.14 |
| | (-5.33) | (-4.03) | (-7.86) | (-7.40) |
| # of Obs. | 6859 | 6859 | 15003 | 15003 |

*Notes*: This table shows results from regressions of forecast errors onto revisions. In columns (1) and (2), we use data from Afrouzi et al. (2023), where the DGP is a Gaussian AR1 process. In columns (3) and (4), we use the experimental data described in the main text, where the DGP is given by equations (14) and (15). Columns (1) and (3) regress errors on revisions. Columns (2) and (4) estimate equation (2). All standard errors are clustered at the participant level. These regressions have fewer than 16,120 observations because computing forecast revisions loses one observation per subject and because we trim the top and bottom 1% of errors and revisions (as in Table 2).

**Forecasting variables.** The SPF is conducted around the end of the second month in each quarter and collects forecasts from individual forecasters that are anonymized and identified by forecaster IDs. We study the same set of outcome variables in the SPF as in BGMS, extending their sample forward to the second quarter in 2025.[19] Following BGMS, for variables that are not stationary in levels (e.g., GDP or CPI), we look at annual growth rates from the most recent realized quarter to three quarters in the future. For the remaining variables, we look at levels three quarters in the future. While macroeconomic variables are released quarterly, they are often subsequently revised. Therefore, we follow BGMS and focus on initial releases from the Real-Time Data Set for Macroeconomists (1968–2025). As an example, actual GDP growth from quarter $t-1$ to $t+3$ is defined as the initial release at $t+4$ of the GDP level in $t+3$ divided by the contemporaneous release of the GDP level at $t-1$. For financial variables, the actual outcomes are not revised, and we use historical data from the Federal Reserve Bank of St. Louis.

**Forecast errors and revisions.** The SPF contains forecasts for the levels of variables, and we focus on an annual forecast horizon. Forecasters report forecasts of outcomes in the current and next

[19]The only variable that we do not include is housing starts, which is only available in a monthly vintage and has many missing observations.

four quarters. Given the timing of this survey, forecasts made in quarter $t$ are made with knowledge of the realized values of variables with quarterly releases in quarter $t-1$ and the realized values of variables with monthly releases up to the previous month. For the variables that we study in growth rates, we transform forecasts of the levels of these variables into growth rates by dividing the level forecasts by the most recent realized level from the previous quarter. For the remaining variables, we use the level forecasts directly. Denoting $y_t$ as a realized outcome in quarter $t$ and $F_{jt}y_{t+h}$ as the forecast made in quarter $t$ for quarter $t+h$ by forecaster $j$, we construct the forecast errors as $ERR_t y_{t+4} = y_{t+4} - F_{jt}y_{t+4}$, and the quarterly forecast revision as $REV_t y_{t+4} = F_{jt}y_{t+4} - F_{jt-1}y_{t+4}$. We winsorize outliers by removing, for each forecast horizon in a given quarter, forecasts that are more than 10 interquartile ranges away from the median. We also require that the forecasters in our sample make at least five forecasts.

**Results.** Table 6 performs the same analysis as Table 2 by estimating equation (2) at the forecaster-level for each of the variables in our SPF sample. As shown in the table, the size of our sample is much smaller, which reflects the fact that we only have one time-series for each variable. Nevertheless, we find patterns that are similar to our first fact at the firm-level: the error-revision relationship is non-linear with more overreaction in the tails of the distribution. Specifically, we find that $\widehat{\beta} > 0$ for most variables, but $\widehat{\gamma} < 0$ for all but one variable. These findings, coupled with our experimental evidence in Section 4.1, suggest that the non-linear error-revision relationship is more general than our sample of firm-level forecasts.

## 4.3 Evidence from Stock Return Predictability

This section examines and tests the predictions that our model of belief formation makes for return predictability. We focus on one type of predictability: momentum, the fact that past returns predict future returns (Jegadeesh and Titman, 2011).

**Model prediction.** To derive predictions about returns, we make two simplifying assumptions: (i) earnings growth is a constant fraction of sales growth, and (ii) the subjective discount rate is constant (as in Bouchaud et al. 2019 and Nagel and Xu 2022), which is consistent with evidence in De la O and Myers (2021). As shown in Appendix B.6, these assumptions coupled with the Campbell (1991) decomposition allow us to link returns with changes in expectations of our measure of Net

Table 6: Non-Linear Error-Revision Relationship: SPF Macro Forecasts

| Error$_{it+1}$ | Revision$_{it}$ | | Revision$_{it}\times$ Top/Bottom 20% of Revision$_{it}$ | | Top 20% of Revision$_{it}$ | | Bottom 20% of Revision$_{it}$ | | |
|---|---|---|---|---|---|---|---|---|---|
| | $\widehat{\beta}$ (1) | $t$-stat (2) | $\widehat{\gamma}$ (3) | $t$-stat (4) | $\widehat{\delta^+}$ (5) | $t$-stat (6) | $\widehat{\delta^-}$ (7) | $t$-stat (8) | # of Obs. (9) |
| Nominal GDP | 0.29 | 2.01 | -0.81 | -3.90 | 0.01 | 3.50 | -0.00 | -1.39 | 6026 |
| Real GDP | 0.43 | 2.49 | -0.96 | -4.32 | 0.01 | 2.72 | -0.01 | -1.75 | 6045 |
| Real Consumption | 0.07 | 0.42 | -0.75 | -2.66 | 0.01 | 3.62 | -0.00 | -0.60 | 4425 |
| Unemployment | 0.48 | 2.55 | -0.98 | -3.46 | 0.46 | 2.35 | -0.52 | -4.14 | 6161 |
| GDP Price Index | 0.42 | 2.02 | -0.70 | -2.93 | 0.01 | 4.03 | -0.00 | -0.82 | 5995 |
| CPI | 0.32 | 1.56 | -0.65 | -3.04 | 0.01 | 2.38 | -0.00 | -1.94 | 4525 |
| Industrial Production | 0.16 | 0.76 | -0.66 | -2.35 | 0.01 | 2.07 | -0.01 | -1.87 | 5567 |
| Real Non-Residential Investment | 0.47 | 2.23 | -1.02 | -3.34 | 0.02 | 2.55 | -0.02 | -1.98 | 4308 |
| Real Residential Inv. | 0.32 | 1.61 | -0.92 | -3.38 | 0.03 | 2.50 | -0.05 | -3.30 | 4302 |
| Real Federal Govt. Consump. | -0.62 | -3.51 | 0.05 | 0.21 | -0.01 | -1.63 | -0.01 | -2.30 | 4158 |
| Real State/Local Govt. Consump. | -0.15 | -0.83 | -0.43 | -2.23 | 0.00 | 1.41 | -0.00 | -1.92 | 4158 |
| 3-Month Treasury Rate | 0.80 | 4.68 | -0.76 | -3.34 | 0.28 | 1.95 | -0.56 | -3.07 | 4491 |
| 10-Year Treasury Rate | 0.14 | 0.67 | -0.46 | -1.81 | 0.28 | 2.78 | -0.12 | -0.91 | 3773 |
| AAA Corporate Bond Rate | -0.15 | -1.15 | -0.13 | -0.78 | 0.08 | 0.91 | -0.09 | -0.77 | 3655 |

*Notes*: This table reports the OLS estimates of equation (2), for each one of the variables in our SPF sample that is constructed following Bordalo et al. (2020a). We report the estimates of $\beta$ (column 1), $\gamma$ (column 3), $\delta^+$ (column 5), and $\delta^-$ (column 7), along with the corresponding $t$-statistics with standard errors clustered by forecaster and time. Variables are ordered following Bordalo et al. (2020a).

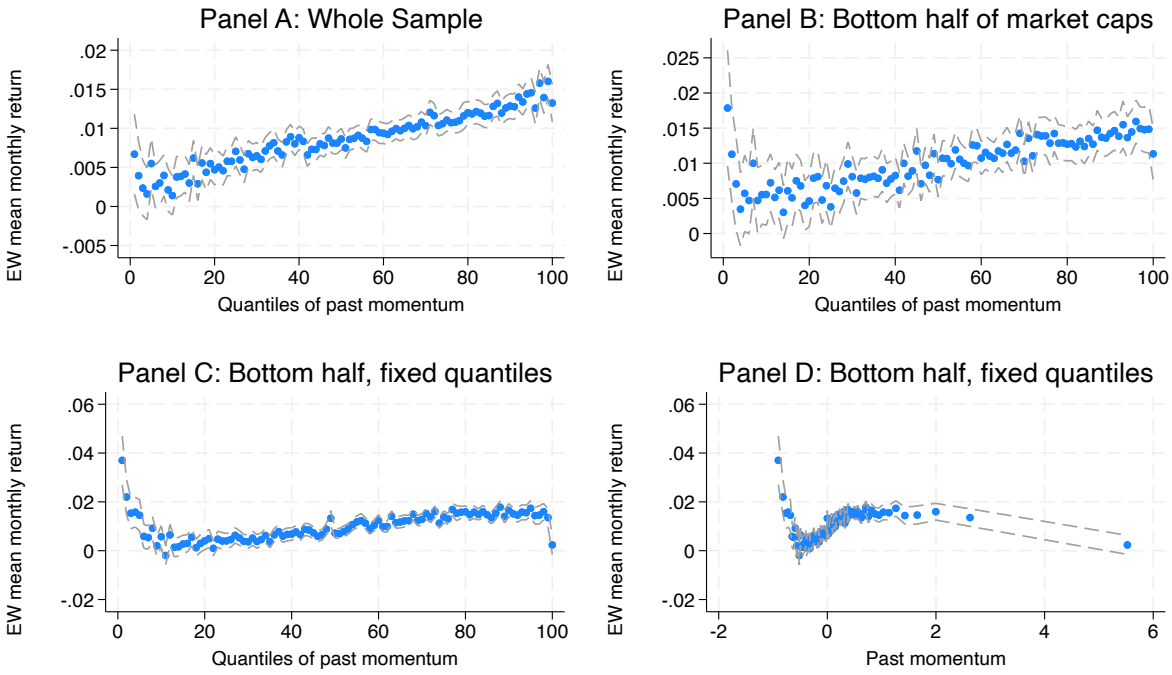
Sales, $g_t$, using the following expression:

$$\log R_t \approx \log(R_f + \pi) + \gamma \left[ (g_t - F_{t-1} g_t) + \sum_{k=1}^{\infty} c^k REV_t g_{t+k} \right], \tag{16}$$

where $R_t$ is the gross return, $\pi$ is the constant risk premium, $R_f$ is the gross risk-free rate, $\gamma$ is the constant relating earnings growth to sales growth, $c$ is a linearization constant, and $REV_t g_{t+k}$ denotes investors' subjective revisions about future growth between $t-1$ and $t$. This expression is intuitive: given that discount rates are fixed, returns are driven by earnings surprises and revisions of future growth. Combining (16) with our model of expectation formation in (9), we show in Appendix B.6 that the correlation of returns across periods inherits the error-revision relationship in Proposition 3 and Corollary 1:

$$\text{Corr}\left(\log R_{t+1}, \log R_t\right) = \text{Corr}\left(ERR_{t+1}, REV_t\right).$$

Figure 4: Relationship Between Current and Past Returns in the Data




*Notes:* This figure shows binned scatterplots of monthly returns, $r_t$, as a function of past annual returns from months $t-12$ through $t-2$, denoted by $r_{t-12,t-2}$. There are 100 bins in each panel. Panels A-C use the standard convention of using bins of past returns $r_{t-12,t-2}$ as the x-axis. Panel D uses past returns, $r_{t-12,t-2}$, directly as the x-axis. Panel A uses the entire CRSP sample of stocks traded on AMEX, NASDAQ and NYSE. Panel B uses only the bottom half of stocks ranked by 12-month lagged market cap. In contrast to Panels A and B, quantiles of returns are defined on the entire sample in Panels C and D, rather than separately for each month. Dashed lines are 95% confidence bands based on standard errors assuming that returns are independent.

Intuitively, forecasters underreact in the bulk of the distribution, so positive current returns should be followed by positive future returns. In the tails, forecasters overreact to large shocks that are unlikely to persist, so extreme returns should partially reverse.

**Empirical results.** We now examine whether momentum indeed exhibits mean-reversion in the tails in the data. We use CRSP monthly returns from 1927 to 2023 adjusted for delisting, and restrict to the sample of firms listed on NYSE, AMEX, and NASDAQ. Our measure of momentum follows the literature (Jegadeesh and Titman, 2011): every month $t$, momentum is the cumulative return from months $t-12$ through $t-2$, thereby excluding month $t-1$, whose returns exhibit reversals. We define firm size as the market capitalization at $t-12$.

We begin by plotting returns against centiles of momentum that are redefined each month, as in standard asset pricing tests. Panel A of Figure 4 shows the results for the entire CRSP sample:

the line is upward sloping, consistent with the presence of momentum. However, we also find that there is a bit of mean-reversion for "super losers", but not for "extreme winners". In Panel B, we show that the non-linearity is more pronounced for smaller firms, which we define as those with below-median size. This is to be expected, as smaller stocks are more expensive to trade and therefore display more predictability (Novy-Marx and Velikov, 2016).

In Panel C of Figure 4, we show results when the centiles of past returns are defined on the entire sample rather than within each month. While this is less standard in asset pricing tests because it implies the portfolios are not tradeable, it allows for the possibility that some of the non-linearity in momentum returns may come from time-series variation, which is more aligned with our model that does not distinguish between time-series and cross-sectional variation. Panel C shows that, using fixed centiles and focusing on firms below the 50% size cutoff, the non-linear relationship starts to appear significantly with the qualitative shape predicted by the model. The differences relative to Panel B suggest that the time series of momentum returns generates some mean-reversion in the tails, consistent with the literature on momentum crashes (Barroso and Santa-Clara, 2015): when volatility is high, losers are more likely to overperform, and winners more likely to underperform. Panel D of Figure 4 reproduces the analysis of Panel C, except that the x-axis is now rescaled to the average values of past returns. This last graph is less consistent with the practice of forming portfolios, as is typically done in asset pricing tests. Nevertheless, there is still significant non-linearity in the return-momentum relationship, in line with the prediction of our model.

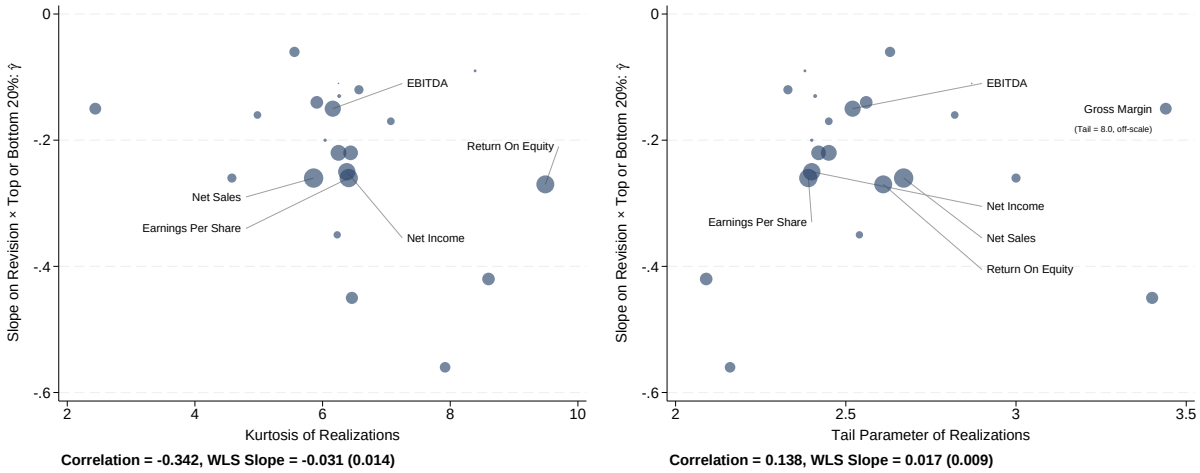
## 4.4 Relationship Between Fat Tails and Error-Revision Slopes

Finally, we examine whether the non-linear error-revision relationship is more pronounced for variables with fatter tails. We measure non-linearity using $\gamma$ in (2), the change in the error-revision slope in the top and bottom quintiles relative to the bulk. More negative values of $\gamma$ therefore indicate greater overreaction in the tails relative to the bulk.

Figure 5 plots $\widehat{\gamma}$ against kurtosis (left) and the power law tail parameter (right) for the 22 firm-level variables in our sample. We find that variables with higher kurtosis or lower tail parameters tend to have more negative estimates of $\gamma$, consistent with more pronounced non-linearity for variables with fatter tails. The relationship is even stronger when we restrict the sample to the 19 variables measured in growth rates, excluding Return on Equity, Return on Assets, and Gross Margin. Figure IA.2 shows that we find a similar pattern among the 14 macroeconomic variables in the SPF sample described in Section 4.2.[20]

[20]We only report results using the kurtosis for these variables because the sample sizes are too small to estimate tail

Figure 5: Fat Tails and Error-Revision Slopes: IBES Firm-Level Variables




*Notes:* This figure shows the relationship between fat tails in realizations and the non-linearity of the error-revision relationship across the 22 firm-level variables in IBES. Each dot corresponds to one variable, and its area is proportional to the number of forecast observations in column (9) of Table 2. The vertical axis in both panels is $\widehat{\gamma}$ from column (3) of Table 2; the horizontal axes show kurtosis (left) and the tail parameter (right) from columns (2) and (3) of Table 3. Each panel reports the unweighted Pearson correlation coefficient and the slope from a weighted least squares regression, with forecast counts as weights and the heteroskedasticity-robust standard error in parentheses. Labels identify the five variables with the most forecast observations, along with off-scale variables. Off-scale points are displayed near the horizontal boundary without changing $\widehat{\gamma}$; regressions and correlations use their actual horizontal coordinates.

This evidence is consistent with our theory, but it is not a direct implication of Proposition 3, which characterizes the error-revision slope for asymptotically large revisions.[21] In contrast, $\gamma$ measures the difference between slopes at fixed quantiles of the distribution. At these fixed quantiles, fatter tails can increase the contribution of large transitory shocks to revisions in the top and bottom quintiles, generating more overreaction relative to the bulk and hence a more negative $\gamma$. Moreover, variables may differ in other features of their DGPs, including persistence and the signal-to-noise ratio, which also affect the error-revision relationship. We therefore view these cross-sectional patterns as suggestive evidence for the key implication of our theory: fat tails can be the source of overreaction to extreme shocks.

# 5 Conclusion

In this paper, we argue that recognizing the complexity of the underlying data generating process, in particular its fat tails and non-Gaussian dynamics, is crucial for understanding the properties of

parameters.

[21]The asymptotic slope $C$ is given in equation (IA.6). Holding persistence $\rho$ and the Kalman gain $K$ fixed, $C$ becomes more negative as the tail parameter $\nu$ increases, even though tails become thinner.

subjective forecasts. We document three facts using data on forecasts of 22 firm-level variables by equity analysts: (i) the relationship between forecast revisions and future forecast errors—the variables used in Coibion and Gorodnichenko (2015) regressions—is strongly non-linear; (ii) the distributions of the underlying processes have fat tails; and (iii) the conditional expectation of future realizations is non-linear in current realizations, with persistence in the bulk and mean reversion in the tails. Next, we build a forecasting model that connects these facts. The key ingredients in our model are that the underlying process is non-Gaussian, but forecasters do not fully recognize this and incorrectly fit a linear model to the data. Finally, we provide additional evidence in support of our theory by showing that it can explain evidence from an online forecasting experiment, that the non-linear relationship between errors and revisions is also present in macroeconomic forecasts, and that it provides an explanation for the presence of non-linearity in the momentum of stock returns.

Our paper raises several questions for further work. First, our paper raises the question of how learning occurs in an environment with fat-tailed processes, which is likely to occur much more slowly and be more difficult. Additionally, many of the driving processes in dynamic models of firm and household behavior have fat tails, such as revenue growth (Stanley et al., 1996; Boar et al., 2022; Jaimovich et al., 2025) and income growth (Guvenen et al., 2014). Our findings suggest that allowing agents to, at least partially, ignore the presence of the fat tails might improve the ability of these models to explain firm and household behavior.

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# Online Appendix to "Expectations Formation with Fat-Tailed Processes: Evidence and Theory"

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# A Additional Tables and Figures

Table IA.1: Summary Statistics: Consensus-Level Forecasts

| | Transformation | Realizations | | Errors | | Revisions | | |
|---|---|---|---|---|---|---|---|---|
| | | Mean | SD | Mean | SD | Mean | SD | # of Obs. |
| Variable | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| Net Sales | Growth | 0.06 | 0.18 | -0.02 | 0.14 | -0.00 | 0.12 | 144567 |
| Return On Equity | Level | 0.12 | 0.15 | -0.01 | 0.09 | -0.01 | 0.06 | 128459 |
| Earnings Per Share | Growth | 0.05 | 0.47 | -0.09 | 0.39 | -0.02 | 0.30 | 126341 |
| Net Income | Growth | 0.06 | 0.47 | -0.08 | 0.39 | -0.01 | 0.31 | 112501 |
| Pre-tax Profit | Growth | 0.05 | 0.45 | -0.09 | 0.38 | -0.01 | 0.28 | 106043 |
| Book Value Per Share | Growth | 0.05 | 0.19 | -0.03 | 0.19 | -0.01 | 0.13 | 99542 |
| EBITDA | Growth | 0.06 | 0.31 | -0.05 | 0.25 | -0.00 | 0.20 | 98311 |
| Return On Assets | Level | 0.05 | 0.06 | -0.01 | 0.04 | -0.01 | 0.03 | 89610 |
| Net Asset Value | Growth | 0.06 | 0.18 | -0.00 | 0.19 | -0.02 | 0.13 | 89572 |
| Dividend Per Share | Growth | 0.06 | 0.31 | -0.01 | 0.28 | -0.01 | 0.22 | 89522 |
| EBIT | Growth | 0.05 | 0.40 | -0.07 | 0.34 | -0.01 | 0.26 | 83365 |
| GAAP EPS | Growth | 0.05 | 0.57 | -0.11 | 0.47 | 0.01 | 0.39 | 82230 |
| Capital Expenditures | Growth | 0.04 | 0.54 | -0.12 | 0.61 | 0.19 | 0.61 | 78556 |
| Gross Margin | Level | 0.40 | 0.22 | -0.01 | 0.07 | -0.01 | 0.04 | 77380 |
| Cash Flow Per Share | Growth | 0.05 | 0.52 | -0.05 | 0.45 | -0.02 | 0.45 | 70384 |
| Net Debt | Growth | 0.04 | 0.42 | 0.09 | 0.43 | 0.07 | 0.38 | 44476 |
| Enterprise Value | Growth | 0.05 | 0.33 | 0.06 | 0.31 | 0.02 | 0.21 | 43661 |
| Operating Profit | Growth | 0.05 | 0.40 | -0.08 | 0.34 | -0.00 | 0.27 | 42824 |
| Cash Earnings Per Share | Growth | 0.04 | 0.32 | -0.04 | 0.30 | -0.03 | 0.26 | 22621 |
| EBITDA Per Share | Growth | 0.04 | 0.31 | -0.04 | 0.26 | -0.02 | 0.22 | 22411 |
| Funds From Operations Per Share | Growth | 0.01 | 0.16 | -0.02 | 0.12 | -0.03 | 0.11 | 3547 |
| EPS Before Goodwill | Growth | 0.08 | 0.35 | -0.01 | 0.29 | -0.04 | 0.25 | 2616 |

*Notes*: This table reports the mean, standard deviation, and number of observations for each one of the 22 firm variables forecasted by analysts in our sample. The sample includes all observations for which the consensus forecast error and revision are available for a given variable. Variables are ranked by decreasing number of observations. 19 variables are transformed into log growth as explained in the text; 3 are left untransformed. Click here to go back to main text.

Table IA.2: Non-Linear Error-Revision Relationship: Allowing for Asymmetric Slopes

| $\text{Error}_{it+1}$ | $\text{Revision}_{it}$ | | $\text{Revision}_{it} \times$ Top 20% of $\text{Revision}_{it}$ | | $\text{Revision}_{it} \times$ Bottom 20% of $\text{Revision}_{it}$ | | Top 20% of $\text{Revision}_{it}$ | | Bottom 20% of $\text{Revision}_{it}$ | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\widehat{\beta}$ | $t$-stat | $\widehat{\gamma^+}$ | $t$-stat | $\widehat{\gamma^-}$ | $t$-stat | $\widehat{\delta^+}$ | $t$-stat | $\widehat{\delta^-}$ | $t$-stat | # of Obs. |
| Variable | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) |
| Net Sales | 0.11 | 6.31 | -0.26 | -11.33 | -0.27 | -12.74 | 0.02 | 12.25 | -0.02 | -5.72 | 399809 |
| Earnings Per Share | 0.08 | 2.97 | -0.24 | -6.65 | -0.27 | -9.64 | 0.01 | 1.48 | -0.08 | -7.58 | 360011 |
| Return On Equity | 0.23 | 9.75 | -0.37 | -8.59 | -0.22 | -11.03 | 0.01 | 11.30 | -0.01 | -6.94 | 345082 |
| Net Income | 0.04 | 1.53 | -0.21 | -6.16 | -0.27 | -12.16 | 0.01 | 1.00 | -0.07 | -8.05 | 325230 |
| EBITDA | -0.04 | -1.51 | -0.11 | -2.29 | -0.17 | -6.55 | 0.00 | 1.43 | -0.03 | -3.58 | 280111 |
| Pre-tax Profit | 0.01 | 0.47 | -0.16 | -4.03 | -0.27 | -11.90 | 0.00 | 1.01 | -0.07 | -6.67 | 269883 |
| EBIT | -0.05 | -1.60 | -0.14 | -3.09 | -0.26 | -11.11 | 0.00 | 0.56 | -0.05 | -4.61 | 231455 |
| GAAP EPS | -0.02 | -0.73 | -0.10 | -2.84 | -0.18 | -7.87 | -0.02 | -1.01 | -0.07 | -3.97 | 171523 |
| Book Value Per Share | -0.06 | -1.52 | -0.49 | -8.97 | -0.37 | -7.56 | 0.02 | 5.10 | -0.03 | -5.08 | 170561 |
| Return On Assets | 0.41 | 10.35 | -0.70 | -11.25 | -0.35 | -9.63 | -0.00 | -0.77 | -0.01 | -6.80 | 157063 |
| Gross Margin | 0.09 | 3.90 | -0.23 | -6.44 | -0.12 | -4.28 | 0.00 | 5.18 | -0.00 | -3.61 | 150220 |
| Net Asset Value | 0.04 | 0.99 | -0.41 | -6.85 | -0.68 | -13.71 | 0.01 | 3.11 | -0.05 | -9.43 | 119885 |
| Cash Flow Per Share | -0.28 | -15.86 | 0.04 | 1.47 | -0.15 | -6.36 | -0.01 | -0.94 | -0.07 | -5.97 | 112323 |
| Net Debt | -0.17 | -8.31 | 0.13 | 5.69 | -0.54 | -21.33 | -0.06 | -5.67 | -0.05 | -4.02 | 90126 |
| Enterprise Value | -0.20 | -5.82 | -0.01 | -0.12 | -0.44 | -11.09 | 0.02 | 1.80 | -0.07 | -7.76 | 88791 |
| Dividend Per Share | -0.12 | -4.54 | -0.18 | -4.51 | -0.16 | -5.66 | -0.01 | -0.61 | -0.03 | -2.89 | 63840 |
| Capital Expenditures | -0.36 | -21.93 | -0.14 | -5.98 | -0.19 | -6.85 | 0.05 | 3.53 | -0.07 | -5.62 | 62099 |
| Operating Profit | 0.12 | 3.07 | -0.31 | -5.97 | -0.37 | -7.95 | 0.02 | 3.59 | -0.07 | -6.42 | 57756 |
| EBITDA Per Share | -0.09 | -2.84 | -0.15 | -2.37 | -0.12 | -2.72 | 0.03 | 3.45 | -0.02 | -1.67 | 16824 |
| Cash Earnings Per Share | -0.19 | -4.95 | -0.15 | -2.12 | -0.24 | -4.17 | 0.01 | 0.79 | -0.04 | -2.11 | 11578 |
| Funds From Operations Per Share | -0.01 | -0.25 | -0.03 | -0.46 | -0.11 | -1.90 | 0.00 | 1.01 | -0.02 | -2.09 | 9279 |
| EPS Before Goodwill | -0.13 | -2.29 | -0.18 | -1.46 | -0.06 | -0.71 | 0.06 | 2.61 | -0.06 | -3.33 | 2558 |

*Notes*: This table reproduces Table 2 allowing for asymmetric slopes. Instead of a single interaction term $\gamma$ for both top and bottom 20% of revisions, we estimate separate interaction coefficients $\gamma^+$ for the top 20% and $\gamma^-$ for the bottom 20%. We report the estimates of $\beta$ (column 1), $\gamma^+$ (column 3), $\gamma^-$ (column 5), $\delta^+$ (column 7), and $\delta^-$ (column 9), along with the corresponding $t$-statistics with standard errors clustered by firm and time. Variables are ranked by number of observations. Click here to go back to main text.

Table IA.3: Non-Linear Error-Revision Relationship: No Outlier Trimming

| $\text{Error}_{it+1}$ | $\text{Revision}_{it}$ | | $\text{Revision}_{it} \times$ Top/Bottom 20% of $\text{Revision}_{it}$ | | Top 20% of $\text{Revision}_{it}$ | | Bottom 20% of $\text{Revision}_{it}$ | | |
|---|---|---|---|---|---|---|---|---|---|
| | $\widehat{\beta}$ (1) | $t$-stat (2) | $\widehat{\gamma}$ (3) | $t$-stat (4) | $\widehat{\delta^+}$ (5) | $t$-stat (6) | $\widehat{\delta^-}$ (7) | $t$-stat (8) | # of Obs. (9) |
| Net Sales | 0.10 | 5.87 | -0.48 | -10.40 | 0.05 | 8.45 | -0.07 | -6.87 | 415349 |
| Earnings Per Share | 0.12 | 3.34 | -0.35 | -9.98 | 0.04 | 2.23 | -0.11 | -5.83 | 373074 |
| Return On Equity | 0.39 | 1.24 | -0.33 | -1.19 | 0.84 | 0.85 | 0.01 | 0.15 | 357925 |
| Net Income | 0.06 | 2.29 | -0.32 | -10.22 | 0.04 | 2.06 | -0.10 | -6.50 | 336930 |
| EBITDA | -0.02 | -0.62 | -0.28 | -10.81 | 0.04 | 2.98 | -0.07 | -6.57 | 290685 |
| Pre-tax Profit | 0.03 | 1.26 | -0.32 | -11.01 | 0.06 | 3.04 | -0.09 | -4.41 | 279453 |
| EBIT | -0.02 | -0.55 | -0.32 | -9.70 | 0.06 | 3.62 | -0.08 | -3.97 | 239652 |
| GAAP EPS | -0.03 | -0.97 | -0.17 | -5.38 | 0.03 | 1.22 | -0.06 | -2.26 | 177906 |
| Book Value Per Share | -0.05 | -1.01 | -0.49 | -7.47 | 0.02 | 2.69 | -0.06 | -5.04 | 176682 |
| Return On Assets | 0.18 | 0.78 | -0.19 | -0.82 | -0.01 | -2.76 | -0.01 | -5.06 | 163967 |
| Gross Margin | 0.03 | 1.16 | -0.13 | -2.22 | 0.00 | 0.90 | -0.01 | -4.02 | 158247 |
| Net Asset Value | 0.05 | 1.17 | -0.54 | -8.77 | 0.04 | 5.03 | -0.03 | -2.28 | 124259 |
| Cash Flow Per Share | -0.28 | -12.58 | -0.15 | -7.54 | 0.11 | 4.42 | -0.06 | -2.90 | 116548 |
| Net Debt | -0.20 | -8.28 | -0.15 | -4.73 | 0.18 | 8.48 | 0.14 | 7.35 | 93173 |
| Enterprise Value | -0.18 | -5.21 | -0.24 | -4.85 | 0.12 | 5.53 | 0.02 | 1.24 | 92133 |
| Dividend Per Share | -0.09 | -3.00 | -0.27 | -7.97 | 0.03 | 1.90 | -0.07 | -5.44 | 66376 |
| Capital Expenditures | -0.37 | -19.65 | -0.21 | -7.42 | 0.10 | 6.66 | -0.07 | -4.58 | 64832 |
| Operating Profit | 0.15 | 2.49 | -0.47 | -6.18 | 0.08 | 4.52 | -0.12 | -5.10 | 59898 |
| EBITDA Per Share | -0.06 | -1.26 | -0.51 | -5.42 | 0.12 | 5.26 | -0.19 | -4.47 | 17460 |
| Cash Earnings Per Share | -0.21 | -4.57 | -0.24 | -2.78 | 0.04 | 1.74 | -0.04 | -1.09 | 12022 |
| Funds From Operations Per Share | 0.02 | 0.41 | -0.17 | -2.09 | 0.01 | 0.69 | -0.03 | -2.17 | 9620 |
| EPS Before Goodwill | -0.13 | -1.50 | -0.14 | -1.39 | 0.04 | 0.90 | -0.08 | -1.94 | 2655 |

*Notes*: This table reproduces Table 2 without removing observations in the top and bottom 1% of the distributions of realizations, forecast errors, and forecast revisions. See Table 2 for additional notes. Click here to go back to main text.

Table IA.4: Non-Linear Error-Revision Relationship: Consensus Forecasts

| $\text{Error}_{it+1}$ | $\text{Revision}_{it}$ | | $\text{Revision}_{it} \times$ Top/Bottom 20% of $\text{Revision}_{it}$ | | Top 20% of $\text{Revision}_{it}$ | | Bottom 20% of $\text{Revision}_{it}$ | | |
|---|---|---|---|---|---|---|---|---|---|
| | $\widehat{\beta}$ (1) | $t$-stat (2) | $\widehat{\gamma}$ (3) | $t$-stat (4) | $\widehat{\delta^+}$ (5) | $t$-stat (6) | $\widehat{\delta^-}$ (7) | $t$-stat (8) | # of Obs. (9) |
| Net Sales | 0.23 | 5.54 | -0.40 | -8.63 | 0.01 | 0.99 | -0.04 | -12.04 | 144567 |
| Return On Equity | 0.40 | 7.59 | -0.44 | -6.83 | 0.00 | 2.08 | -0.03 | -10.01 | 128459 |
| Earnings Per Share | 0.03 | 0.54 | -0.22 | -4.57 | -0.03 | -2.51 | -0.06 | -5.91 | 126341 |
| Net Income | -0.08 | -1.83 | -0.11 | -2.74 | -0.03 | -1.75 | -0.05 | -4.65 | 112501 |
| Pre-tax Profit | -0.07 | -1.27 | -0.13 | -2.72 | -0.02 | -1.56 | -0.04 | -3.15 | 106043 |
| Book Value Per Share | -0.21 | -7.50 | -0.28 | -9.27 | -0.01 | -2.79 | -0.03 | -10.86 | 99542 |
| EBITDA | -0.03 | -1.03 | -0.19 | -5.28 | -0.01 | -0.90 | -0.05 | -4.07 | 98311 |
| Return On Assets | 0.46 | 11.38 | -0.50 | -11.22 | -0.01 | -9.85 | -0.02 | -14.51 | 89610 |
| Net Asset Value | -0.30 | -10.01 | -0.24 | -6.96 | 0.01 | 2.70 | -0.01 | -1.01 | 89572 |
| Dividend Per Share | -0.00 | -0.04 | -0.30 | -4.01 | -0.00 | -0.03 | -0.04 | -4.69 | 89522 |
| EBIT | -0.16 | -4.70 | -0.08 | -2.16 | -0.02 | -1.77 | -0.03 | -2.60 | 83365 |
| GAAP EPS | -0.20 | -4.19 | 0.01 | 0.29 | -0.02 | -0.98 | -0.00 | -0.17 | 82230 |
| Capital Expenditures | -0.40 | -24.74 | -0.23 | -9.68 | 0.10 | 6.00 | -0.09 | -5.34 | 78556 |
| Gross Margin | 0.08 | 2.46 | -0.27 | -6.58 | 0.00 | 3.17 | -0.01 | -7.67 | 77380 |
| Cash Flow Per Share | -0.32 | -22.43 | 0.03 | 1.58 | 0.01 | 1.15 | 0.03 | 4.34 | 70384 |
| Net Debt | -0.22 | -7.28 | -0.01 | -0.42 | 0.11 | 10.86 | 0.14 | 13.29 | 44476 |
| Enterprise Value | -0.09 | -1.51 | -0.35 | -6.00 | 0.09 | 7.89 | -0.01 | -1.03 | 43661 |
| Operating Profit | 0.00 | 0.00 | -0.32 | -4.50 | -0.01 | -0.54 | -0.07 | -4.00 | 42824 |
| Cash Earnings Per Share | -0.39 | -11.97 | 0.02 | 0.51 | -0.02 | -1.82 | -0.00 | -0.18 | 22621 |
| EBITDA Per Share | -0.09 | -1.94 | -0.17 | -2.87 | -0.00 | -0.08 | -0.03 | -2.42 | 22411 |
| Funds From Operations Per Share | -0.01 | -0.08 | -0.22 | -2.22 | -0.00 | -0.09 | -0.04 | -4.58 | 3547 |
| EPS Before Goodwill | -0.30 | -2.11 | 0.12 | 0.76 | 0.02 | 1.06 | -0.00 | -0.00 | 2616 |

*Notes*: This table reproduces Table 2 using consensus-level forecasts. See Table 2 for additional notes. Click here to go back to main text.

Table IA.5: Non-Linear Error-Revision Relationship: Percent Growth

| $\text{Error}_{it+1}$ | $\text{Revision}_{it}$ | | $\text{Revision}_{it} \times$ Top/Bottom 20% of $\text{Revision}_{it}$ | | Top 20% of $\text{Revision}_{it}$ | | Bottom 20% of $\text{Revision}_{it}$ | | |
|---|---|---|---|---|---|---|---|---|---|
| | $\widehat{\beta}$ (1) | $t$-stat (2) | $\widehat{\gamma}$ (3) | $t$-stat (4) | $\widehat{\delta^+}$ (5) | $t$-stat (6) | $\widehat{\delta^-}$ (7) | $t$-stat (8) | # of Obs. (9) |
| Net Sales | 0.10 | 5.66 | -0.27 | -13.31 | 0.03 | 7.76 | -0.02 | -3.63 | 399891 |
| Earnings Per Share | 0.08 | 2.49 | -0.25 | -6.04 | -0.03 | -1.42 | -0.15 | -5.08 | 382991 |
| Net Income | 0.04 | 1.38 | -0.23 | -5.84 | -0.03 | -1.37 | -0.14 | -5.05 | 344383 |
| EBITDA | -0.07 | -2.73 | -0.09 | -2.58 | 0.01 | 0.85 | -0.01 | -1.02 | 285367 |
| Pre-tax Profit | -0.02 | -0.61 | -0.16 | -4.56 | -0.01 | -0.73 | -0.07 | -2.76 | 284258 |
| EBIT | -0.09 | -2.42 | -0.10 | -2.16 | -0.01 | -0.80 | -0.03 | -1.35 | 241663 |
| GAAP EPS | -0.01 | -0.41 | -0.23 | -4.86 | -0.10 | -2.27 | -0.30 | -7.60 | 187644 |
| Book Value Per Share | -0.11 | -2.78 | -0.39 | -8.71 | 0.01 | 3.02 | -0.04 | -7.26 | 171474 |
| Net Asset Value | -0.00 | -0.13 | -0.46 | -8.97 | 0.03 | 5.61 | -0.02 | -2.54 | 120994 |
| Cash Flow Per Share | -0.29 | -17.49 | -0.03 | -1.53 | -0.02 | -2.03 | -0.04 | -2.53 | 116942 |
| Net Debt | -0.14 | -6.03 | 0.02 | 0.67 | 0.03 | 1.91 | 0.21 | 11.72 | 99662 |
| Enterprise Value | -0.17 | -4.51 | -0.04 | -0.85 | 0.05 | 5.14 | 0.06 | 4.60 | 88749 |
| Dividend Per Share | -0.14 | -4.42 | -0.11 | -2.88 | -0.03 | -2.45 | -0.04 | -2.06 | 66317 |
| Capital Expenditures | -0.38 | -25.06 | -0.46 | -24.16 | 0.29 | 15.56 | -0.21 | -16.03 | 62096 |
| Operating Profit | 0.05 | 1.38 | -0.25 | -5.70 | 0.02 | 1.25 | -0.09 | -4.07 | 60329 |
| EBITDA Per Share | -0.11 | -2.95 | -0.12 | -2.35 | 0.03 | 1.91 | -0.02 | -1.41 | 17120 |
| Cash Earnings Per Share | -0.22 | -5.83 | -0.13 | -2.30 | -0.00 | -0.30 | -0.02 | -0.64 | 11828 |
| Funds From Operations Per Share | 0.01 | 0.36 | -0.08 | -1.49 | 0.01 | 1.02 | -0.01 | -1.20 | 9379 |
| EPS Before Goodwill | -0.08 | -1.21 | -0.24 | -2.87 | 0.05 | 2.18 | -0.17 | -2.75 | 2636 |

*Notes*: This table reproduces Table 2 where we change our transformation of the variables that are converted into growth rates from log growth rates to percent growth rates. We compute percent growth rates as $g_{it} = \frac{X_{it} - X_{it-1}}{X_{it-1}}$, and redefine forecasts for these variables as $F_t^j g_{it+h} = \frac{F_t^j X_{it+h} - F_t^j X_{it+h-1}}{F_t^j X_{it+h-1}}$, with the convention that $F_t^j X_{it} = X_{it}$. See Table 2 for additional notes. Click here to go back to main text.

Table IA.6: Non-Linear Error-Revision Relationship: Adjusting for Firm-Level Volatility

| $\text{Error}_{it+1}$ | $\text{Revision}_{it}$ | | $\text{Revision}_{it} \times$ Top/Bottom 20% of $\text{Revision}_{it}$ | | Top 20% of $\text{Revision}_{it}$ | | Bottom 20% of $\text{Revision}_{it}$ | | |
|---|---|---|---|---|---|---|---|---|---|
| | $\widehat{\beta}$ (1) | $t$-stat (2) | $\widehat{\gamma}$ (3) | $t$-stat (4) | $\widehat{\delta^+}$ (5) | $t$-stat (6) | $\widehat{\delta^-}$ (7) | $t$-stat (8) | # of Obs. (9) |
| Net Sales | 0.11 | 6.81 | -0.33 | -16.89 | 0.20 | 9.95 | -0.16 | -9.31 | 303623 |
| Earnings Per Share | 0.03 | 1.49 | -0.26 | -7.67 | 0.15 | 7.12 | -0.12 | -3.85 | 272283 |
| Return On Equity | 0.18 | 11.53 | -0.31 | -15.35 | 0.13 | 6.05 | -0.17 | -8.09 | 248053 |
| Net Income | -0.02 | -1.11 | -0.24 | -7.07 | 0.17 | 7.12 | -0.10 | -2.90 | 242091 |
| EBITDA | -0.03 | -1.67 | -0.29 | -11.04 | 0.19 | 7.35 | -0.13 | -4.85 | 205224 |
| Pre-tax Profit | -0.00 | -0.18 | -0.32 | -11.37 | 0.20 | 8.94 | -0.13 | -3.43 | 196802 |
| EBIT | -0.06 | -2.21 | -0.41 | -10.22 | 0.30 | 12.92 | -0.21 | -4.35 | 159223 |
| Book Value Per Share | -0.12 | -2.92 | -0.40 | -8.27 | 0.16 | 5.64 | -0.17 | -5.36 | 119693 |
| GAAP EPS | -0.03 | -1.06 | -0.20 | -5.21 | 0.15 | 4.30 | -0.06 | -1.97 | 110490 |
| Return On Assets | 0.23 | 8.95 | -0.39 | -11.66 | 0.00 | 0.07 | -0.38 | -9.18 | 95610 |
| Gross Margin | 0.11 | 5.23 | -0.25 | -11.33 | 0.12 | 5.29 | -0.17 | -9.62 | 85574 |
| Cash Flow Per Share | -0.27 | -16.27 | -0.21 | -9.07 | 0.27 | 11.26 | -0.11 | -2.68 | 74669 |
| Net Asset Value | -0.07 | -2.03 | -0.62 | -10.91 | 0.41 | 16.04 | -0.24 | -4.08 | 70815 |
| Net Debt | -0.15 | -5.53 | -0.22 | -5.64 | 0.38 | 8.35 | 0.25 | 6.64 | 58422 |
| Enterprise Value | -0.15 | -3.65 | -0.46 | -11.48 | 0.55 | 16.25 | -0.11 | -2.99 | 43748 |
| Dividend Per Share | -0.21 | -8.59 | -0.15 | -4.54 | 0.12 | 4.02 | -0.07 | -2.22 | 38038 |
| Capital Expenditures | -0.33 | -16.59 | -0.13 | -5.00 | 0.10 | 1.66 | -0.08 | -2.57 | 33460 |
| Operating Profit | 0.15 | 2.85 | -0.46 | -7.14 | 0.26 | 6.52 | -0.12 | -2.47 | 22761 |
| Funds From Operations Per Share | 0.01 | 0.18 | -0.11 | -1.86 | 0.11 | 3.03 | -0.11 | -2.27 | 6941 |
| Cash Earnings Per Share | -0.25 | -5.81 | -0.26 | -4.16 | 0.23 | 3.90 | -0.28 | -3.31 | 4483 |
| EBITDA Per Share | -0.04 | -1.08 | -0.36 | -6.07 | 0.25 | 4.43 | -0.24 | -5.92 | 4068 |

*Notes*: This table reproduces Table 2 after normalizing forecasts and realizations by estimates of the firm-level standard deviation of the realizations. The sample here consists of firms for which we have at least 10 observations. See Table 2 for additional notes. This table does not include EPS Before Goodwill because we do not have enough observations. Click here to go back to main text.

Table IA.7: Non-Linear Error-Revision Relationship: Adjusting for Aggregate Volatility

| $\text{Error}_{it+1}$ | $\text{Revision}_{it}$ | | $\text{Revision}_{it} \times$ Top/Bottom 20% of $\text{Revision}_{it}$ | | Top 20% of $\text{Revision}_{it}$ | | Bottom 20% of $\text{Revision}_{it}$ | | |
|---|---|---|---|---|---|---|---|---|---|
| | $\widehat{\beta}$ (1) | $t$-stat (2) | $\widehat{\gamma}$ (3) | $t$-stat (4) | $\widehat{\delta^+}$ (5) | $t$-stat (6) | $\widehat{\delta^-}$ (7) | $t$-stat (8) | # of Obs. (9) |
| Net Sales | 0.11 | 5.46 | -0.27 | -12.06 | 0.06 | 4.90 | -0.09 | -7.87 | 399555 |
| Earnings Per Share | 0.08 | 2.53 | -0.25 | -10.00 | 0.02 | 1.26 | -0.11 | -5.65 | 359817 |
| Return On Equity | 0.21 | 8.94 | -0.22 | -8.29 | 0.00 | 1.05 | -0.03 | -5.32 | 344222 |
| Net Income | 0.03 | 1.42 | -0.24 | -13.95 | 0.02 | 1.25 | -0.10 | -5.90 | 325134 |
| EBITDA | -0.04 | -1.34 | -0.14 | -5.99 | 0.02 | 1.18 | -0.05 | -2.64 | 279943 |
| Pre-tax Profit | 0.00 | 0.09 | -0.21 | -10.90 | 0.03 | 1.79 | -0.08 | -3.97 | 269692 |
| EBIT | -0.04 | -1.37 | -0.22 | -9.49 | 0.04 | 2.50 | -0.06 | -2.93 | 231398 |
| GAAP EPS | -0.02 | -0.91 | -0.13 | -4.67 | -0.00 | -0.03 | -0.06 | -2.21 | 171482 |
| Book Value Per Share | -0.07 | -1.71 | -0.41 | -7.92 | 0.03 | 2.49 | -0.12 | -6.63 | 170367 |
| Return On Assets | 0.50 | 8.27 | -0.52 | -8.59 | -0.08 | -4.80 | -0.14 | -8.02 | 156935 |
| Gross Margin | 0.10 | 4.23 | -0.15 | -5.62 | 0.00 | 1.33 | -0.02 | -5.75 | 150124 |
| Net Asset Value | 0.02 | 0.62 | -0.55 | -11.89 | 0.11 | 8.67 | -0.11 | -5.41 | 119865 |
| Cash Flow Per Share | -0.28 | -15.30 | -0.08 | -4.21 | 0.08 | 4.39 | -0.01 | -0.52 | 112303 |
| Net Debt | -0.17 | -8.09 | -0.12 | -5.44 | 0.20 | 10.25 | 0.20 | 8.31 | 90095 |
| Enterprise Value | -0.19 | -5.20 | -0.28 | -8.53 | 0.26 | 12.27 | -0.03 | -1.03 | 88775 |
| Dividend Per Share | -0.14 | -5.46 | -0.19 | -7.40 | 0.00 | 0.01 | -0.08 | -4.87 | 63788 |
| Capital Expenditures | -0.37 | -23.67 | -0.16 | -6.75 | 0.12 | 4.93 | -0.10 | -4.19 | 62079 |
| Operating Profit | 0.12 | 3.29 | -0.35 | -8.27 | 0.07 | 3.57 | -0.11 | -4.80 | 57718 |
| EBITDA Per Share | -0.09 | -3.02 | -0.13 | -3.47 | 0.04 | 1.88 | -0.06 | -1.98 | 16816 |
| Cash Earnings Per Share | -0.18 | -4.54 | -0.19 | -3.41 | 0.04 | 1.20 | -0.04 | -0.90 | 11563 |
| Funds From Operations Per Share | 0.05 | 1.62 | -0.15 | -2.72 | 0.03 | 1.19 | -0.06 | -1.67 | 9231 |
| EPS Before Goodwill | -0.11 | -2.23 | -0.10 | -1.19 | 0.06 | 1.26 | -0.13 | -2.30 | 2552 |

*Notes*: This table reproduces Table 2 after normalizing forecasts and realizations by estimates of the cross-sectional standard deviation of the realizations in each year. See Table 2 for additional notes. Click here to go back to main text.

Table IA.8: Non-Linear Error-Revision Relationship: Firm Fixed Effects

| $\text{Error}_{it+1}$ | $\text{Revision}_{it}$ | | $\text{Revision}_{it} \times$ Top/Bottom 20% of $\text{Revision}_{it}$ | | Top 20% of $\text{Revision}_{it}$ | | Bottom 20% of $\text{Revision}_{it}$ | | |
|---|---|---|---|---|---|---|---|---|---|
| | $\widehat{\beta}$ (1) | $t$-stat (2) | $\widehat{\gamma}$ (3) | $t$-stat (4) | $\widehat{\delta^+}$ (5) | $t$-stat (6) | $\widehat{\delta^-}$ (7) | $t$-stat (8) | # of Obs. (9) |
| Net Sales | 0.08 | 5.01 | -0.24 | -16.86 | 0.02 | 9.81 | -0.01 | -5.46 | 399809 |
| Earnings Per Share | 0.07 | 2.99 | -0.21 | -7.66 | 0.06 | 5.73 | -0.03 | -2.63 | 360011 |
| Return On Equity | 0.11 | 5.58 | -0.21 | -11.04 | 0.01 | 5.40 | -0.01 | -5.01 | 345082 |
| Net Income | 0.04 | 2.25 | -0.22 | -9.89 | 0.06 | 5.96 | -0.03 | -2.60 | 325230 |
| EBITDA | -0.05 | -2.43 | -0.11 | -5.51 | 0.04 | 5.63 | -0.00 | -0.27 | 280111 |
| Pre-tax Profit | 0.02 | 0.88 | -0.19 | -8.58 | 0.07 | 7.52 | -0.01 | -1.05 | 269883 |
| EBIT | -0.06 | -2.75 | -0.17 | -8.28 | 0.07 | 9.25 | -0.00 | -0.39 | 231455 |
| GAAP EPS | 0.05 | 2.83 | -0.14 | -4.76 | 0.06 | 3.44 | -0.00 | -0.21 | 171523 |
| Book Value Per Share | -0.13 | -3.65 | -0.31 | -8.30 | 0.02 | 5.83 | -0.02 | -5.95 | 170561 |
| Return On Assets | 0.07 | 3.30 | -0.18 | -7.84 | -0.00 | -0.99 | -0.01 | -6.53 | 157063 |
| Gross Margin | -0.00 | -0.09 | -0.12 | -4.52 | 0.00 | 5.80 | -0.00 | -2.33 | 150220 |
| Net Asset Value | -0.08 | -2.37 | -0.43 | -11.86 | 0.04 | 10.68 | -0.02 | -4.70 | 119885 |
| Cash Flow Per Share | -0.23 | -15.39 | -0.06 | -3.20 | 0.09 | 8.26 | 0.02 | 1.50 | 112323 |
| Net Debt | -0.18 | -9.96 | -0.09 | -3.94 | 0.11 | 8.77 | 0.12 | 13.19 | 90126 |
| Enterprise Value | -0.16 | -4.50 | -0.22 | -6.97 | 0.11 | 10.30 | -0.00 | -0.00 | 88791 |
| Dividend Per Share | -0.18 | -6.70 | -0.08 | -2.80 | 0.01 | 1.50 | 0.00 | 0.63 | 63840 |
| Capital Expenditures | -0.28 | -17.75 | -0.15 | -7.27 | 0.06 | 5.47 | -0.05 | -5.33 | 62099 |
| Operating Profit | 0.13 | 3.54 | -0.32 | -6.26 | 0.09 | 6.45 | -0.02 | -1.39 | 57756 |
| EBITDA Per Share | -0.08 | -2.16 | -0.09 | -2.07 | 0.05 | 4.39 | -0.01 | -0.39 | 16824 |
| Cash Earnings Per Share | -0.13 | -2.65 | -0.17 | -3.01 | 0.06 | 4.82 | -0.01 | -0.57 | 11578 |
| Funds From Operations Per Share | -0.00 | -0.06 | -0.08 | -1.57 | 0.01 | 2.29 | -0.01 | -1.76 | 9279 |
| EPS Before Goodwill | -0.10 | -0.97 | 0.00 | 0.04 | 0.03 | 1.32 | -0.01 | -0.13 | 2558 |

*Notes*: This table reproduces Table 2 after including firm fixed effects. See Table 2 for additional notes. Click here to go back to main text.

Table IA.9: Non-Linear Error-Revision Relationship: Split by Analyst Experience

| Quartile of Past Experience | 1 | | 2 | | 3 | | 4 | |
|---|---|---|---|---|---|---|---|---|
| Variable | $\widehat{\gamma}$ (1) | $t$-stat (2) | $\widehat{\gamma}$ (3) | $t$-stat (4) | $\widehat{\gamma}$ (5) | $t$-stat (6) | $\widehat{\gamma}$ (7) | $t$-stat (8) |
| Net Sales | -0.26 | -18.90 | -0.25 | -12.60 | -0.27 | -11.22 | -0.29 | -9.47 |
| Earnings Per Share | -0.28 | -8.51 | -0.25 | -8.85 | -0.28 | -8.19 | -0.23 | -5.87 |
| Return On Equity | -0.28 | -10.44 | -0.25 | -9.98 | -0.28 | -8.08 | -0.29 | -6.79 |
| Net Income | -0.24 | -10.79 | -0.25 | -8.79 | -0.25 | -8.02 | -0.24 | -8.00 |
| EBITDA | -0.14 | -5.08 | -0.15 | -5.06 | -0.16 | -5.31 | -0.15 | -3.73 |
| Pre-tax Profit | -0.23 | -6.86 | -0.23 | -7.35 | -0.19 | -5.83 | -0.24 | -7.09 |
| EBIT | -0.20 | -6.98 | -0.21 | -6.20 | -0.23 | -6.73 | -0.24 | -5.10 |
| GAAP EPS | -0.12 | -4.74 | -0.15 | -4.01 | -0.16 | -3.81 | -0.12 | -2.76 |
| Book Value Per Share | -0.43 | -10.99 | -0.42 | -6.70 | -0.41 | -7.36 | -0.43 | -5.63 |
| Return On Assets | -0.51 | -9.87 | -0.43 | -8.36 | -0.42 | -7.79 | -0.40 | -6.37 |
| Gross Margin | -0.14 | -3.67 | -0.12 | -4.08 | -0.20 | -4.72 | -0.20 | -5.13 |
| Net Asset Value | -0.54 | -9.52 | -0.56 | -9.03 | -0.60 | -8.39 | -0.58 | -8.50 |
| Cash Flow Per Share | -0.09 | -2.58 | -0.04 | -2.15 | -0.07 | -2.67 | -0.07 | -1.75 |
| Net Debt | -0.14 | -4.93 | -0.08 | -1.88 | -0.17 | -4.19 | -0.08 | -2.10 |
| Enterprise Value | -0.24 | -5.26 | -0.26 | -7.76 | -0.27 | -5.33 | -0.32 | -7.80 |
| Dividend Per Share | -0.18 | -4.32 | -0.24 | -6.80 | -0.07 | -1.25 | -0.12 | -1.87 |
| Capital Expenditures | -0.16 | -6.84 | -0.13 | -2.10 | -0.18 | -5.69 | -0.18 | -4.70 |
| Operating Profit | -0.34 | -6.13 | -0.34 | -6.34 | -0.36 | -4.89 | -0.37 | -5.81 |
| EBITDA Per Share | -0.14 | -2.17 | -0.12 | -1.71 | -0.07 | -1.27 | -0.20 | -3.11 |
| Cash Earnings Per Share | -0.25 | -4.05 | -0.09 | -1.02 | -0.17 | -2.00 | -0.19 | -2.33 |
| Funds From Operations Per Share | -0.09 | -0.88 | 0.00 | 0.00 | -0.12 | -1.27 | -0.20 | -1.82 |
| EPS Before Goodwill | -0.11 | -1.18 | -0.17 | -0.98 | -0.29 | -0.93 | 0.26 | 1.67 |

*Notes*: Each pair of columns in this table reports the OLS estimates of equation (2), for each one of the 22 variables on a different subsample based on analyst-firm experience. For each observation, the experience of analyst $j$ is defined at date $t$ as the number of prior firm-years in which analyst $j$ issued forecasts for the same firm, excluding the current firm-year. Experience quartiles are computed separately within each variable's estimation sample. The table reports estimates of $\gamma$, along with the corresponding $t$-statistics with standard errors clustered by firm and time. Variables are ranked by number of observations. Click here to go back to main text.

Table IA.10: Fat Tails in Realizations: Adjusting for Firm-Level Volatility

| | Transformation | Kurtosis | Log-Log regression | | # of Obs. |
|---|---|---|---|---|---|
| | | | Tail Parameter | $R^2$ | |
| Variable | (1) | (2) | (3) | (4) | (5) |
| Net Sales | Growth | 3.40 | 4.14 | 0.99 | 87500 |
| Earnings Per Share | Growth | 3.70 | 4.17 | 0.98 | 80346 |
| Return On Equity | Level | 3.76 | 4.15 | 1.00 | 73100 |
| Net Income | Growth | 3.60 | 4.25 | 0.98 | 69126 |
| Pre-tax Profit | Growth | 3.66 | 4.13 | 0.98 | 59902 |
| EBITDA | Growth | 3.40 | 4.40 | 0.98 | 59770 |
| Book Value Per Share | Growth | 3.64 | 3.94 | 0.99 | 50930 |
| EBIT | Growth | 3.49 | 4.46 | 0.98 | 44622 |
| Return On Assets | Level | 4.01 | 3.92 | 0.99 | 38226 |
| GAAP EPS | Growth | 3.47 | 4.59 | 0.98 | 36745 |
| Cash Flow Per Share | Growth | 3.23 | 4.58 | 0.98 | 33233 |
| Net Asset Value | Growth | 3.32 | 3.86 | 0.99 | 33099 |
| Gross Margin | Level | 5.38 | 3.17 | 0.99 | 27309 |
| Net Debt | Growth | 3.80 | 3.76 | 0.98 | 24395 |
| Dividend Per Share | Growth | 4.19 | 3.93 | 0.99 | 23469 |
| Enterprise Value | Growth | 2.80 | 5.07 | 0.98 | 18837 |
| Capital Expenditures | Growth | 2.80 | 5.26 | 0.98 | 17345 |
| Operating Profit | Growth | 3.60 | 4.24 | 0.98 | 10142 |
| EBITDA Per Share | Growth | 3.19 | 4.92 | 0.99 | 3089 |
| Cash Earnings Per Share | Growth | 3.10 | 5.08 | 0.97 | 2975 |
| Funds From Operations Per Share | Growth | 4.02 | 3.88 | 0.97 | 2250 |
| Normal Distribution | · | 3 | $\infty$ | · | · |

*Notes*: This table reproduces Table 3 after normalizing realizations by estimates of the firm-level standard deviation of the realizations. The sample here consists of firms for which we have at least 10 observations. See Table 3 for additional notes. Click here to go back to main text.

Table IA.11: Fat Tails in Realizations: Adjusting for Aggregate Volatility

| | Transformation | Kurtosis | Log-Log regression | | # of Obs. |
|---|---|---|---|---|---|
| | | | Tail Parameter | $R^2$ | |
| Variable | (1) | (2) | (3) | (4) | (5) |
| Net Sales | Growth | 5.84 | 2.70 | 0.99 | 140481 |
| Earnings Per Share | Growth | 6.34 | 2.42 | 0.99 | 129735 |
| Return On Equity | Level | 7.62 | 2.76 | 1.00 | 127268 |
| Net Income | Growth | 6.28 | 2.42 | 0.99 | 113861 |
| Pre-tax Profit | Growth | 6.18 | 2.47 | 0.99 | 102386 |
| EBITDA | Growth | 6.07 | 2.53 | 0.99 | 99731 |
| Book Value Per Share | Growth | 8.38 | 2.13 | 1.00 | 89882 |
| EBIT | Growth | 6.36 | 2.48 | 0.99 | 82099 |
| Return On Assets | Level | 6.02 | 3.16 | 0.99 | 79477 |
| GAAP EPS | Growth | 5.88 | 2.57 | 0.99 | 74212 |
| Net Asset Value | Growth | 7.84 | 2.17 | 0.99 | 71494 |
| Gross Margin | Level | 2.47 | 7.21 | 0.98 | 64481 |
| Cash Flow Per Share | Growth | 5.52 | 2.64 | 0.99 | 62446 |
| Dividend Per Share | Growth | 6.87 | 2.50 | 0.99 | 49401 |
| Net Debt | Growth | 6.57 | 2.33 | 0.99 | 48338 |
| Enterprise Value | Growth | 4.44 | 3.12 | 0.99 | 48140 |
| Capital Expenditures | Growth | 4.96 | 2.82 | 0.99 | 44623 |
| Operating Profit | Growth | 6.08 | 2.54 | 0.99 | 33033 |
| EBITDA Per Share | Growth | 6.19 | 2.38 | 0.99 | 15344 |
| Cash Earnings Per Share | Growth | 6.01 | 2.44 | 0.99 | 10012 |
| Funds From Operations Per Share | Growth | 7.11 | 2.33 | 1.00 | 3399 |
| EPS Before Goodwill | Growth | 5.70 | 3.07 | 0.98 | 2413 |
| Normal Distribution | · | 3 | $\infty$ | · | · |

*Notes*: This table reproduces Table 3 after normalizing realizations by the yearly cross-sectional standard deviation of the realizations. See Table 2 for additional notes. Click here to go back to main text.

Table IA.12: Fat Tails in Realizations: Percent Growth

| | Transformation | Kurtosis | Log-Log regression | | # of Obs. |
|---|---|---|---|---|---|
| | | | Tail Parameter | $R^2$ | |
| Variable | (1) | (2) | (3) | (4) | (5) |
| Net Sales | Growth | 8.24 | 2.36 | 1.00 | 140550 |
| Earnings Per Share | Growth | 16.25 | 1.45 | 1.00 | 138262 |
| Net Income | Growth | 17.04 | 1.44 | 1.00 | 120986 |
| Pre-tax Profit | Growth | 15.30 | 1.56 | 1.00 | 107989 |
| EBITDA | Growth | 12.58 | 2.01 | 1.00 | 101576 |
| Book Value Per Share | Growth | 9.49 | 2.11 | 1.00 | 90354 |
| EBIT | Growth | 13.88 | 1.64 | 0.99 | 85525 |
| GAAP EPS | Growth | 17.78 | 1.38 | 0.99 | 81453 |
| Net Asset Value | Growth | 11.27 | 1.99 | 0.99 | 71923 |
| Cash Flow Per Share | Growth | 18.83 | 1.70 | 1.00 | 64934 |
| Dividend Per Share | Growth | 11.97 | 2.70 | 0.93 | 51536 |
| Net Debt | Growth | 19.80 | 1.43 | 1.00 | 50938 |
| Enterprise Value | Growth | 8.98 | 2.08 | 1.00 | 48191 |
| Capital Expenditures | Growth | 16.18 | 1.60 | 1.00 | 44652 |
| Operating Profit | Growth | 14.17 | 1.76 | 0.99 | 34292 |
| EBITDA Per Share | Growth | 12.03 | 2.11 | 1.00 | 15563 |
| Cash Earnings Per Share | Growth | 10.21 | 2.11 | 1.00 | 10196 |
| Funds From Operations Per Share | Growth | 8.49 | 2.00 | 0.99 | 3434 |
| EPS Before Goodwill | Growth | 15.94 | 1.70 | 0.99 | 2509 |
| Normal Distribution | · | 3 | $\infty$ | · | · |

*Notes*: This table reproduces Table 3 where we change our transformation of the variables that are converted into growth rates from log growth rates to percent growth rates. We compute percent growth rates as $g_{it} = \frac{X_{it}-X_{it-1}}{X_{it-1}}$. We do not report results for the three "level variables" which are not transformed anyway. See Table 3 for additional notes. Click here to go back to main text.

Table IA.13: Non-Linear Conditional Expectation of Realizations: Adjusting for Firm-Level Volatility

| $g_{it+1}$ | $g_{it}$ | | $g_{it} \times$ Top/Bottom 20% of $g_{it}$ | | Top 20% of $g_{it}$ | | Bottom 20% of $g_{it}$ | | |
|---|---|---|---|---|---|---|---|---|---|
| Variable | $\widehat{\beta}_g$ (1) | $t$-stat (2) | $\widehat{\gamma}_g$ (3) | $t$-stat (4) | $\widehat{\delta_g^+}$ (5) | $t$-stat (6) | $\widehat{\delta_g^-}$ (7) | $t$-stat (8) | # of Obs. (9) |
| Net Sales | 0.43 | 11.87 | -0.36 | -6.89 | 0.63 | 6.90 | 0.10 | 3.80 | 87500 |
| Earnings Per Share | 0.30 | 9.77 | -0.53 | -12.58 | 0.77 | 12.90 | 0.10 | 2.42 | 80346 |
| Return On Equity | 0.89 | 74.58 | -0.03 | -2.00 | 0.22 | 2.73 | 0.13 | 6.02 | 73100 |
| Net Income | 0.30 | 10.46 | -0.51 | -11.63 | 0.75 | 10.10 | 0.12 | 2.72 | 69126 |
| Pre-tax Profit | 0.30 | 9.28 | -0.54 | -13.47 | 0.79 | 12.29 | 0.09 | 2.39 | 59902 |
| EBITDA | 0.28 | 9.90 | -0.40 | -10.64 | 0.64 | 9.13 | 0.10 | 3.04 | 59770 |
| Book Value Per Share | 0.51 | 17.75 | -0.45 | -13.20 | 0.76 | 14.56 | 0.16 | 6.28 | 50930 |
| EBIT | 0.33 | 10.13 | -0.53 | -10.57 | 0.81 | 9.82 | 0.08 | 2.00 | 44622 |
| Return On Assets | 0.87 | 51.63 | 0.06 | 3.60 | -0.20 | -2.41 | 0.06 | 2.29 | 38226 |
| GAAP EPS | 0.12 | 3.74 | -0.46 | -11.98 | 0.72 | 11.63 | 0.11 | 2.33 | 36745 |
| Cash Flow Per Share | -0.13 | -3.89 | -0.16 | -4.44 | 0.36 | 9.43 | 0.14 | 5.13 | 33233 |
| Net Asset Value | 0.49 | 15.54 | -0.40 | -9.22 | 0.74 | 10.05 | 0.06 | 1.71 | 33099 |
| Gross Margin | 1.00 | 293.41 | -0.01 | -2.07 | 0.07 | 2.78 | 0.21 | 6.66 | 27309 |
| Net Debt | 0.12 | 4.18 | -0.23 | -6.82 | 0.34 | 9.23 | 0.09 | 2.23 | 24395 |
| Dividend Per Share | 0.44 | 10.73 | -0.51 | -9.41 | 0.85 | 9.68 | 0.15 | 4.85 | 23469 |
| Enterprise Value | -0.03 | -0.66 | -0.16 | -2.64 | 0.28 | 3.48 | 0.02 | 0.31 | 18837 |
| Capital Expenditures | 0.03 | 0.87 | -0.11 | -2.27 | 0.15 | 2.10 | -0.07 | -1.05 | 17345 |
| Operating Profit | 0.22 | 4.03 | -0.48 | -6.28 | 0.77 | 7.66 | 0.06 | 1.03 | 10142 |
| EBITDA Per Share | 0.30 | 4.89 | -0.29 | -2.81 | 0.55 | 2.64 | 0.22 | 1.88 | 3089 |
| Cash Earnings Per Share | 0.12 | 1.97 | -0.29 | -2.53 | 0.52 | 2.88 | 0.19 | 1.81 | 2975 |
| Funds From Operations Per Share | 0.39 | 5.53 | -0.39 | -3.50 | 0.51 | 3.84 | -0.10 | -1.01 | 2250 |

*Notes*: This table reproduces Table 4 after normalizing realizations by estimates of the firm-level standard deviation of the realizations. The sample here consists of firms for which we have at least 10 observations. This explains why "EPS Before Goodwill" is not reported here (there are not enough observations). See Table 4 for additional notes. Click here to go back to main text.

Table IA.14: Non-Linear Conditional Expectation of Realizations: Adjusting for Aggregate Volatility

| $g_{it+1}$ | $g_{it}$ | | $g_{it}$ × Top/Bottom 20% of $g_{it}$ | | Top 20% of $g_{it}$ | | Bottom 20% of $g_{it}$ | | |
|---|---|---|---|---|---|---|---|---|---|
| Variable | $\widehat{\beta}_g$ (1) | $t$-stat (2) | $\widehat{\gamma}_g$ (3) | $t$-stat (4) | $\widehat{\delta_g^+}$ (5) | $t$-stat (6) | $\widehat{\delta_g^-}$ (7) | $t$-stat (8) | # of Obs. (9) |
| Net Sales | 0.42 | 12.19 | -0.41 | -8.60 | 0.32 | 6.58 | 0.01 | 0.56 | 140481 |
| Earnings Per Share | 0.20 | 5.75 | -0.47 | -15.52 | 0.32 | 10.57 | -0.03 | -1.04 | 129735 |
| Return On Equity | 0.44 | 2.30 | 0.08 | 0.41 | 0.02 | 0.31 | -0.03 | -0.68 | 127268 |
| Net Income | 0.19 | 5.96 | -0.45 | -13.33 | 0.33 | 9.35 | -0.02 | -0.63 | 113861 |
| Pre-tax Profit | 0.19 | 5.25 | -0.43 | -10.98 | 0.34 | 10.26 | -0.01 | -0.28 | 102386 |
| EBITDA | 0.22 | 7.38 | -0.39 | -11.49 | 0.31 | 8.89 | -0.01 | -0.38 | 99731 |
| Book Value Per Share | 0.47 | 14.84 | -0.51 | -13.69 | 0.23 | 10.52 | -0.08 | -5.45 | 89882 |
| EBIT | 0.24 | 5.58 | -0.44 | -10.61 | 0.30 | 8.28 | -0.01 | -0.36 | 82099 |
| Return On Assets | 0.64 | 4.03 | -0.04 | -0.20 | 0.10 | 0.72 | 0.04 | 0.77 | 79477 |
| GAAP EPS | 0.02 | 0.34 | -0.37 | -8.35 | 0.28 | 10.25 | 0.00 | 0.01 | 74212 |
| Net Asset Value | 0.47 | 13.30 | -0.45 | -12.15 | 0.24 | 12.54 | -0.08 | -4.83 | 71494 |
| Gross Margin | 0.90 | 16.42 | -0.23 | -1.65 | 0.69 | 1.73 | 0.53 | 1.80 | 64481 |
| Cash Flow Per Share | -0.18 | -6.21 | -0.13 | -4.34 | 0.18 | 7.33 | 0.07 | 3.56 | 62446 |
| Dividend Per Share | 0.43 | 7.51 | -0.65 | -11.34 | 0.35 | 9.69 | -0.06 | -2.07 | 49401 |
| Net Debt | 0.08 | 4.18 | -0.27 | -8.77 | 0.24 | 10.35 | 0.00 | 0.18 | 48338 |
| Enterprise Value | -0.06 | -1.19 | -0.09 | -1.43 | 0.13 | 2.35 | -0.03 | -0.98 | 48140 |
| Capital Expenditures | 0.06 | 1.67 | -0.22 | -5.86 | 0.11 | 4.71 | -0.10 | -3.42 | 44623 |
| Operating Profit | 0.24 | 4.77 | -0.49 | -10.77 | 0.43 | 10.51 | -0.02 | -0.46 | 33033 |
| EBITDA Per Share | 0.12 | 2.67 | -0.24 | -3.11 | 0.13 | 2.12 | 0.01 | 0.09 | 15344 |
| Cash Earnings Per Share | 0.08 | 2.12 | -0.34 | -5.68 | 0.21 | 4.36 | -0.03 | -0.86 | 10012 |
| Funds From Operations Per Share | 0.46 | 4.25 | -0.46 | -3.72 | 0.22 | 2.09 | -0.09 | -1.08 | 3399 |
| EPS Before Goodwill | 0.21 | 2.09 | -0.54 | -3.97 | 0.40 | 3.45 | -0.08 | -1.00 | 2413 |

*Notes*: This table reproduces Table 4 after normalizing realizations by yearly cross-sectional dispersion of realizations. See Table 4 for additional notes. Click here to go back to main text.

Table IA.15: Non-Linear Conditional Expectation of Realizations: Percent Growth

| $g_{it+1}$ | $g_{it}$ | | $g_{it} \times$ Top/Bottom 20% of $g_{it}$ | | Top 20% of $g_{it}$ | | Bottom 20% of $g_{it}$ | | |
|---|---|---|---|---|---|---|---|---|---|
| Variable | $\widehat{\beta}_g$ (1) | $t$-stat (2) | $\widehat{\gamma}_g$ (3) | $t$-stat (4) | $\widehat{\delta_g^+}$ (5) | $t$-stat (6) | $\widehat{\delta_g^-}$ (7) | $t$-stat (8) | # of Obs. (9) |
| Net Sales | 0.44 | 12.75 | -0.39 | -8.74 | 0.11 | 7.34 | 0.02 | 2.13 | 140550 |
| Earnings Per Share | 0.21 | 4.88 | -0.24 | -5.58 | 0.10 | 5.19 | 0.12 | 2.28 | 138262 |
| Net Income | 0.21 | 5.77 | -0.25 | -7.10 | 0.12 | 5.80 | 0.14 | 2.33 | 120986 |
| Pre-tax Profit | 0.22 | 5.65 | -0.26 | -5.98 | 0.12 | 5.79 | 0.18 | 3.25 | 107989 |
| EBITDA | 0.22 | 6.80 | -0.28 | -7.55 | 0.13 | 6.81 | 0.11 | 4.23 | 101576 |
| Book Value Per Share | 0.46 | 14.70 | -0.52 | -16.33 | 0.10 | 14.20 | -0.02 | -3.44 | 90354 |
| EBIT | 0.26 | 6.36 | -0.31 | -7.06 | 0.12 | 6.10 | 0.13 | 2.61 | 85525 |
| GAAP EPS | 0.08 | 1.32 | -0.14 | -2.46 | 0.05 | 2.79 | 0.20 | 2.24 | 81453 |
| Net Asset Value | 0.46 | 14.54 | -0.46 | -16.01 | 0.10 | 18.63 | -0.02 | -2.09 | 71923 |
| Cash Flow Per Share | -0.20 | -5.86 | 0.15 | 4.69 | -0.00 | -0.04 | 0.46 | 13.38 | 64934 |
| Dividend Per Share | 0.48 | 6.27 | -0.56 | -7.17 | 0.16 | 8.88 | 0.09 | 2.88 | 51536 |
| Net Debt | 0.06 | 1.62 | -0.10 | -2.31 | 0.08 | 5.09 | 0.16 | 5.06 | 50938 |
| Enterprise Value | -0.06 | -1.21 | -0.00 | -0.02 | 0.05 | 2.06 | 0.09 | 6.04 | 48191 |
| Capital Expenditures | 0.04 | 1.31 | -0.10 | -3.39 | 0.09 | 8.12 | 0.11 | 4.53 | 44652 |
| Operating Profit | 0.26 | 3.89 | -0.30 | -4.89 | 0.16 | 6.91 | 0.17 | 3.26 | 34292 |
| EBITDA Per Share | 0.09 | 2.85 | -0.14 | -2.32 | 0.08 | 2.42 | 0.11 | 2.47 | 15563 |
| Cash Earnings Per Share | 0.07 | 1.60 | -0.22 | -3.70 | 0.12 | 4.96 | 0.13 | 4.56 | 10196 |
| Funds From Operations Per Share | 0.36 | 3.45 | -0.32 | -2.38 | 0.04 | 1.78 | -0.02 | -0.93 | 3434 |
| EPS Before Goodwill | 0.14 | 1.22 | -0.24 | -1.70 | 0.09 | 1.15 | 0.17 | 2.49 | 2509 |

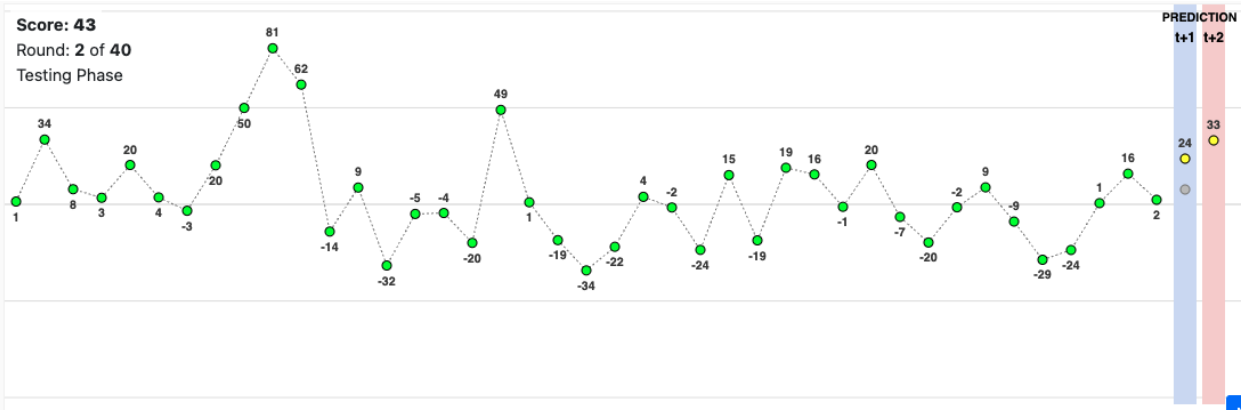
*Notes*: This table reproduces Table 4 where we change our transformation of the variables that are converted into growth rates from log growth rates to percent growth rates. We compute percent growth rates as $g_{it} = \frac{X_{it}-X_{it-1}}{X_{it-1}}$. We do not report results for the three "level variables" which are not transformed anyway. See Table 4 for additional notes. Click here to go back to main text.

Table IA.16: Fit of a Gaussian Approximation in the Bulk of the Distribution

| Variable | $R^2$ (1) | # of Obs. (2) |
|---|---|---|
| Net Sales | 0.954 | 87500 |
| Earnings Per Share | 0.939 | 80345 |
| Return On Equity | 0.831 | 73100 |
| Net Income | 0.949 | 69126 |
| Pre-tax Profit | 0.940 | 59903 |
| EBITDA | 0.951 | 59771 |
| Book Value Per Share | 0.950 | 50929 |
| EBIT | 0.941 | 44620 |
| Return On Assets | 0.717 | 38226 |
| GAAP EPS | 0.908 | 36745 |
| Cash Flow Per Share | 0.929 | 33233 |
| Net Asset Value | 0.902 | 33099 |
| Gross Margin | 0.508 | 27309 |
| Net Debt | 0.894 | 24396 |
| Dividend Per Share | 0.893 | 23469 |
| Enterprise Value | 0.776 | 18838 |
| Capital Expenditures | 0.826 | 17344 |
| Operating Profit | 0.771 | 10142 |
| EBITDA Per Share | 0.631 | 3089 |
| Cash Earnings Per Share | 0.600 | 2976 |
| Funds From Operations Per Share | 0.583 | 2249 |

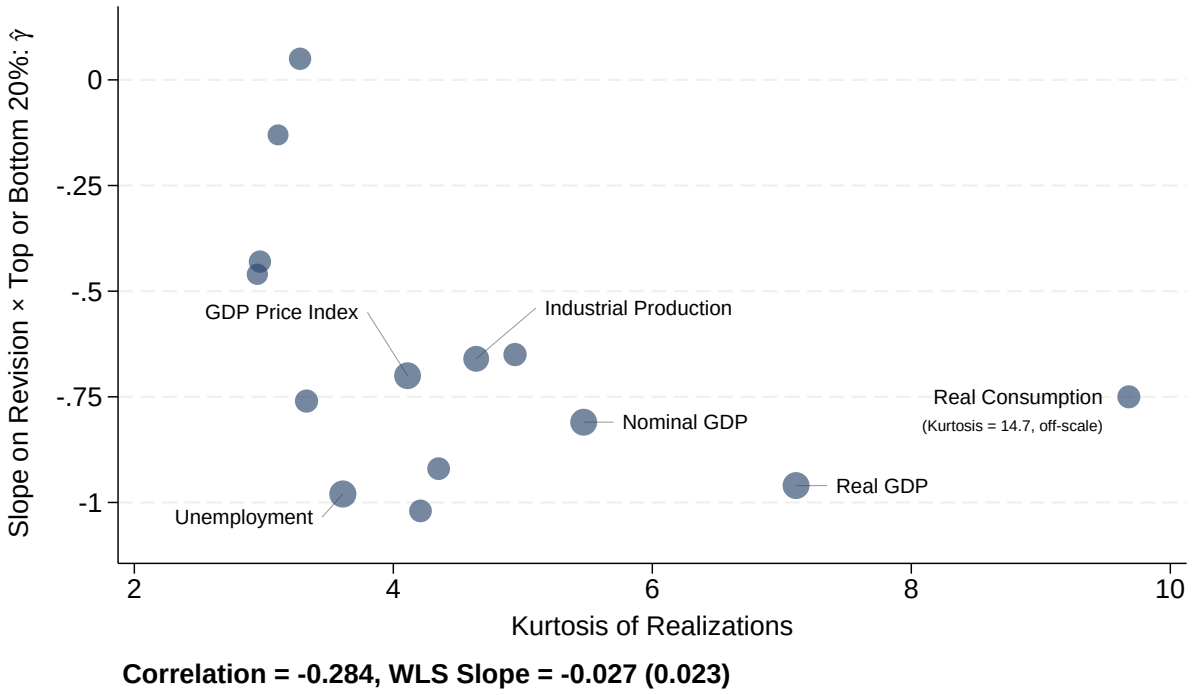
*Notes*: This table tests whether the central portion of the distribution of realizations is approximately Gaussian. For each variable, $x$, we keep one observation per firm-year. We restrict the sample to observations between the 20th and 80th percentiles. We then compute the empirical log-PDF for each centile bin $c$ as $p_c = \log\left(\frac{1}{100}\frac{1}{x_c^+ - x_c^-}\right)$, where $x_c^+$ is the upper bound of $c$ and $x_c^-$ is the lower bound of $c$. We compute $\widehat{x_c}$ as the mean value of $x$ in centile $c$. We then regress $p_c$ on $\widehat{x_c}$ and $\widehat{x_c}^2$. For a Gaussian distribution, the $R^2$ of this regression should equal to 1. Variables are ranked by number of observations. Click here to go back to main text.

Figure IA.1: Screenshot of the Experiment Interface




*Notes*: This figure shows a screenshot of our experiment. Click here to go back to main text.

Figure IA.2: Fat Tails and Error-Revision Slopes: SPF Macro Variables




*Notes:* This figure reproduces the left panel of Figure 5 for the SPF variables described in Section 4.2. Each dot corresponds to a macroeconomic variable, with $\widehat{\gamma}$ from column (3) of Table 6 on the vertical axis. Dot areas and regression weights use the observation counts from column (9) of Table 6. See Figure 5 for additional details. Go back.

# B Proofs and Additional Derivations

In this appendix, we prove all of the results in the main text. We prove our results under the following version of Assumption 1, which generalizes the tail condition by replacing the constant $\Theta$ with a slowly varying function. Additionally, we require two mild regularity conditions on the density $f_\epsilon(\cdot)$.

**Assumption IA.1.** *The distribution of $\epsilon_t$ is symmetric and regularly varying:*

$$P(\epsilon_t > x) = \int_x^\infty f_\epsilon(\epsilon)d\epsilon \sim x^{-\nu}L(x) \text{ as } x \longrightarrow \infty,$$

*where $\nu > 2$ is the index and $L(x)$ is slowly varying in the sense that, for all $c$, $\frac{L(cx)}{L(x)} \xrightarrow{x\to\infty} 1$. In addition, (i) $f_\epsilon(\cdot)$ is unimodal, and (ii) $f_\epsilon(\cdot)$ is differentiable for large $x$ with $\frac{xf'_\epsilon(x)}{f_\epsilon(x)} \longrightarrow -(\nu+1)$ as $x \longrightarrow \infty$.*

Condition (ii) states that the density varies smoothly in the tail: it implies that $f_\epsilon(x) = x^{-(\nu+1)}L_1(x)$ for large $x$, where $L_1(x) \equiv x^{\nu+1}f_\epsilon(x)$ is slowly varying with $\frac{xL'_1(x)}{L_1(x)} \longrightarrow 0$, and Karamata's theorem (Bingham et al., 1987, Proposition 1.5.10) then implies $L_1(x) \sim \nu L(x)$. Conditions (i) and (ii) discipline the local behavior of $f_\epsilon(\cdot)$ in the tails, which we require because our proofs characterize conditional expectations given exact realizations in the tails, while the tail condition above only restricts the counter-CDF. Both conditions are satisfied by standard fat-tailed distributions: for example, for the $t$-distribution referenced in the main text, $\frac{xf'_\epsilon(x)}{f_\epsilon(x)} = -(\nu+1)\frac{x^2}{\nu+x^2} \longrightarrow -(\nu+1)$.

## B.1 Proof of Proposition 1

*Proof.* From the definition of the process, we obtain:

$$E\left(g_{t+1}|g_t\right) = \rho E\left(g_t^*|g_t\right)$$

We now apply Tweedie's formula (Robbins, 1956; Efron, 2011), which states that, for $y = x + \eta$, where $x$ is Gaussian and $x$ and $\eta$ are independent, then:

$$E(x|y) = -\sigma_x^2 \frac{d\log h(y)}{dy}$$

where $h$ is the marginal distribution of $y$ and $\sigma_x^2$ the variance of $x$. Combining the last two equations with $x = g_t^*$ and $\eta = \epsilon_t$ delivers the result. □

## B.2 Proof of Proposition 2

*Proof.* By the definition of a conditional expectation, we have:

$$E(g^*|g) = \frac{\int x f_\epsilon\left(\frac{g-x}{\sigma_\epsilon}\right)\Phi(x)dx}{\int f_\epsilon\left(\frac{g-x}{\sigma_\epsilon}\right)\Phi(x)dx} = \frac{\int x\frac{f_\epsilon\left(\frac{g-x}{\sigma_\epsilon}\right)}{f_\epsilon\left(\frac{g}{\sigma_\epsilon}\right)}\Phi(x)dx}{\int \frac{f_\epsilon\left(\frac{g-x}{\sigma_\epsilon}\right)}{f_\epsilon\left(\frac{g}{\sigma_\epsilon}\right)}\Phi(x)dx}. \tag{IA.1}$$

Given Assumption IA.1, we have

$$\frac{f_\epsilon\left(\frac{g-x}{\sigma_\epsilon}\right)}{f_\epsilon\left(\frac{g}{\sigma_\epsilon}\right)} = \frac{(g-x)^{-\nu-1}L_1\left(\frac{g-x}{\sigma_\epsilon}\right)}{g^{-\nu-1}L_1\left(\frac{g}{\sigma_\epsilon}\right)} \tag{IA.2}$$

$$= \left(1-\frac{x}{g}\right)^{-\nu-1}\frac{L_1\left(\frac{g-x}{\sigma_\epsilon}\right)}{L_1\left(\frac{g}{\sigma_\epsilon}\right)}. \tag{IA.3}$$

where the first equality follows from condition (ii) of Assumption IA.1, which implies that $f_\epsilon(y) = y^{-(\nu+1)}L_1(y)$ for the slowly varying function $L_1(\cdot)$ defined there. Now we will derive some bounds on each component of the previous expression, with the goal of deriving bounds on $\frac{f_\epsilon\left(\frac{g-x}{\sigma_\epsilon}\right)}{f_\epsilon\left(\frac{g}{\sigma_\epsilon}\right)}$. By the binomial expansion, we have:

$$\left(1-\frac{x}{g}\right)^{-\nu-1} = 1+(\nu+1)\frac{x}{g}+O\left(\frac{x^2}{g^2}\right).$$

When $|x| \le |g|^\beta$ and $\beta \in (0, 1/2)$, this remainder term is $O(g^{2\beta-2}) = o\left(\frac{1}{g}\right)$. Next, we bound the ratio of slowly varying functions. The Uniform Convergence Theorem for slowly varying functions (Bingham et al., 1987, Theorem 1.2.1) implies that this ratio converges to one uniformly over $|x| \le |g|^\beta$, but it provides no rate of convergence, and is therefore not sufficient to obtain a remainder term that is uniformly $o\left(\frac{1}{g}\right)$ over this region. Instead, we use condition (ii) of Assumption IA.1, which provides the required local control. Define $\eta(t) \equiv \frac{tL_1'(t)}{L_1(t)} = (\nu+1) + \frac{tf_\epsilon'(t)}{f_\epsilon(t)}$, which converges to zero by condition (ii). Writing the logarithm of the ratio as an integral, we have, for $|x| \le |g|^\beta$ and $|g|$ sufficiently large,

$$\left|\log\frac{L_1\left(\frac{g-x}{\sigma_\epsilon}\right)}{L_1\left(\frac{g}{\sigma_\epsilon}\right)}\right| = \left|\int_{\frac{g-x}{\sigma_\epsilon}}^{\frac{g}{\sigma_\epsilon}}\frac{\eta(t)}{t}dt\right| \le \eta_g \log\left(\frac{|g|}{|g|-|x|}\right) \le 2\eta_g\frac{|x|}{|g|}, \quad \text{where } \eta_g \equiv \sup_{t\ge\frac{|g|-|g|^\beta}{\sigma_\epsilon}}|\eta(t)| \longrightarrow 0.$$

Exponentiating, we have, uniformly for $|x| \le |g|^\beta$,

$$\frac{L_1\left(\frac{g-x}{\sigma_\epsilon}\right)}{L_1\left(\frac{g}{\sigma_\epsilon}\right)} = 1+O\left(\eta_g\frac{|x|}{|g|}\right).$$

Multiplying the previous two results, we have, uniformly for $|x| \leq |g|^{\beta}$,

$$\frac{f_\epsilon\left(\frac{g-x}{\sigma_\epsilon}\right)}{f_\epsilon\left(\frac{g}{\sigma_\epsilon}\right)} = 1 + (\nu+1)\frac{x}{g} + R_g(x), \quad \text{where } |R_g(x)| \leq C\left(\frac{x^2}{g^2} + \eta_g \frac{|x|}{|g|}\right)$$

for some constant $C > 0$. Note that the remainder $R_g(x)$ is not uniformly $o\left(\frac{1}{g}\right)$ over this region because $|x|$ can be as large as $|g|^{\beta}$. Instead, when we return to the integrals in (IA.1) below, we integrate the bound on $|R_g(x)|$ against the Gaussian density $\Phi(\cdot)$, whose finite moments deliver the required $o\left(\frac{1}{g}\right)$ rate.

Next, consider the region $|g/2| > |x| > |g|^{\beta}$. Taylor's theorem implies that, for some constant $C$,

$$\left| \left(1 - \frac{x}{g}\right)^{-\nu-1} - 1 - (\nu+1)\frac{x}{g} \right| \leq C\left(\frac{x}{g}\right)^2$$

Since $L_1$ is slowly varying, the Uniform Convergence Theorem implies that

$$\sup_{|x| \leq |g|/2} \left| \frac{L_1\left(\frac{g-x}{\sigma_\epsilon}\right)}{L_1\left(\frac{g}{\sigma_\epsilon}\right)} - 1 \right| \longrightarrow 0 \quad \text{as } |g| \to \infty.$$

Hence, uniformly for $|x| \leq |g|/2$,

$$\frac{L_1\left(\frac{g-x}{\sigma_\epsilon}\right)}{L_1\left(\frac{g}{\sigma_\epsilon}\right)} = 1 + o(1).$$

Combining these results, we obtain

$$\begin{aligned} \frac{f_\epsilon\left(\frac{g-x}{\sigma_\epsilon}\right)}{f_\epsilon\left(\frac{g}{\sigma_\epsilon}\right)} &= \left(1 - \frac{x}{g}\right)^{-\nu-1} \frac{L_1\left(\frac{g-x}{\sigma_\epsilon}\right)}{L_1\left(\frac{g}{\sigma_\epsilon}\right)} \\ &= 1 + (\nu+1)\frac{x}{g} + O\left(\frac{x^2}{g^2}\right) + o(1), \end{aligned}$$

where the $o(1)$ term is uniform over $|x| \leq |g|/2$. Therefore, for $|g|^{\beta} < |x| < |g|/2$,

$$\left| \frac{f_\epsilon\left(\frac{g-x}{\sigma_\epsilon}\right)}{f_\epsilon\left(\frac{g}{\sigma_\epsilon}\right)} - 1 - (\nu+1)\frac{x}{g} \right| \leq C'\frac{x^2}{g^2} + o(1),$$

for some constant $C' > 0$.

Finally, we consider the region $|x| \geq |g|/2$. In this region, Taylor's theorem does not give a uniform bound because the remainder term approaches $\infty$ as $|x| \longrightarrow |g|$. However, since $\epsilon$ is regularly varying and has a

bounded density, we have

$$\frac{f_\epsilon\left(\frac{g-x}{\sigma_\epsilon}\right)}{f_\epsilon\left(\frac{g}{\sigma_\epsilon}\right)} \leq B|g|^{\nu+1},$$

for some $B > 0$.

Next, return to the integrals in (IA.1). Focusing on the denominator, we have

$$\begin{aligned}
\int \frac{f_\epsilon\left(\frac{g-x}{\sigma_\epsilon}\right)}{f_\epsilon\left(\frac{g}{\sigma_\epsilon}\right)}\Phi(x)dx &= \int \left(1+(\nu+1)\frac{x}{g}\right)\Phi(x)dx + \int \left(\frac{f_\epsilon\left(\frac{g-x}{\sigma_\epsilon}\right)}{f_\epsilon\left(\frac{g}{\sigma_\epsilon}\right)} - 1 - (\nu+1)\frac{x}{g}\right)\Phi(x)dx \\
&= \int \left(1+(\nu+1)\frac{x}{g}\right)\Phi(x)dx + \int_{|x|\leq g^\beta} \left(\frac{f_\epsilon\left(\frac{g-x}{\sigma_\epsilon}\right)}{f_\epsilon\left(\frac{g}{\sigma_\epsilon}\right)} - 1 - (\nu+1)\frac{x}{g}\right)\Phi(x)dx \\
&\quad + \int_{|x|>g^\beta} \left(\frac{f_\epsilon\left(\frac{g-x}{\sigma_\epsilon}\right)}{f_\epsilon\left(\frac{g}{\sigma_\epsilon}\right)} - 1 - (\nu+1)\frac{x}{g}\right)\Phi(x)dx
\end{aligned}$$

where the final equality just breaks up the integral. Focusing on the first integral, the triangle inequality and the bound on $|R_g(x)|$ established above imply

$$\begin{aligned}
\left|\int_{|x|\leq g^\beta} \left(\frac{f_\epsilon\left(\frac{g-x}{\sigma_\epsilon}\right)}{f_\epsilon\left(\frac{g}{\sigma_\epsilon}\right)} - 1 - (\nu+1)\frac{x}{g}\right)\Phi(x)dx\right| &\leq \int_{|x|\leq g^\beta} |R_g(x)|\Phi(x)dx \\
&\leq \frac{C}{g^2}\int x^2\Phi(x)dx + \frac{C\eta_g}{|g|}\int |x|\Phi(x)dx \\
&= o\left(\frac{1}{g}\right)
\end{aligned}$$

where the final equality follows from the fact that $\Phi(\cdot)$ has finite first and second moments and $\eta_g \longrightarrow 0$. Turning to the second integral, we split it into two regions based on the bounds above. The first of these two regions is $|x| \in \left(g^\beta, |g|/2\right) \equiv \mathcal{R}$. In this region, we have

$$\begin{aligned}
\left|\int_{|x|\in\mathcal{R}} \left(\frac{f_\epsilon\left(\frac{g-x}{\sigma_\epsilon}\right)}{f_\epsilon\left(\frac{g}{\sigma_\epsilon}\right)} - 1 - (\nu+1)\frac{x}{g}\right)\Phi(x)\,dx\right| &\leq \int_{|x|\in\mathcal{R}} \left|\frac{f_\epsilon\left(\frac{g-x}{\sigma_\epsilon}\right)}{f_\epsilon\left(\frac{g}{\sigma_\epsilon}\right)} - 1 - (\nu+1)\frac{x}{g}\right|\Phi(x)\,dx \\
&\leq \frac{C'}{g^2}\int_{|x|\in\mathcal{R}} x^2\Phi(x)\,dx \;+\; o(1)\int_{|x|\in\mathcal{R}}\Phi(x)\,dx \\
&= o\left(\frac{1}{g}\right)
\end{aligned}$$

where the first line follows from the triangle inequality, the second inequality follows from the bound established above, and the final line follows because a Gaussian PDF implies $\int_{|x|>g^\beta} x^k\Phi(x)\,dx \to 0$ for any finite $k$ as $|g| \to \infty$.

For the second region, $|x| \geq |g|/2$, we have

$$\begin{aligned}\left|\int_{|x|\geq|g|/2}\left(\frac{f_\epsilon\left(\frac{g-x}{\sigma_\epsilon}\right)}{f_\epsilon\left(\frac{g}{\sigma_\epsilon}\right)}-1-(\nu+1)\frac{x}{g}\right)\Phi(x)dx\right| &\leq \int_{|x|\geq|g|/2}\left(B|g|^{\nu+1}+1+(\nu+1)\frac{|x|}{|g|}\right)\Phi(x)dx \\ &= O\left(|g|^{\nu+1}e^{-g^2/(8\sigma_{g^*}^2)}\right) \\ &= o\left(\frac{1}{g}\right)\end{aligned}$$

where the first line follows from the bounds above, the second line follows from Gaussian tail bounds, and the third line follows from the fact that exponential decay dominates polynomial growth.

Putting together the expressions for these two integrals, we have

$$\int\frac{f_\epsilon\left(\frac{g-x}{\sigma_\epsilon}\right)}{f_\epsilon\left(\frac{g}{\sigma_\epsilon}\right)}\Phi(x)dx=\int\left(1+(\nu+1)\frac{x}{g}\right)\Phi(x)dx+o\left(\frac{1}{g}\right).$$

We can then use the same arguments to show that

$$\int x\frac{f_\epsilon\left(\frac{g-x}{\sigma_\epsilon}\right)}{f_\epsilon\left(\frac{g}{\sigma_\epsilon}\right)}\Phi(x)dx=\int x\left(1+(\nu+1)\frac{x}{g}\right)\Phi(x)dx+o\left(\frac{1}{g}\right).$$

Combining the previous two results and using the fact that $\int\Phi(x)dx=1$, $\int x\Phi(x)dx=0$, and $\int x^2\Phi(x)dx=\sigma_{g^*}^2$, we have, as $|g|\to\infty$,

$$E(g^*\mid g)\sim\frac{(\nu+1)\sigma_{g^*}^2}{g}.$$

Applying the data-generating process delivers the desired result. □

## B.3 Proof of Proposition 3

*Proof.* By definition of subjective expectations:

$$ERR_{t+1}=\frac{1-K}{K}REV_t+\underbrace{(g_{t+1}-\rho g_t)}_{\sigma_\epsilon\epsilon_{t+1}+\sigma_u u_{t+1}-\rho\sigma_\epsilon\epsilon_t} \tag{IA.4}$$

This implies:

$$E\left(ERR_{t+1}|REV_t\right)=\frac{1-K}{K}REV_t-\rho\sigma_\epsilon E\left(\epsilon_t|REV_t\right) \tag{IA.5}$$

Now, we focus on the second term on the right-hand side. Given the definition of forecasts in equation (9), we can express forecast revisions using the following orthogonal decomposition:

$$REV_t = K\rho\sigma_\epsilon \left[ \epsilon_t - \underbrace{K\rho \sum_{s\geq 0} ((1-K)\rho)^s \epsilon_{t-s-1}}_{\equiv E_t} + \underbrace{\frac{\sigma_u}{\sigma_\epsilon} \sum_{s\geq 0} ((1-K)\rho)^s u_{t-s}}_{\equiv U_t} \right]$$

Under Assumption IA.1, we can make use of the following result for the asymptotic behavior of regularly varying random variables to characterize the tail behavior of $E_t$:

**Lemma IA.1** (Geluk and de Vries 2006, Theorem 1)**.** *Suppose $X_0, X_1, ...$ are IID regularly varying random variables and $c_0, c_1, ...$ are weights that represent a convergent geometric series. Then*

$$P\left(\sum_{s\geq 0} c_i X_i > x\right) \sim \sum_{s\geq 0} P(c_i X_i > x) \text{ as } x \to \infty$$

Applying Lemma IA.1 to the definition of $E_t$, we have, as $E \to +\infty$,

$$\begin{aligned}
P\left(E_t > E\right) &= P\left(K\rho \sum_{s\geq 0} ((1-K)\rho)^s \epsilon_{t-s-1} > E\right) \\
&\sim \sum_{s\geq 0} P\left(K\rho\left((1-K)\rho\right)^s \epsilon_{t-s-1} > E\right) \\
&\sim \sum_{s\geq 0} L(E) \left(\frac{K\rho\left((1-K)\rho\right)^s}{E}\right)^\nu \\
&= L(E)\frac{1}{E^\nu} \times \frac{K^\nu\rho^\nu}{1-\rho^\nu(1-K)^\nu} \\
&\sim P(\epsilon_t > E) \times \frac{K^\nu\rho^\nu}{1-\rho^\nu(1-K)^\nu}
\end{aligned}$$

The final expression shows that $E_t$ is also regularly varying with index $\nu$ but a different scale given by the formula. Given this result, we can characterize the $E\left(\epsilon_t | REV_t\right)$ for large revisions as follows:

$$\begin{aligned}
E\left(\epsilon_t | REV_t = r\right) &= E\left(\epsilon_t \mid K\rho\sigma_\epsilon\left(\epsilon_t - E_t + U_t\right) = r\right) \\
&= E\left(\epsilon_t \mid \epsilon_t - E_t + U_t = \frac{r}{K\rho\sigma_\epsilon}\right) \\
&\sim E\left(\epsilon_t \mid \epsilon_t - E_t = \frac{r}{K\rho\sigma_\epsilon}\right) \text{ as } r \to +\infty
\end{aligned}$$

where the last line follows from the fact that $E_t$ has the same tail parameter as $\epsilon_t$, which is thicker than that of $U_t$ by Assumption 2. Because we condition on the exact value of the sum of two regularly varying variables,

rather than on a tail event, computing this conditional expectation requires knowing the relative scale of the *densities* of these two variables in the tails, not only the relative scale of their counter-CDFs (Denuit et al., 2025). As shown above, $E_t$ has the same index as $\epsilon_t$, but its counter-CDF differs by a scale of $\frac{K^\nu \rho^\nu}{1-\rho^\nu(1-K)^\nu}$ in the tails. We now show that this scale carries over to the densities.

Since $\epsilon_t$ is symmetric and unimodal by Assumption IA.1, and $E_t$ is an almost surely convergent weighted sum of independent draws of $\epsilon_t$, $E_t$ is also symmetric and unimodal: the class of symmetric and unimodal distributions is closed under scaling, convolution, and weak limits (Wintner, 1938; Dharmadhikari and Joag-Dev, 1988). The densities of $\epsilon_t$ and $E_t$ are therefore non-increasing for large arguments, so applying the Monotone Density Theorem (Bingham et al., 1987, Theorem 1.7.2) to the counter-CDF asymptotics in Assumption IA.1 and to those derived for $E_t$ above yields, as $x \to +\infty$:

$$f_\epsilon(x) \sim \nu x^{-\nu-1} L(x) \quad \text{and} \quad f_{E_t}(x) \sim \nu x^{-\nu-1} L(x) \times \frac{K^\nu \rho^\nu}{1-\rho^\nu(1-K)^\nu}.$$

Hence the densities of $\epsilon_t$ and $E_t$ are regularly varying with the same index, and their relative scale in the tails equals the relative scale of their counter-CDFs.

Applying Proposition 4.9 from Denuit et al. (2025) along with the fact that $\epsilon_t$ (and hence $E_t$) is symmetric, we have, as $r \to +\infty$:

$$\begin{aligned} E\left(\epsilon_t | \epsilon_t - E_t = \frac{r}{K\rho\sigma_\epsilon}\right) &\sim \frac{L(\frac{r}{K\rho\sigma_\epsilon})}{L(\frac{r}{K\rho\sigma_\epsilon}) + L(\frac{r}{K\rho\sigma_\epsilon})\frac{K^\nu\rho^\nu}{1-\rho^\nu(1-K)^\nu}} \frac{r}{K\rho\sigma_\epsilon} \\ &= \frac{1}{1+\frac{K^\nu\rho^\nu}{1-\rho^\nu(1-K)^\nu}} \frac{r}{K\rho\sigma_\epsilon} \end{aligned}$$

Now, we plug this result into equation (IA.5) and obtain that:

$$E\left(ERR_{t+1} | REV_t\right) \sim \underbrace{\left[\frac{1-K}{K} - \frac{1}{K}\frac{1}{1+\frac{K^\nu\rho^\nu}{1-(1-K)^\nu\rho^\nu}}\right]}_{\equiv C} REV_t. \tag{IA.6}$$

Repeating the analogous argument for the case in which $REV_t \to -\infty$ and using the fact that $\epsilon_t$ and $u_t$ are symmetric delivers the first desired result.

To show that $C < 0$, we use the definition of the steady state Kalman gain, $K$, and prior variance, $P$:

$$\begin{aligned} K &= \frac{P}{P+\sigma_\epsilon^2} \\ P(1-\rho^2(1-K)) &= \sigma_u^2 \end{aligned}$$

whose combination shows that

$$1 - K = \frac{K}{\frac{\sigma_u^2}{\sigma_\epsilon^2} + K\rho^2}.$$

For $C < 0$, we need

$$\frac{1-K}{K} < \frac{1}{K} \cdot \frac{1}{1 + \frac{K^\nu \rho^\nu}{1-(1-K)^\nu \rho^\nu}}.$$

Multiplying both sides by $K$ and cross-multiplying yields

$$(1-K)K^\nu \rho^\nu + K(1-K)^\nu \rho^\nu < K.$$

Dividing by $K$ and factoring

$$(1-K)\rho^\nu \left[K^{\nu-1} + (1-K)^{\nu-1}\right] < 1.$$

Since $\nu > 2$, we have $\nu - 1 > 1$. For any $x \in (0,1)$, this implies $x^{\nu-1} < x$. Applying this to both $K$ and $1-K$:

$$K^{\nu-1} + (1-K)^{\nu-1} < K + (1-K) = 1.$$

Therefore:

$$(1-K)\rho^\nu \left[K^{\nu-1} + (1-K)^{\nu-1}\right] < (1-K)\rho^\nu < 1$$

where the final inequality comes from the fact that $\rho, K \in (0,1)^2$. This is the desired inequality two lines above, which concludes the proof. □

## B.4 Proof of Corollary 1

*Proof.* The definition of the Kalman filter implies that it is the minimum linear mean squared error predictor. Combined with the law of iterated expectations, this implies that:

$$\begin{aligned} E\left(ERR_{t+1}REV_t\right) = E\Big[&E\left(ERR_{t+1}REV_t||REV_t| \geq r\right) P\left(|REV_t| \geq r\right) \\ &+ E\left(ERR_{t+1}REV_t||REV_t| < r\right) P\left(|REV_t| < r\right)\Big] = 0. \end{aligned}$$

By Proposition 3, there exists an $\overline{R}$ such that $E\left(ERR_{t+1}REV_t||REV_t| \geq \overline{R}\right) < 0$. Therefore, the fact that $E\left(ERR_{t+1}REV_t\right) = 0$ implies that $E\left(ERR_{t+1}REV_t||REV_t| < \overline{R}\right) > 0$, which is the desired result. □

## B.5 Signal Strength in the Dynamic Model

This section discusses how the results in Section 3.5 generalize to the dynamic model. While we cannot characterize signal strength as explicitly as in the one-period problem, we can still derive a similar intuition for why insensitivity to signal strength provides a microfoundation for ignoring fat tails. We define signal

strength as

$$\mathcal{S}(REV_t) \equiv \frac{E\left(g_t^* - F_{t-1}g_t^* \mid REV_t\right)}{g_t - F_{t-1}g_t}, \quad g_t \neq F_{t-1}g_t. \tag{IA.7}$$

This is the natural generalization of $S(g_t)$ in (11) to a dynamic setting because the rational revision relative to the forecaster's prior forecast is $\rho\,\mathcal{S}(REV_t)\,(g_t - F_{t-1}g_t)$, just as the rational forecast in the one-period problem is $\rho S(g_t)g_t$.[22] Unlike in the one-period problem, the signal is now the forecast innovation, $g_t - F_{t-1}g_t$, and $\mathcal{S}(REV_t)$ is the share of this innovation that rationally reflects news about the persistent component. The forecaster's actual revision instead applies the constant strength $K$ to every innovation, $REV_t = \rho K\,(g_t - F_{t-1}g_t)$, mirroring the one-period forecaster, who applies the default $\bar{S}$ to every signal. Therefore, we obtain an exact analogue of (13):

$$E\left(ERR_{t+1} \mid REV_t\right) = \rho\left(\mathcal{S}(REV_t) - K\right)\left(g_t - F_{t-1}g_t\right). \tag{IA.8}$$

To derive these expressions, note that equation (9) implies $F_t g_{t+1} = \rho K g_t + \rho(1-K)F_{t-1}g_t$ and $F_{t-1}g_{t+1} = \rho F_{t-1}g_t$, which together deliver $REV_t = \rho K\,(g_t - F_{t-1}g_t)$; in particular, the denominator of (IA.7) is pinned down by the conditioning variable. The expression for the rational revision follows because $g_{t+1} = \rho g_t^* + \sigma_u u_{t+1} + \sigma_\epsilon \epsilon_{t+1}$, where the time-$(t+1)$ shocks are mean zero and independent of $REV_t$, and because $F_{t-1}g_{t+1} = \rho F_{t-1}g_t = \rho F_{t-1}g_t^*$, given that the forecaster's expectation of $\epsilon_t$ based on period-$(t-1)$ information is zero. Together, these results imply:

$$E\left(g_{t+1} - F_{t-1}g_{t+1} \mid REV_t\right) = \rho E\left(g_t^* - F_{t-1}g_t^* \mid REV_t\right) = \rho\,\mathcal{S}(REV_t)\left(g_t - F_{t-1}g_t\right).$$

Equation (IA.8) then follows by combining this expression with $g_{t+1} - F_{t-1}g_{t+1} = ERR_{t+1} + REV_t$, which implies that $E\left(ERR_{t+1} \mid REV_t\right)$ equals the rational revision minus $REV_t$, and substituting $REV_t = \rho K\,(g_t - F_{t-1}g_t)$. Expressed in units of revisions, (IA.8) reads $E\left(ERR_{t+1} \mid REV_t\right) = \frac{\mathcal{S}(REV_t)-K}{K} \times REV_t$, and it implies that the condition characterizing $\overline{R}$ in Section 3.4, $E\left(ERR_{t+1} \mid REV_t = \overline{R}\right) = 0$, is equivalent to $\mathcal{S}(\overline{R}) = K$.

Equation (IA.8) provides another way of understanding Proposition 3 and Corollary 1. Below, we show that $\mathcal{S}(\cdot)$ inherits the two properties of signal strength that generate underreaction and overreaction in the one-period problem. First, $K$ equals the weighted average of $\mathcal{S}(REV_t)$, so that the Kalman filter is correct on average. Second, consistent with Proposition 3, $\mathcal{S}(REV_t)$ converges to a limit that lies strictly below $K$ for large revisions, resulting in overreaction.[23] Finally, consistent with Corollary 1, there is a range of moderate revisions for which $\mathcal{S}(REV_t) > K$ and forecasters underreact. The switch between underreaction and overreaction occurs at the revision where the two strengths cross, $\mathcal{S}(\overline{R}) = K$.

[22] This differs from the standard definition of signal strength in forecasting problems discussed in Section 3.5 (Bastianello and Imas, 2026), which is not tractable in our setting.

[23] Unlike in the one-period problem, where $S(g_t) \longrightarrow 0$ in the tails, this limit is strictly positive: a large revision may reflect not only a large transitory shock today, but also the unwinding of past transitory shocks, which is informative about the gap between the forecaster's prior and the persistent component. See below for the exact expression.

We now establish the properties of $\mathcal{S}(\cdot)$ stated above. Throughout, we work with the normalized forecast innovation, $s_t \equiv \frac{g_t - F_{t-1}g_t}{\sigma_\epsilon} = \frac{REV_t}{K\rho\sigma_\epsilon}$, which satisfies $s_t = \epsilon_t - E_t + U_t$ by the orthogonal decomposition in the proof of Proposition 3, and, with a slight abuse of notation, we write $\mathcal{S}(s)$ for the value of $\mathcal{S}(\cdot)$ at the revision $K\rho\sigma_\epsilon s$. Defining $\psi(s) \equiv E(\epsilon_t \mid s_t = s)$ and using $g_t^* - F_{t-1}g_t^* = \sigma_\epsilon(s_t - \epsilon_t)$, the definition (IA.7) implies $\mathcal{S}(s) = 1 - \frac{\psi(s)}{s}$.

First, the weighted average of $\mathcal{S}(\cdot)$ equals $K$. Computing the variances of the geometric sums $E_t$ and $U_t$ defined in the proof of Proposition 3, the variance of $s_t$ is:

$$E\left(s_t^2\right) = 1 + \frac{K^2\rho^2 + \frac{\sigma_u^2}{\sigma_\epsilon^2}}{1 - (1-K)^2\rho^2} = \frac{1}{1-K},$$

where the second equality uses $\frac{\sigma_u^2}{\sigma_\epsilon^2} = \frac{K}{1-K}\left(1 - \rho^2(1-K)\right)$, which follows from combining the two steady-state conditions in the proof of Proposition 3. By the law of iterated expectations, $E\left[s_t\psi(s_t)\right] = E\left(\epsilon_t s_t\right) = 1$, where the final equality uses the independence of $\epsilon_t$ from $E_t$ and $U_t$. Therefore:

$$E\left[s_t^2\,\mathcal{S}(s_t)\right] = E\left(s_t^2\right) - E\left[s_t\psi(s_t)\right] = \frac{1}{1-K} - 1 = K\,E\left(s_t^2\right),$$

which is the dynamic analogue of the identity $\bar{S} = E\left(\frac{g_t^2}{Eg_t^2}S(g_t)\right)$ in Section 3.5.

Second, the proof of Proposition 3 shows that $\psi(s) \sim \frac{s}{1+c_E}$ as $s \longrightarrow +\infty$, where $c_E \equiv \frac{K^\nu\rho^\nu}{1-(1-K)^\nu\rho^\nu}$, and the same limit applies to $\frac{\psi(s)}{s}$ as $s \longrightarrow -\infty$ by the symmetry of the shocks. Therefore:

$$\lim_{|s|\to\infty}\mathcal{S}(s) = \frac{c_E}{1+c_E} \equiv \mathcal{S}_\infty > 0 \quad \text{and} \quad C = \frac{\mathcal{S}_\infty - K}{K},$$

where the second expression follows from comparing (IA.8), expressed in units of revisions, with Proposition 3. The inequality $C < 0$ established in the proof of Proposition 3 is therefore equivalent to $\mathcal{S}_\infty < K$.

Finally, we establish that there is a range of revisions for which $\mathcal{S}(\cdot) > K$, along with the crossing point $\overline{R}$. Note that $\mathcal{S}(\cdot)$ is even and continuous away from zero: $\psi(\cdot)$ is odd because the shocks are symmetric, and $\psi(\cdot)$ is continuous because $f_\epsilon(\cdot)$ is bounded, which implies that the density of $s_t$, denoted $f_{s_t}(\cdot)$, is continuous and strictly positive. Suppose, toward a contradiction, that $\mathcal{S}(s) \leq K$ for all $s > 0$. The tail limit implies that there exists an $s_2$ such that $\mathcal{S}(s) - K \leq \frac{\mathcal{S}_\infty - K}{2} < 0$ for all $s \geq s_2$, so the evenness of $\mathcal{S}(\cdot)$, the symmetry of $s_t$, and the weighted-average identity imply:

$$0 = E\left[s_t^2\left(\mathcal{S}(s_t) - K\right)\right] = 2\int_0^\infty s^2\left(\mathcal{S}(s) - K\right)f_{s_t}(s)ds \leq \left(\mathcal{S}_\infty - K\right)\int_{s_2}^\infty s^2 f_{s_t}(s)ds < 0,$$

which is a contradiction. Therefore, there exists an $s_1 > 0$ with $\mathcal{S}(s_1) > K$, which by continuity holds over a range of revisions, and the intermediate value theorem implies that there exists $\bar{s}$ between $s_1$ and $s_2$ such that $\mathcal{S}(\bar{s}) = K$. Therefore, $\overline{R} = K\rho\sigma_\epsilon\bar{s}$, which satisfies $\mathcal{S}(\overline{R}) = K$.

## B.6 Additional Derivations in Section 4.3

We first derive (16). Define the gross return on a stock as $R_{t+1} = \frac{P_{t+1}+D_{t+1}}{P_t}$, which has sales $S_t$, earnings $\mathcal{E}_t$, and a payout ratio $DE_t = \frac{D_t}{\mathcal{E}_t}$. Consistent with our variable definitions and transformations in Section 1, denote the (log) growth rate of sales as $g_t = \log S_t - \log S_{t-1}$. Assume that earnings growth is a constant fraction of sales growth: $\log\left(\frac{\mathcal{E}_t}{\mathcal{E}_{t-1}}\right) = \gamma g_t$. Denoting lower-case letters as logs and following Campbell and Shiller (1988), we can approximate the price-earnings ratio to first order around the mean price-dividend ratio as:

$$p_t - e_t = \kappa + \gamma g_{t+1} - r_{t+1} + (1-c)(d_{t+1} - e_{t+1}) + c(p_{t+1} - e_{t+1}),$$

where $c = \frac{e^{\overline{pd}}}{1+e^{\overline{pd}}}$, $\overline{pd}$ is the mean of the log price-dividend ratio, and $\kappa$ is a constant. Assuming a constant log payout ratio of $\overline{de}$, we can iterate the first equation forward to obtain:

$$p_t - e_t = \frac{\kappa}{1-c} + \overline{de} + \sum_{k=1}^{\infty} c^{k-1}\left(\gamma g_{it+k} - r_{it+k}\right) + \lim_{k\to\infty} c^k (p_{it+k} - e_{it+k}).$$

Imposing the usual transversality condition, the previous equation becomes:

$$p_t - e_t = \frac{\kappa}{1-c} + \overline{de} + \sum_{k=1}^{\infty} c^{k-1}\left(\gamma g_{it+k} - r_{it+k}\right).$$

Now, rearranging the first equation above, we have

$$r_{t+1} = \kappa + \gamma g_{t+1} + (1-c)\overline{de} + c(p_{t+1} - e_{t+1}) - (p_t - e_t).$$

Letting $F_t$ denote investors' subjective beliefs, the previous two equations imply:

$$F_{t+1}r_{t+1} - F_t r_{t+1} = \gamma F_{t+1} g_{t+1} - \gamma F_t g_{t+1} + c\left[(p_{t+1} - e_{t+1}) - F_t(p_{t+1} - e_{t+1})\right].$$

Rewriting, we obtain:

$$r_{t+1} - F_t r_{t+1} = \gamma\left(F_{t+1} - F_t\right)\sum_{k=0}^{\infty} c^k g_{t+1+k} - \left(F_{t+1} - F_t\right)\sum_{k=1}^{\infty} c^k r_{t+1+k}.$$

To derive predictions based on our model of subjective expectations, we follow Bouchaud et al. (2019) and Nagel and Xu (2022) and assume that subjective risk premia are constant and equal to the risk-free rate, $R_f$, plus a constant risk premium, $\pi$. Under this assumption, the final term in the previous equation is zero. Since $F_{t+1}g_{t+1} = g_{t+1}$, shifting the time index back one period gives (16):

$$r_t = \log(R_f + \pi) + \gamma\left[(g_t - F_{t-1}g_t) + \sum_{k=1}^{\infty} c^k REV_t g_{t+k}\right].$$

We now derive the link between return predictability and the error-revision relationship. Under the linear forecasting rule in (9), the revision at horizon $k$ is:

$$\begin{aligned} REV_t g_{t+k} &= F_t g_{t+k} - F_{t-1} g_{t+k} \\ &= \rho^k K (g_t - F_{t-1} g_t). \end{aligned}$$

For the one-period revision $REV_t \equiv REV_t g_{t+1}$, the return equation above therefore implies:

$$\begin{aligned} r_t - \log(R_f + \pi) &= \gamma \left(1 + \frac{Kc\rho}{1 - c\rho}\right) (g_t - F_{t-1} g_t) \\ &= \frac{\gamma}{\rho K} \left(1 + \frac{Kc\rho}{1 - c\rho}\right) REV_t. \end{aligned}$$

Applying the same expression one period forward gives:

$$r_{t+1} - \log(R_f + \pi) = \gamma \left(1 + \frac{Kc\rho}{1 - c\rho}\right) ERR_{t+1}.$$

Since the constants multiplying $ERR_{t+1}$ and $REV_t$ in the previous two equations have the same sign, taking correlations gives:

$$\text{Corr}\left(r_{t+1}, r_t\right) = \text{Corr}\left(ERR_{t+1}, REV_t\right).$$